

SDD V1 — Section 1

Document Control & Purpose

1.1 Document Information

Field	Value
Document Name	System Design Document (SDD)
Product Name	TODP (The Opportunity Discovery Platform)
Version	V1.0
Status	Draft
Owner	Engineering Team
Reviewers	Architecture Board
Approval Authority	Product Governance Board
Last Updated	TBD

1.2 Purpose

This System Design Document (SDD) defines the technical architecture, system components, interfaces, deployment model, operational characteristics, and implementation blueprint for TODP.

The SDD translates approved requirements from:

- Product Requirements Document (PRD V1)
- Architecture Decision Document (ADD V1)
- Domain Design Document (DDD V1)

into an implementable system design.

The SDD serves as the authoritative engineering reference for:

- Software development
 - Infrastructure deployment
 - Security implementation
 - Testing
 - Observability
 - Operations
 - Future maintenance
-

1.3 Scope

This document covers:

Application Layer

- React Frontend
- FastAPI Backend
- API Gateway Layer
- Authentication Layer
- Search Layer
- Recommendation Layer
- Ranking Engine
- Authority Verification Engine
- Evidence Collection Engine

Data Layer

- PostgreSQL
- pgvector
- Redis

Retrieval Layer

- BM25 Retrieval
- Vector Retrieval
- Hybrid Retrieval

Observability Layer

- OpenTelemetry
- Prometheus
- Grafana

Infrastructure Layer

- Containerization
- Deployment Architecture
- Scaling Strategy
- Security Controls
- Disaster Recovery

1.4 Out Of Scope

The following items are not defined in this document:

Business Requirements

Covered by:

- PRD V1

Domain Models

Covered by:

- DDD V1

Architecture Decisions

Covered by:

- ADD V1

Organizational Processes

- Hiring plans
- Team structure
- Budget approvals
- Vendor negotiations

1.5 Design Goals

The system design shall satisfy the following goals.

G1 — Evidence First

Every recommendation must be supported by verifiable evidence.

G2 — Authority Before Recommendation

Authorities must be identified before opportunities are recommended.

G3 — Ranking Before Recommendation

Ranking must occur before recommendation generation.

G4 — Traceability

Every result must be traceable to source evidence.

G5 — Auditability

All ranking and recommendation decisions must be auditable.

G6 — Reproducibility

Identical inputs must produce reproducible outputs.

G7 — Transparency

No hidden ranking logic shall exist.

G8 — Vendor Neutrality

No sponsored placement or paid prioritization shall influence ranking.

G9 — Scalability

System shall support future expansion from local discovery to international discovery.

G10 — Extensibility

New opportunity types, authority types, and ranking dimensions shall be addable without redesigning core architecture.

1.6 Intended Audience

This document is intended for:

Engineering

- Backend Engineers
- Frontend Engineers
- Platform Engineers
- DevOps Engineers

Architecture

- Solution Architects
- Enterprise Architects

Operations

- SRE Teams
- Support Teams

Governance

- Product Owners
- Review Boards
- Auditors

1.7 Relationship To Other Documents

Document	Purpose
PRD V1	Product requirements
ADD V1	Architecture decisions
DDD V1	Domain model and business rules
SDD V1	Technical implementation design

1.8 Success Criteria

The system design shall enable implementation of a platform capable of:

1. Discovering opportunities across all hierarchy levels:
 - Institution
 - Local
 - District
 - State
 - National
 - International
2. Supporting all user journeys:
 - Discover Online
 - Discover Physical
 - Create Something Similar
 - Create Something New
3. Applying the universal ranking model:

Total Cost =
Money Cost +
Time Cost +
Distance Cost

4. Producing transparent, reproducible, evidence-backed opportunity recommendations.
-

SDD V1 — Section 2

System Context & High-Level Architecture

2.1 Purpose

This section defines the overall system context, architectural boundaries, primary actors, major subsystems, and high-level data flows of TODP.

The objective is to provide a common architectural view before detailed component design is introduced in subsequent sections.

2.2 System Overview

TODP (The Opportunity Discovery Platform) is an evidence-first opportunity discovery system that enables users to:

- Discover opportunities
- Evaluate opportunities
- Compare opportunities
- Create similar opportunities
- Create novel opportunities

across the following hierarchy:

- Institution
- Local
- District
- State
- National
- International

The platform applies transparent ranking based on:

Total Cost =
Money Cost +
Time Cost +
Distance Cost

before any recommendation is presented.

2.3 Architectural Principles

The architecture shall enforce the following principles:

AP-01 Evidence First

Evidence must exist before recommendations are generated.

AP-02 Authority Before Recommendation

Authorities must be identified before opportunities are evaluated.

AP-03 Ranking Before Recommendation

Ranking calculations must execute before recommendation generation.

AP-04 Explainability

Every recommendation shall provide traceable justification.

AP-05 Auditability

All ranking and recommendation decisions shall be reproducible.

AP-06 Stateless Services

Application services shall remain stateless wherever possible.

AP-07 API First

All functionality shall be exposed through versioned APIs.

AP-08 Horizontal Scalability

Core services shall support horizontal scaling.

2.4 Primary Actors

User

Consumes opportunity discovery services.

Capabilities:

- Search opportunities
- Explore authorities
- Compare opportunities
- Create opportunities

- View evidence
 - View rankings
-

System Administrator

Maintains operational health.

Capabilities:

- Monitor system
 - Configure services
 - Manage indexing
 - Audit operations
-

Data Sources

Provide opportunity and authority information.

Examples:

- Government portals
 - Educational institutions
 - Research organizations
 - Public databases
 - Open data repositories
-

2.5 High-Level Architecture

TODP consists of five logical layers.

Layer 1 — Presentation Layer

Components:

- React Web Application

Responsibilities:

- User interaction
- Search interfaces
- Opportunity exploration
- Ranking visualization
- Evidence visualization

Layer 2 — API Layer

Components:

- FastAPI Gateway

Responsibilities:

- Request validation
 - Routing
 - Authentication
 - Authorization
 - Rate limiting
-

Layer 3 — Application Services Layer

Components:

- Discovery Service
- Authority Service
- Evidence Service
- Ranking Service
- Recommendation Service
- Opportunity Creation Service

Responsibilities:

- Business workflows
 - Domain orchestration
 - Retrieval coordination
-

Layer 4 — Retrieval & Intelligence Layer

Components:

- BM25 Retrieval Engine
- Vector Retrieval Engine
- Hybrid Retrieval Engine

Responsibilities:

- Document retrieval
- Similarity search

- Context generation
-

Layer 5 — Data Layer

Components:

- PostgreSQL
- pgvector
- Redis

Responsibilities:

- Transactional storage
 - Vector storage
 - Caching
 - Session storage
-

2.6 External Systems

The following external systems may interact with TODP.

Authority Sources

Provide authoritative information.

Examples:

- Government agencies
 - Universities
 - Institutions
-

Opportunity Sources

Provide opportunity data.

Examples:

- Public programs
- Research grants
- Courses
- Competitions
- Funding opportunities

Mapping Services

Provide distance calculations.

Responsibilities:

- Route estimation
- Travel distance calculation

Identity Providers

Provide authentication.

Examples:

- Internal authentication
- OAuth providers

2.7 Core System Flow

Discovery Flow

1. User submits query.
2. Discovery Service receives request.
3. Retrieval Layer executes search.
4. Authority Service validates authorities.
5. Evidence Service gathers evidence.
6. Ranking Service calculates rankings.
7. Recommendation Service generates results.
8. Results returned to user.

Creation Flow

1. User selects opportunity.
2. Opportunity analyzed.
3. Similarity search performed.
4. Opportunity patterns extracted.
5. User receives:
 - Similar opportunity suggestions
 - Novel opportunity insights

2.8 Bounded Context Mapping

The following DDD bounded contexts map to deployable services.

Bounded Context	Service
Discovery	Discovery Service
Authority	Authority Service
Evidence	Evidence Service
Ranking	Ranking Service
Recommendation	Recommendation Service
Opportunity Creation	Opportunity Creation Service

2.9 Non-Functional Targets

Availability

99.9%

API Response Time

P95 < 500 ms

Search Response Time

P95 < 2 seconds

Ranking Response Time

P95 < 1 second

Scalability Target

1 million opportunities

100,000 authorities

10 million evidence records

2.10 Architecture Summary

TODP follows a layered architecture consisting of:

- Presentation Layer
- API Layer
- Application Services Layer
- Retrieval Layer
- Data Layer

The architecture ensures:

- Evidence-first discovery
- Authority validation
- Transparent ranking
- Traceability
- Auditability
- Reproducibility

while remaining horizontally scalable and extensible for future opportunity domains.

SDD Progress

- ☐ Section 1 — Document Control & Purpose
- ☐ Section 2 — System Context & High-Level Architecture
- ☐ Section 3 — Logical Architecture & Service Decomposition

SDD V1 — Section 3

Logical Architecture & Service Decomposition

3.1 Purpose

This section defines the logical architecture of TODP and the decomposition of the system into logical services, bounded contexts, responsibilities, ownership boundaries, and interaction patterns.

The purpose of this decomposition is to:

- Maintain clear separation of concerns.
- Preserve domain integrity.

- Enable independent evolution of system capabilities.
- Support future scalability.
- Ensure traceability and auditability.
- Align implementation with DDD V1.
- Prevent business logic duplication.

This section describes logical services, not deployment topology. Physical deployment architecture is defined in later sections.

3.2 Architectural Philosophy

TODP is fundamentally a Discovery Platform rather than a Content Platform.

The primary responsibility of the platform is not to create information but to:

- Discover information.
- Verify information.
- Rank information.
- Explain information.
- Trace information.

The platform therefore follows the principle:

Evidence → Authority → Ranking → Recommendation

and never:

Recommendation → Ranking → Evidence

This ordering is a non-negotiable architectural constraint.

3.3 Architectural Style

TODP adopts a:

Modular Layered Service-Oriented Architecture

The architecture combines:

- Layered Architecture
- Domain-Driven Design
- Service-Oriented Architecture
- API-First Design

while avoiding unnecessary microservice complexity during V1.

Why Not Pure Monolith?

A pure monolith would:

- Increase coupling
 - Reduce ownership clarity
 - Complicate future scaling
 - Increase deployment risk
-

Why Not Full Microservices?

A full microservice architecture introduces:

- Network complexity
- Distributed transactions
- Operational overhead
- Increased deployment complexity

before product-market validation.

Chosen Approach

TODP V1 shall implement:

Modular Monolith

with:

- Clear service boundaries
- Independent modules
- Internal APIs
- Shared database

allowing future migration to microservices without domain redesign.

3.4 Architectural Layers

TODP is organized into five logical layers.

Layer 1 — Presentation Layer

Purpose

Provide user interaction capabilities.

Components

- React Web Application

Responsibilities

- User input
- User navigation
- Search interfaces
- Ranking visualization
- Evidence visualization
- Recommendation presentation

Constraints

The presentation layer shall not contain:

- Ranking logic
 - Retrieval logic
 - Authority validation logic
 - Business rules
-

Layer 2 — API Layer

Purpose

Expose platform functionality through stable contracts.

Components

- FastAPI API Gateway

Responsibilities

- Request validation
- Authentication

- Authorization
- Rate limiting
- Routing
- API versioning

Constraints

Business decisions shall not be implemented in the API layer.

Layer 3 — Application Layer

Purpose

Orchestrate business workflows.

Responsibilities

- Service coordination
- Workflow execution
- Domain orchestration

Constraints

Application services orchestrate domain operations but do not own domain rules.

Layer 4 — Domain Layer

Purpose

Implement business logic.

Responsibilities

- Opportunity rules
- Authority rules
- Ranking rules
- Evidence rules

Characteristics

This is the most critical layer of TODP.

All business decisions originate here.

Layer 5 — Infrastructure Layer

Purpose

Provide technical capabilities.

Components

- PostgreSQL
- pgvector
- Redis
- Retrieval engines
- Observability stack

Responsibilities

- Persistence
- Caching
- Search
- Telemetry
- Monitoring

3.5 Service Decomposition Principles

Service decomposition follows these rules.

SDP-01 Single Business Purpose

Each service shall own one primary business capability.

SDP-02 Domain Ownership

Each service owns its domain logic.

No other service may duplicate that logic.

SDP-03 Explicit Interfaces

Services communicate only through defined contracts.

SDP-04 Stateless Processing

Application services shall remain stateless whenever possible.

SDP-05 Auditability

All service decisions shall be traceable.

SDP-06 Future Extraction Readiness

A service should be extractable into an independent microservice without redesign.

3.6 Core Logical Services

The following services represent logical ownership boundaries.

Discovery Service

Mission

Find opportunities relevant to user intent.

Responsibilities

- Query interpretation
- Search orchestration
- Retrieval execution
- Candidate generation

Inputs

- User query

- User filters
- User location

Outputs

- Candidate opportunities

Does Not Own

- Ranking
- Recommendation
- Authority validation

Authority Service

Mission

Identify and verify authoritative entities.

Responsibilities

- Authority registration
- Authority validation
- Authority metadata management
- Authority confidence assessment

Examples

Authorities may include:

- Institutions
- Government bodies
- Educational organizations
- Research organizations

Outputs

Verified authority records.

Evidence Service

Mission

Manage evidence supporting opportunities.

Responsibilities

- Evidence ingestion
- Evidence normalization
- Evidence linkage
- Evidence traceability

Outputs

Evidence packages linked to opportunities.

Ranking Service

Mission

Determine opportunity ordering.

Responsibilities

- Cost calculation
- Ranking computation
- Tie resolution
- Ranking explanation

Governing Formula

Total Cost =
Money Cost +
Time Cost +
Distance Cost

Outputs

Ordered opportunity list.

Recommendation Service

Mission

Convert ranked opportunities into user-facing recommendations.

Responsibilities

- Recommendation generation

- Recommendation explanation
- Recommendation packaging

Constraint

Recommendations may not override rankings.

Opportunity Creation Service

Mission

Support opportunity creation journeys.

Responsibilities

- Similarity analysis
- Pattern extraction
- Opportunity gap detection
- Opportunity synthesis support

User Journeys Supported

- Create Something Similar
 - Create Something New
-

3.7 Bounded Context Mapping

Logical services are derived directly from DDD bounded contexts.

Bounded Context	Logical Service
Discovery	Discovery Service
Authority	Authority Service
Evidence	Evidence Service
Ranking	Ranking Service
Recommendation	Recommendation Service
Opportunity Creation	Opportunity Creation Service

This mapping shall remain stable unless DDD changes.

3.8 Service Ownership Matrix

Capability	Owning Service
Search	Discovery
Retrieval	Discovery
Authority Validation	Authority
Evidence Management	Evidence
Cost Calculation	Ranking
Ranking	Ranking
Recommendation Generation	Recommendation
Similarity Analysis	Opportunity Creation
Novel Opportunity Support	Opportunity Creation

No capability may have multiple owners.

3.9 Service Interaction Model

The canonical discovery workflow is:

User Request



Discovery Service



Authority Service



Evidence Service



Ranking Service



Recommendation Service



User Response

Each downstream service enriches the request.

No service may bypass the ranking stage.

3.10 Decision Ownership Rules

To prevent ambiguity, ownership of major decisions is explicitly assigned.

Decision	Owner
Opportunity Eligibility	Discovery
Authority Validity	Authority
Evidence Validity	Evidence
Opportunity Ranking	Ranking
Recommendation Packaging	Recommendation
Similarity Assessment	Opportunity Creation

Only the owning service may make the decision.

3.11 Future Service Expansion

Future versions may introduce:

Analytics Service

Usage analytics and insights.

Notification Service

Alerts and subscriptions.

Workflow Service

Complex multi-step opportunity workflows.

Collaboration Service

Shared discovery and planning.

These services are intentionally excluded from V1.

3.12 Architectural Constraints

The following constraints are mandatory.

AC-01

AI shall not be considered an authority.

AC-02

Recommendations shall not bypass ranking.

AC-03

Evidence shall exist before recommendation.

AC-04

Authority verification shall occur before ranking.

AC-05

No hidden ranking factors may exist.

AC-06

All rankings shall be reproducible.

AC-07

All recommendation outputs shall be traceable.

3.13 Architecture Summary

TODP adopts a Modular Layered Service-Oriented Architecture with clearly defined ownership boundaries.

The architecture is designed to:

- Support Evidence First principles.
- Support Authority Before Recommendation.
- Preserve auditability.
- Preserve traceability.
- Enable future scaling.
- Enable future microservice extraction.

This logical architecture serves as the foundation for all subsequent physical, deployment, data, retrieval, and security architecture decisions.

SDD Progress

- ☐ Section 1 — Document Control & Purpose
- ☐ Section 2 — System Context & High-Level Architecture
- ☐ Section 3 — Logical Architecture & Service Decomposition
- ☐ Section 4 — Physical Architecture & Deployment Topology

SDD V1 — Section 4

Physical Architecture & Deployment Topology

4.1 Purpose

This section defines the physical architecture of TODP, including deployment topology, runtime components, infrastructure boundaries, network flows, environment strategy, scalability approach, and operational deployment model.

While Section 3 defined the logical architecture of the platform, this section defines how those logical components are physically deployed and executed.

The objective is to ensure:

- Scalability
- Reliability
- Operational simplicity
- Security
- Observability
- Future extensibility

without introducing unnecessary infrastructure complexity during V1.

4.2 Physical Architecture Principles

The deployment architecture shall adhere to the following principles.

PAP-01 Simplicity First

Infrastructure complexity shall be minimized until justified by measurable system needs.

PAP-02 Stateless Compute

Application instances shall remain stateless whenever possible.

State shall be stored only in dedicated persistence layers.

PAP-03 Horizontal Scalability

All application components shall support horizontal scaling.

PAP-04 Infrastructure Observability

Every runtime component shall expose telemetry.

PAP-05 Infrastructure Reproducibility

Infrastructure shall be deployable through version-controlled definitions.

PAP-06 Environment Consistency

Development, staging, and production environments shall be structurally similar.

4.3 High-Level Deployment Topology

TODP V1 shall be deployed as a multi-tier architecture.

Client Layer



Presentation Layer



API Layer



Application Layer



Data Layer



Observability Layer

The architecture separates user interaction, application execution, persistence, and monitoring concerns.

4.4 Runtime Components

The following runtime components constitute the production deployment.

Component 1 — React Frontend

Technology

React

Purpose

User interface delivery.

Responsibilities

- Search interface
- Opportunity exploration
- Ranking visualization
- Evidence visualization
- Recommendation presentation

Scaling Strategy

Horizontal

Multiple frontend instances may be deployed.

Component 2 — FastAPI Backend

Technology

FastAPI

Purpose

Business workflow execution.

Responsibilities

- API execution
- Service orchestration
- Authentication enforcement
- Ranking execution
- Recommendation generation

Scaling Strategy

Horizontal

Multiple backend replicas may run simultaneously.

Component 3 — PostgreSQL Database

Technology

PostgreSQL

Purpose

Primary transactional datastore.

Responsibilities

- Opportunity storage
- Authority storage
- Evidence storage
- Audit storage
- User storage

Scaling Strategy

Primary-replica architecture.

Component 4 — pgvector Extension

Technology

pgvector

Purpose

Vector similarity storage and retrieval.

Responsibilities

- Embedding storage
- Similarity search
- Semantic retrieval

Scaling Strategy

Co-located with PostgreSQL in V1.

Component 5 — Redis

Technology

Redis

Purpose

In-memory performance layer.

Responsibilities

- Query caching
- Session caching
- Temporary state management
- Rate limiting support

Scaling Strategy

Single node initially.

Cluster-capable in future versions.

Component 6 — Observability Stack

Technologies

- OpenTelemetry
- Prometheus
- Grafana

Purpose

System monitoring and diagnostics.

Responsibilities

- Metrics collection
 - Tracing
 - Alerting
 - Dashboarding
-

4.5 Network Topology

The platform shall operate within a controlled network boundary.

External Traffic

Internet

↓

Frontend

↓

API Gateway

↓

Backend Services

↓

Databases

Direct database access from the public internet shall be prohibited.

4.6 Environment Strategy

TODP shall maintain separate environments.

Development Environment

Purpose:

Developer productivity.

Characteristics:

- Local execution
 - Mock integrations
 - Test datasets
-

Staging Environment

Purpose:

Pre-production validation.

Characteristics:

- Production-like configuration
- Integration testing

- Performance testing
-

Production Environment

Purpose:

Live user traffic.

Characteristics:

- Hardened security
 - Monitoring enabled
 - Backup enabled
 - Restricted access
-

4.7 Deployment Units

The system shall be deployed using containerized workloads.

Frontend Deployment Unit

Contains:

- React application
- Static assets

Deployable independently.

Backend Deployment Unit

Contains:

- FastAPI application
- Domain services
- Retrieval orchestration

Deployable independently.

Database Deployment Unit

Contains:

- PostgreSQL
- pgvector

Managed separately from application workloads.

Cache Deployment Unit

Contains:

- Redis

Managed independently.

Observability Deployment Unit

Contains:

- Prometheus
- Grafana
- Telemetry collectors

Managed independently.

4.8 Containerization Strategy

All application workloads shall be packaged as containers.

Objectives

- Consistency
 - Portability
 - Reproducibility
 - Isolation
-

Container Categories

Frontend Container

Hosts React application.

Backend Container

Hosts FastAPI services.

Telemetry Container

Hosts observability agents.

4.9 Scaling Model

TODP shall initially scale horizontally at the application layer.

Frontend Scaling

Additional frontend instances may be added behind a load balancer.

Backend Scaling

Additional FastAPI instances may be added behind a load balancer.

Database Scaling

Scaling progression:

Stage 1

Single PostgreSQL instance

↓

Stage 2

Read replicas

↓

Stage 3

Partitioning

↓

Stage 4

Distributed architecture

Redis Scaling

Scaling progression:

Single Node

↓

Replica

↓

Cluster

4.10 Availability Strategy

The platform shall target:

Availability: 99.9%

Planned Downtime

Allowed during:

- Maintenance windows
 - Infrastructure upgrades
-

Unplanned Downtime Reduction

Achieved through:

- Health checks
 - Monitoring
 - Automated restart policies
 - Backup procedures
-

4.11 Load Balancing Strategy

Traffic distribution shall occur through load balancing.

Frontend Load Balancing

Purpose:

Distribute user traffic.

Backend Load Balancing

Purpose:

Distribute API requests.

Benefits

- High availability
 - Scalability
 - Fault isolation
-

4.12 Infrastructure Security Boundaries

Infrastructure shall be segmented into security zones.

Public Zone

Components:

- Frontend

Accessible from internet.

Application Zone

Components:

- FastAPI services

Accessible only through controlled interfaces.

Data Zone

Components:

- PostgreSQL
- pgvector
- Redis

Not publicly accessible.

Monitoring Zone

Components:

- Prometheus
- Grafana

Restricted administrative access only.

4.13 Infrastructure Observability Topology

Telemetry shall flow through the following path.

Application

↓

OpenTelemetry

↓

Prometheus

↓

Grafana

Collected telemetry includes:

- Request metrics
 - Error metrics
 - Latency metrics
 - Resource utilization
 - Distributed traces
-

4.14 Backup Architecture

The platform shall support backup and recovery operations.

Database Backups

Frequency:

Daily

Retention:

Configurable

Configuration Backups

Frequency:

On deployment changes

Audit Record Preservation

Audit records shall be preserved according to governance requirements.

4.15 Disaster Recovery Strategy

The disaster recovery objective is restoration of critical platform capabilities.

Recovery Priorities

Priority 1

Database recovery

Priority 2

Backend recovery

Priority 3

Frontend recovery

Priority 4

Observability recovery

Recovery Objectives

RTO (Recovery Time Objective)

To be defined during production readiness review.

RPO (Recovery Point Objective)

To be defined during production readiness review.

4.16 Future Deployment Evolution

The deployment architecture shall support future migration toward:

- Kubernetes
- Service mesh
- Independent microservices
- Multi-region deployment
- Distributed search infrastructure

without requiring domain redesign.

4.17 Architectural Decisions Derived From ADD

The following approved decisions directly influence deployment topology.

Decision	Impact
React Frontend	Separate frontend deployment
FastAPI Backend	Separate API deployment
PostgreSQL	Primary transactional datastore
pgvector	Semantic retrieval support
Redis	High-speed cache layer
OpenTelemetry	Unified telemetry
Prometheus	Metrics platform
Grafana	Observability dashboards

These decisions are considered frozen unless ADD V1 changes.

4.18 Architecture Summary

TODP V1 adopts a containerized multi-tier deployment architecture consisting of:

- React Frontend
- FastAPI Backend
- PostgreSQL
- pgvector
- Redis
- OpenTelemetry
- Prometheus
- Grafana

The architecture prioritizes:

- Simplicity
- Scalability
- Reliability
- Security
- Observability
- Future evolution

while avoiding premature operational complexity.

SDD Progress

- ☐ Section 1 — Document Control & Purpose
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- ☐ Section 4 — Physical Architecture & Deployment Topology
- ☐ Section 5 — Infrastructure Architecture & Environment Design

SDD V1 — Section 5

Infrastructure Architecture & Environment Design

5.1 Purpose

This section defines the infrastructure architecture required to support TODP in development, staging, and production environments.

The purpose of this architecture is to provide:

- Secure execution environments
- Consistent deployments
- Scalable infrastructure
- Operational reliability
- Disaster recovery readiness
- Environment reproducibility

This section focuses on infrastructure design rather than application logic.

5.2 Infrastructure Objectives

The infrastructure architecture shall achieve the following objectives.

IO-01 Reliability

Infrastructure shall provide stable and predictable execution of platform services.

IO-02 Scalability

Infrastructure shall support increasing user traffic and data volume without architectural redesign.

IO-03 Security

Infrastructure shall enforce strong isolation and access controls.

IO-04 Observability

Infrastructure shall expose telemetry necessary for monitoring and troubleshooting.

IO-05 Reproducibility

Infrastructure shall be provisioned using version-controlled definitions.

IO-06 Recoverability

Infrastructure failures shall be recoverable using documented procedures.

5.3 Infrastructure Domains

The infrastructure is divided into five major domains.

Domain 1 — Compute Infrastructure

Provides execution environments for platform workloads.

Responsibilities:

- Container execution
 - Runtime resource allocation
 - Horizontal scaling
-

Domain 2 — Data Infrastructure

Provides persistent storage.

Responsibilities:

- Relational storage
 - Vector storage
 - Backup management
-

Domain 3 — Network Infrastructure

Provides secure communication.

Responsibilities:

- Traffic routing
 - Segmentation
 - Load balancing
-

Domain 4 — Observability Infrastructure

Provides monitoring capabilities.

Responsibilities:

- Metrics
 - Logging
 - Tracing
 - Alerting
-

Domain 5 — Delivery Infrastructure

Provides deployment automation.

Responsibilities:

- Build pipelines
 - Release pipelines
 - Environment promotion
-

5.4 Environment Strategy

TODP shall maintain strict environment separation.

Environment 1 — Development

Purpose:

Local developer productivity.

Characteristics:

- Individual developer environments
- Mock integrations allowed
- Test datasets permitted
- Reduced security constraints

Primary Users:

- Developers
-

Environment 2 — Integration

Purpose:

Cross-service integration testing.

Characteristics:

- Shared environment
- Real service interactions
- Automated testing

Primary Users:

- Engineering teams
-

Environment 3 — Staging

Purpose:

Pre-production validation.

Characteristics:

- Production-like configuration
- Production-like infrastructure
- Performance testing
- Security validation

Primary Users:

- QA
 - Product teams
 - Release teams
-

Environment 4 — Production

Purpose:

Serve live users.

Characteristics:

- Hardened security
- Monitoring enabled
- Backup enabled
- Restricted access

Primary Users:

- End users
-

5.5 Environment Isolation Rules

The following rules shall apply.

EIR-01

Production data shall never be copied into development environments without approved sanitization.

EIR-02

Development credentials shall not function in production.

EIR-03

Environment-specific secrets shall remain isolated.

EIR-04

Production access shall be restricted to authorized personnel.

EIR-05

Each environment shall maintain separate databases.

5.6 Compute Architecture

Compute Model

TODP V1 shall execute workloads using containerized services.

The compute layer shall host:

- Frontend workloads
 - Backend workloads
 - Observability workloads
-

Frontend Compute

Hosts:

- React application

Characteristics:

- Stateless
 - Horizontally scalable
-

Backend Compute

Hosts:

- FastAPI application
- Domain services
- Retrieval orchestration

Characteristics:

- Stateless
 - Horizontally scalable
-

Telemetry Compute

Hosts:

- OpenTelemetry collectors
- Monitoring agents

Characteristics:

- Independent scaling
-

5.7 Resource Allocation Principles

Resources shall be allocated according to workload characteristics.

CPU Intensive Workloads

Examples:

- Ranking calculations
- Retrieval operations

Priority:

High CPU allocation

Memory Intensive Workloads

Examples:

- Redis caching
- Vector search operations

Priority:

High memory allocation

Storage Intensive Workloads

Examples:

- Evidence storage
- Audit storage

Priority:

High persistent storage allocation

5.8 Network Architecture

The network architecture shall enforce security boundaries.

Network Zone 1 — Public Zone

Accessible from the internet.

Contains:

- Frontend endpoints
-

Network Zone 2 — Application Zone

Restricted internal access.

Contains:

- FastAPI services
-

Network Zone 3 — Data Zone

Highly restricted access.

Contains:

- PostgreSQL
- pgvector
- Redis

Network Zone 4 — Operations Zone

Administrative access only.

Contains:

- Prometheus
- Grafana
- Monitoring systems

5.9 Internal Traffic Rules

Allowed traffic:

Frontend

↓

Backend

↓

Database

Monitoring

↓

All Components

Blocked traffic:

Internet

× Database

Internet

× Redis

Internet

× Monitoring Systems

5.10 Load Balancing Architecture

Load balancing shall distribute traffic across application instances.

Frontend Load Balancer

Responsibilities:

- Traffic distribution
 - Availability management
-

Backend Load Balancer

Responsibilities:

- API request distribution
 - Instance failover
-

5.11 Secrets Management

Secrets shall never be stored in source code.

Managed Secrets

Examples:

- Database passwords
- API keys

- Service credentials
 - Authentication secrets
-

Secret Storage Requirements

Secrets shall be:

- Encrypted at rest
 - Access controlled
 - Auditable
-

Secret Rotation

Secrets shall support periodic rotation without application redesign.

5.12 Configuration Management

Configuration shall be externalized from application code.

Configuration Categories

Infrastructure Configuration

Examples:

- Network settings
 - Storage settings
-

Application Configuration

Examples:

- Feature flags
 - Timeouts
 - Limits
-

Environment Configuration

Examples:

- URLs
 - Environment variables
-

5.13 Infrastructure as Code

Infrastructure provisioning shall be automated.

Objectives

- Consistency
 - Auditability
 - Repeatability
-

Infrastructure Definition Requirements

All infrastructure definitions shall be:

- Version controlled
 - Reviewable
 - Reproducible
-

5.14 CI/CD Architecture

Deployment shall be automated through controlled pipelines.

Continuous Integration

Responsibilities:

- Build validation
- Unit testing
- Static analysis

Continuous Delivery

Responsibilities:

- Packaging
- Deployment preparation
- Artifact publishing

Continuous Deployment

Future capability.

Not mandatory for V1.

5.15 Deployment Promotion Model

Deployments shall move through environments in sequence.

Development

↓

Integration

↓

Staging

↓

Production

Direct deployment from development to production is prohibited.

5.16 Infrastructure Security Controls

Infrastructure security controls shall include:

- Network segmentation
 - Access controls
 - Secret management
 - Audit logging
 - Environment isolation
-

Least Privilege Principle

Every component shall operate with the minimum permissions required.

Administrative Access Control

Administrative actions shall be logged and auditable.

5.17 Logging Infrastructure

Infrastructure logs shall be collected centrally.

Log Categories

Application Logs

Generated by application services.

System Logs

Generated by infrastructure.

Security Logs

Generated by authentication and authorization systems.

Audit Logs

Generated by business actions.

5.18 Monitoring Infrastructure

The monitoring platform shall collect:

Availability Metrics

Examples:

- Uptime
 - Health status
-

Performance Metrics

Examples:

- Latency
 - Throughput
-

Resource Metrics

Examples:

- CPU utilization
 - Memory utilization
 - Storage utilization
-

Business Metrics

Examples:

- Search volume
 - Ranking volume
 - Recommendation volume
-

5.19 Infrastructure Backup Design

The infrastructure shall support recovery of critical assets.

Backup Scope

Includes:

- Databases
 - Configurations
 - Audit records
-

Backup Requirements

Backups shall be:

- Automated
 - Verified
 - Retained according to policy
-

5.20 Disaster Recovery Infrastructure

The disaster recovery architecture shall support restoration of:

1. Data
2. Application services
3. User access
4. Monitoring systems

in that order.

Disaster Recovery Principles

- Recover critical data first.
 - Restore core business functionality second.
 - Restore operational tooling third.
-

5.21 Future Infrastructure Evolution

The infrastructure architecture shall support future migration toward:

- Kubernetes orchestration
- Multi-region deployment
- Service mesh architecture
- Dedicated vector databases
- Distributed caching
- Global traffic routing

without requiring redesign of domain services.

5.22 Infrastructure Summary

TODP V1 infrastructure architecture provides:

- Environment isolation
- Secure networking
- Scalable compute
- Persistent data storage
- Automated delivery
- Centralized observability
- Disaster recovery readiness

while maintaining operational simplicity appropriate for V1 deployment.

The infrastructure is designed to support future growth without introducing premature operational complexity.

SDD Progress

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- ☐ Section 5 — Infrastructure Architecture & Environment Design
- ☐ Section 6 — Data Architecture & Data Management Strategy

SDD V1 — Section 6

Data Architecture & Data Management Strategy

6.1 Purpose

This section defines the data architecture of TODP, including data ownership, storage strategy, persistence principles, schema organization, data lifecycle management, auditability requirements, and retrieval support mechanisms.

The purpose of the data architecture is to ensure:

- Data integrity
- Data consistency
- Data traceability
- Data auditability
- Data scalability
- Retrieval efficiency
- Reproducibility of results

This section serves as the authoritative reference for all persistence-related design decisions.

6.2 Data Architecture Principles

The TODP data architecture shall be governed by the following principles.

DAP-01 Single Source of Truth

Every business entity shall have a single authoritative source of record.

Duplicate ownership is prohibited.

DAP-02 Evidence Preservation

Evidence records shall never be silently modified or overwritten.

Historical evidence must remain recoverable.

DAP-03 Traceability

Every recommendation shall be traceable to:

- Evidence
- Authority
- Ranking calculation

DAP-04 Auditability

All business-critical decisions shall leave auditable records.

DAP-05 Reproducibility

Historical rankings must be reproducible using historical inputs.

DAP-06 Storage Independence

Business logic shall not depend on storage implementation details.

DAP-07 Retrieval Optimization

Data architecture shall support both:

- Structured retrieval
- Semantic retrieval

6.3 Data Classification

TODP manages multiple categories of data.

Category 1 — Master Data

Represents authoritative platform entities.

Examples:

- Opportunities
- Authorities
- Institutions
- Locations

Characteristics:

- Long-lived
 - Highly referenced
 - Business critical
-

Category 2 — Transactional Data

Represents user and system activity.

Examples:

- Searches
- Rankings
- Recommendations
- User actions

Characteristics:

- High volume
 - Frequently generated
-

Category 3 — Reference Data

Provides supporting context.

Examples:

- Geographic data
- Administrative regions
- Opportunity classifications

Characteristics:

- Low change frequency

Category 4 — Audit Data

Provides governance records.

Examples:

- Ranking decisions
- Evidence decisions
- Administrative actions

Characteristics:

- Immutable
 - Long retention
-

Category 5 — Retrieval Data

Supports search operations.

Examples:

- Embeddings
- Search indexes
- Retrieval metadata

Characteristics:

- Optimized for search performance
-

6.4 Data Ownership Model

Every major domain owns its data.

Discovery Domain

Owns:

- Search records
- Search metadata

Authority Domain

Owns:

- Authority records
- Authority verification records

Evidence Domain

Owns:

- Evidence records
- Evidence links

Ranking Domain

Owns:

- Ranking results
- Ranking calculations

Recommendation Domain

Owns:

- Recommendation outputs

Opportunity Creation Domain

Owns:

- Similarity analyses
 - Opportunity generation records
-

6.5 Canonical Data Entities

The following entities represent the core platform model.

Opportunity

Represents a discoverable opportunity.

Examples:

- Course
 - Grant
 - Scholarship
 - Competition
 - Program
 - Job
 - Research Opportunity
-

Core Attributes

- Opportunity ID
 - Name
 - Description
 - Opportunity Type
 - Authority ID
 - Location
 - Status
 - Creation Timestamp
 - Update Timestamp
-

Authority

Represents an authoritative source.

Core Attributes

- Authority ID
- Name
- Authority Type
- Verification Status

- Confidence Score
-

Evidence

Represents supporting evidence.

Core Attributes

- Evidence ID
 - Evidence Type
 - Source Reference
 - Authority Reference
 - Collection Timestamp
-

Ranking Record

Represents ranking execution output.

Core Attributes

- Ranking ID
 - Opportunity ID
 - Money Cost
 - Time Cost
 - Distance Cost
 - Total Cost
 - Ranking Position
-

Recommendation Record

Represents recommendation output.

Core Attributes

- Recommendation ID
- User ID

- Opportunity ID
 - Ranking ID
 - Recommendation Timestamp
-

6.6 Relational Storage Architecture

TODP shall use PostgreSQL as the primary system of record.

Responsibilities

PostgreSQL stores:

- Business entities
 - Audit records
 - Transactional records
 - Metadata
-

Design Characteristics

- ACID compliance
 - Strong consistency
 - Referential integrity
 - Transaction support
-

6.7 Schema Organization Strategy

Schemas shall be organized by domain ownership.

discovery_schema

Owns discovery-related tables.

authority_schema

Owns authority-related tables.

evidence_schema

Owns evidence-related tables.

ranking_schema

Owns ranking-related tables.

recommendation_schema

Owns recommendation-related tables.

opportunity_creation_schema

Owns opportunity creation data.

audit_schema

Owns audit records.

system_schema

Owns infrastructure and operational metadata.

6.8 Primary Key Strategy

Every major entity shall use globally unique identifiers.

Requirements

Identifiers shall be:

- Unique
 - Stable
 - Non-reusable
-

Benefits

- Easier distributed evolution
 - Reduced collision risk
 - Simplified integration
-

6.9 Referential Integrity Strategy

Relationships shall be enforced through foreign key constraints.

Examples

Opportunity

↓

Authority

Evidence

↓

Authority

Recommendation

↓

Ranking

Ranking

↓

Opportunity

Objectives

- Data consistency
 - Integrity enforcement
 - Orphan prevention
-

6.10 Transaction Management

Business-critical operations shall execute within transactions.

Examples

Opportunity Creation

Create Opportunity

-

Link Authority

-

Link Evidence

=

Single Transaction

Ranking Generation

Store Ranking

-

Store Explanation

-

Store Audit Record

=

Single Transaction

6.11 Data Versioning Strategy

Certain entities require historical preservation.

Versioned Entities

- Opportunity
 - Authority
 - Evidence
-

Purpose

Support:

- Auditing
 - Historical analysis
 - Reproducibility
-

Requirements

Historical versions shall remain accessible.

6.12 Audit Data Architecture

Audit records are first-class entities.

Audit Scope

The following actions shall be auditable:

- Opportunity creation
 - Authority verification
 - Evidence validation
 - Ranking execution
 - Recommendation generation
 - Administrative actions
-

Audit Requirements

Audit records shall contain:

- Actor
 - Action
 - Timestamp
 - Previous State
 - New State
-

6.13 Search Data Architecture

Search operations require specialized storage.

Search Metadata

Stores:

- Search terms
 - Filters
 - Query statistics
-

Search Analytics

Stores:

- Query frequency
 - Search success rates
 - Retrieval metrics
-

6.14 Vector Data Architecture

Semantic retrieval requires vector storage.

Storage Technology

pgvector

Vectorized Entities

Initially:

- Opportunities
- Evidence

Future:

- Authorities
 - Recommendations
-

Stored Components

- Entity ID
 - Embedding Vector
 - Embedding Version
 - Generation Timestamp
-

6.15 Embedding Lifecycle Management

Embeddings must remain manageable over time.

Trigger Events

Embedding regeneration may occur when:

- Opportunity changes
- Evidence changes
- Embedding model changes

Version Tracking

Each embedding shall contain:

- Model version
- Generation timestamp

6.16 Caching Data Architecture

Redis shall provide temporary storage.

Cached Data Types

- Search results
- Ranking results
- Frequently accessed records

Cache Principles

Cache is not a system of record.

Persistent storage remains PostgreSQL.

6.17 Data Retention Strategy

Different data categories require different retention periods.

Master Data

Retain indefinitely.

Audit Data

Retain according to governance requirements.

Search Analytics

Retention policy configurable.

Temporary Cache Data

Expire automatically.

6.18 Data Archival Strategy

Historical records may be archived.

Candidate Data

- Historical rankings
 - Historical recommendations
 - Historical searches
-

Objectives

- Reduce operational storage load
 - Preserve historical records
-

6.19 Data Security Architecture

Sensitive data shall be protected.

Protection Mechanisms

- Encryption at rest
 - Encryption in transit
 - Access controls
 - Audit logging
-

Access Principle

Least privilege access shall be enforced.

6.20 Data Quality Controls

Data quality shall be monitored continuously.

Validation Categories

Completeness

Required fields present.

Consistency

Relationships remain valid.

Accuracy

Data reflects authoritative sources.

Timeliness

Data remains current.

Uniqueness

Duplicate records minimized.

6.21 Data Recovery Strategy

Data recovery capabilities shall include:

- Backup restoration
 - Point-in-time recovery
 - Audit recovery
-

Recovery Priority

1. Master Data
 2. Audit Data
 3. Retrieval Data
 4. Analytics Data
-

6.22 Data Scalability Strategy

The architecture shall support future growth.

Growth Dimensions

- Opportunity count
 - Authority count
 - Evidence count
 - Search volume
 - Vector volume
-

Scaling Path

Stage 1

Single PostgreSQL instance

↓

Stage 2

Read replicas

↓

Stage 3

Partitioning

↓

Stage 4

Distributed storage architecture

6.23 Data Governance Rules

The following governance rules are mandatory.

DG-01

Every entity must have an owner.

DG-02

Evidence shall remain traceable.

DG-03

Historical rankings shall remain reproducible.

DG-04

Audit records shall not be silently deleted.

DG-05

No recommendation shall exist without ranking lineage.

DG-06

No ranking shall exist without evidence lineage.

6.24 Data Architecture Summary

The TODP data architecture combines:

- PostgreSQL for transactional persistence
- pgvector for semantic retrieval
- Redis for caching

while enforcing:

- Traceability
- Auditability
- Reproducibility
- Data ownership
- Governance

The architecture supports both operational workloads and future large-scale opportunity discovery without requiring fundamental redesign.

SDD Progress

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- □ Section 7 — Database Design & Persistence Architecture

SDD V1 — Section 7

Database Design & Persistence Architecture

7.1 Purpose

This section defines the database architecture and persistence design for TODP.

While Section 6 established data architecture principles, ownership boundaries, and governance requirements, this section defines how those principles are physically implemented within PostgreSQL and pgvector.

The objectives of the database architecture are:

- Reliable persistence
- High data integrity
- Strong consistency
- Efficient retrieval
- Auditability
- Scalability
- Maintainability

This section serves as the authoritative reference for database implementation.

7.2 Database Technology Stack

Primary Database

Technology:

PostgreSQL

Purpose:

Primary transactional datastore.

Responsibilities:

- Business entities
- Relationships

- Transactions
 - Audit records
 - Metadata
-

Vector Database Layer

Technology:

pgvector

Purpose:

Semantic retrieval support.

Responsibilities:

- Embedding storage
 - Similarity search
 - Hybrid retrieval support
-

Cache Layer

Technology:

Redis

Purpose:

Performance optimization.

Responsibilities:

- Query caching
- Session caching
- Temporary state storage

Redis shall not be treated as a persistent datastore.

7.3 Database Design Principles

DDP-01 Database as System of Record

PostgreSQL shall remain the authoritative source of truth.

No external component may become the primary record source.

DDP-02 Domain Ownership

Tables shall be organized according to domain ownership.

DDP-03 Explicit Relationships

Relationships shall be represented explicitly through foreign keys.

DDP-04 Strong Consistency

Business-critical operations shall prioritize consistency over eventual consistency.

DDP-05 Audit Preservation

Historical records shall remain recoverable.

DDP-06 Future Scalability

Schema design shall support future partitioning and distribution.

7.4 Database Logical Structure

The database shall be organized into domain-oriented schemas.

discovery_schema

Purpose:

Discovery-related persistence.

Examples:

- Searches
 - Search metadata
 - Search analytics
-

authority_schema

Purpose:

Authority management.

Examples:

- Authorities
 - Verification records
 - Authority classifications
-

evidence_schema

Purpose:

Evidence management.

Examples:

- Evidence
 - Evidence sources
 - Evidence mappings
-

ranking_schema

Purpose:

Ranking operations.

Examples:

- Ranking results
 - Ranking calculations
 - Ranking explanations
-

recommendation_schema

Purpose:

Recommendation operations.

Examples:

- Recommendations
 - Recommendation history
-

opportunity_creation_schema

Purpose:

Opportunity generation workflows.

Examples:

- Similarity analysis
 - Pattern analysis
 - Opportunity derivations
-

audit_schema

Purpose:

System auditability.

Examples:

- Audit events
 - Change history
 - Administrative actions
-

system_schema

Purpose:

Operational metadata.

Examples:

- Configuration
 - Feature flags
 - Internal settings
-

7.5 Table Classification Model

Tables shall be classified into functional categories.

Master Tables

Store core business entities.

Examples:

- opportunities
- authorities
- evidence

Characteristics:

- Long-lived
 - Highly referenced
 - Business critical
-

Transaction Tables

Store business activities.

Examples:

- searches
- rankings
- recommendations

Characteristics:

- High growth
 - Frequent writes
-

Mapping Tables

Store relationships.

Examples:

- opportunity_evidence
- authority_evidence

Characteristics:

- Many-to-many relationships
-

Audit Tables

Store governance records.

Examples:

- audit_events
- ranking_history

Characteristics:

- Immutable
-

Analytics Tables

Store reporting data.

Examples:

- search_statistics
- ranking_statistics

Characteristics:

- Aggregated data

7.6 Entity Persistence Standards

Every persisted business entity shall contain standardized metadata.

Required Columns

id

created_at

updated_at

created_by

updated_by

status

version

Objectives

- Consistency
 - Auditing
 - Change tracking
-

7.7 Identifier Strategy

Primary Key Standard

All business entities shall use globally unique identifiers.

Characteristics:

- Stable
- Non-reusable
- Environment independent

Identifier Requirements

Identifiers shall never:

- Change
 - Be recycled
 - Depend on business meaning
-

7.8 Foreign Key Strategy

Relationships shall be enforced at the database level.

Example

Opportunity

↓

Authority

Authority

↓

Evidence

Recommendation

↓

Ranking

Ranking

↓

Opportunity

Benefits

- Referential integrity
 - Data consistency
 - Error prevention
-

7.9 Transaction Architecture

Transactions shall protect business consistency.

Transaction Rules

Transactions shall be:

- Atomic
 - Consistent
 - Isolated
 - Durable
-

Example

Create Opportunity

-

Create Authority Link

-

Create Evidence Link

=

Single Transaction

Rollback Rule

Failure of any step shall rollback the entire transaction.

7.10 Concurrency Control

The database shall support concurrent operations safely.

Concurrency Objectives

Prevent:

- Lost updates
 - Dirty reads
 - Inconsistent state
-

Strategy

Use PostgreSQL transaction isolation mechanisms.

Version Columns

Version tracking shall support optimistic locking where required.

7.11 Indexing Architecture

Indexes shall be created based on access patterns.

Index Categories

Primary Key Indexes

Automatically generated.

Foreign Key Indexes

Required for relationship navigation.

Search Indexes

Required for retrieval operations.

Audit Indexes

Required for governance queries.

7.12 Search Optimization Strategy

Frequently searched fields shall be indexed.

Candidate Fields

Opportunity Name

Opportunity Type

Authority Name

Location

Status

Creation Timestamp

Objectives

- Low latency retrieval
- Reduced table scans

7.13 Full Text Search Architecture

PostgreSQL full-text search capabilities shall support BM25-style retrieval workflows.

Searchable Content

Opportunity descriptions

Evidence descriptions

Authority descriptions

Search Goals

- Keyword retrieval
 - Relevance ranking
 - Hybrid retrieval support
-

7.14 Vector Storage Architecture

pgvector shall store semantic representations.

Vectorized Entities

Opportunities

Primary semantic search target.

Evidence

Supporting semantic retrieval.

Future Candidates

Authorities

Recommendations

Opportunity patterns

Stored Components

entity_id

embedding

embedding_version

generated_at

7.15 Embedding Persistence Rules

Embeddings must remain traceable.

Required Metadata

Embedding Model

Embedding Version

Generation Time

Source Entity

Regeneration Triggers

Opportunity update

Evidence update

Model update

7.16 Partitioning Strategy

Initial deployment shall not require partitioning.

However, schemas shall be partition-ready.

Future Partition Candidates

searches

rankings

recommendations

audit_events

Partition Dimensions

Time-based

Region-based

Organization-based

7.17 Database Scaling Strategy

Scaling progression:

Stage 1

Single Database

↓

Stage 2

Read Replicas

↓

Stage 3

Partitioned Tables



Stage 4

Distributed Architecture

Scaling Objective

Avoid redesign during growth.

7.18 Persistence Performance Targets

Target metrics:

Query Latency

P95 < 100 ms

Transaction Latency

P95 < 250 ms

Search Latency

P95 < 500 ms

Vector Retrieval Latency

P95 < 300 ms

7.19 Audit Persistence Architecture

Auditability is a first-class requirement.

Audited Events

Opportunity creation

Authority verification

Evidence validation

Ranking execution

Recommendation generation

Administrative actions

Audit Record Contents

Actor

Timestamp

Action

Entity

Previous State

New State

Correlation ID

7.20 Historical Data Architecture

Historical preservation supports reproducibility.

Historical Records

Opportunity versions

Authority versions

Evidence versions

Ranking versions

Recommendation versions

Objective

Enable reconstruction of historical platform state.

7.21 Data Retention Implementation

Retention policies shall be enforced through automated processes.

Long-Term Retention

Master Data

Audit Data

Governance Data

Medium-Term Retention

Search Analytics

Operational Metrics

Short-Term Retention

Temporary Processing Data

7.22 Backup Persistence Strategy

Database backups shall support:

Full Backups

Entire database state.

Incremental Backups

Changed records only.

Point-in-Time Recovery

Restore specific moments in history.

7.23 Recovery Architecture

Recovery priorities:

1. PostgreSQL
 2. Audit Records
 3. Vector Storage
 4. Analytics Data
-

Recovery Objective

Restore business continuity while preserving auditability.

7.24 Database Security Architecture

Security controls shall include:

Authentication

Database access authentication required.

Authorization

Role-based access controls enforced.

Encryption

Encryption at rest.

Encryption in transit.

Auditing

Administrative database actions logged.

7.25 Database Governance Rules

The following rules are mandatory.

DG-DB-01

Every table must have an owner.

DG-DB-02

Every entity must have traceable lineage.

DG-DB-03

Audit records must remain immutable.

DG-DB-04

No recommendation record may exist without ranking lineage.

DG-DB-05

No ranking record may exist without evidence lineage.

DG-DB-06

Database changes must be version controlled.

7.26 Future Evolution

The database architecture shall support future migration toward:

- Dedicated vector databases
- Distributed PostgreSQL clusters
- Data lake integration
- Real-time analytics pipelines
- Multi-region replication

without requiring redesign of the domain model.

7.27 Database Architecture Summary

TODP utilizes PostgreSQL as the primary system of record and pgvector as the semantic retrieval layer.

The architecture is designed to provide:

- Strong consistency
- Referential integrity
- Auditability
- Traceability
- Reproducibility
- Retrieval performance
- Future scalability

while maintaining clear domain ownership and governance controls.

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SDD V1 — Section 8

Retrieval Architecture & Search Strategy

8.1 Purpose

This section defines the retrieval architecture used by TODP to discover, retrieve, rank, and present opportunities.

Retrieval is the foundational capability of TODP because every user journey begins with discovering relevant information.

The retrieval architecture must support:

- Evidence First
- Authority Before Recommendation
- Ranking Before Recommendation
- Traceability
- Auditability
- Reproducibility

The retrieval architecture shall never operate as a black box.

Every retrieval decision must be explainable and traceable.

8.2 Retrieval Philosophy

TODP is not a recommendation-first platform.

TODP is a discovery-first platform.

Therefore retrieval must prioritize:

1. Relevant opportunities
2. Verified authorities
3. Supporting evidence
4. Transparent ranking
5. Recommendation generation

The system shall not generate recommendations from information that has not first been retrieved and validated.

8.3 Retrieval Objectives

The retrieval architecture shall satisfy the following objectives.

RO-01 Recall

The system should retrieve all potentially relevant opportunities.

RO-02 Precision

The system should minimize irrelevant opportunities.

RO-03 Authority Awareness

The system should identify authoritative sources during retrieval.

RO-04 Evidence Awareness

Retrieved opportunities should contain evidence lineage.

RO-05 Explainability

Retrieved results should explain why they were returned.

RO-06 Scalability

Retrieval performance should remain predictable as data volume grows.

8.4 Retrieval Architecture Overview

TODP shall implement a Hybrid Retrieval Architecture.

The architecture combines:

- Structured Retrieval
- Keyword Retrieval
- Semantic Retrieval
- Authority Retrieval
- Evidence Retrieval

to produce candidate opportunities.

High-Level Retrieval Flow

User Query

↓

Query Processing

↓

Candidate Retrieval

↓

Authority Retrieval



Evidence Retrieval



Ranking



Recommendation



Response

8.5 Retrieval Layers

The retrieval system is divided into five logical layers.

Layer 1 — Query Understanding Layer

Purpose:

Interpret user intent.

Responsibilities:

- Query normalization
- Filter extraction
- Intent detection
- Location detection

Input:

User query

Output:

Structured retrieval request

Layer 2 — Candidate Retrieval Layer

Purpose:

Find potentially relevant opportunities.

Responsibilities:

- BM25 retrieval
- Vector retrieval
- Metadata retrieval

Output:

Candidate opportunity pool

Layer 3 — Authority Retrieval Layer

Purpose:

Identify authoritative entities.

Responsibilities:

- Authority lookup
- Authority validation
- Authority enrichment

Output:

Verified authority information

Layer 4 — Evidence Retrieval Layer

Purpose:

Collect supporting evidence.

Responsibilities:

- Evidence lookup
- Evidence aggregation
- Evidence linking

Output:

Evidence package

Layer 5 — Ranking Preparation Layer

Purpose:

Prepare ranking inputs.

Responsibilities:

- Data aggregation
- Context assembly
- Cost calculation preparation

Output:

Ranking-ready opportunity set

8.6 Query Understanding Architecture

The query understanding layer converts user language into retrieval instructions.

Query Components

A query may contain:

Intent

Examples:

- Discover
 - Compare
 - Explore
 - Create Similar
 - Create New
-

Domain

Examples:

- Scholarship
 - Grant
 - Course
 - Competition
-

Geography

Examples:

- Institution
 - Local
 - District
 - State
 - National
 - International
-

Constraints

Examples:

- Budget
 - Distance
 - Time
-

8.7 Query Normalization

Queries shall be normalized before retrieval.

Normalization Activities

- Lowercasing
 - Stop-word removal
 - Synonym mapping
 - Typo correction
 - Entity normalization
-

Objective

Improve retrieval consistency.

8.8 Retrieval Modes

TODP supports multiple retrieval modes.

Mode 1 — Keyword Retrieval

Uses BM25-style ranking.

Best for:

- Exact matches
 - Named entities
 - Structured searches
-

Mode 2 — Semantic Retrieval

Uses embeddings and vector similarity.

Best for:

- Intent discovery
 - Conceptual similarity
 - Opportunity exploration
-

Mode 3 — Metadata Retrieval

Uses structured filters.

Best for:

- Geography
 - Opportunity type
 - Authority filtering
-

Mode 4 — Hybrid Retrieval

Combines:

Keyword

-

Semantic

-

Metadata

This shall be the default retrieval mode.

8.9 BM25 Retrieval Architecture

BM25 retrieval supports lexical matching.

Indexed Content

Opportunity Name

Opportunity Description

Evidence Description

Authority Name

Authority Description

Advantages

- Fast
 - Explainable
 - Deterministic
-

Limitations

- Cannot fully capture meaning
 - Sensitive to wording
-

8.10 Vector Retrieval Architecture

Vector retrieval supports semantic matching.

Process

Document



Embedding Generation



Vector Storage



Similarity Search



Candidate Retrieval

Advantages

- Captures meaning
 - Handles paraphrasing
 - Discovers related opportunities
-

Limitations

- Less transparent than lexical search
 - Requires embedding maintenance
-

8.11 Hybrid Retrieval Strategy

Hybrid retrieval combines multiple signals.

Inputs

BM25 Score

-

Vector Score

-

Metadata Match

Output

Unified Candidate Set

Objective

Maximize both:

Recall

and

Precision

8.12 Authority Retrieval Architecture

Authority retrieval is a unique TODP capability.

Purpose

Identify authoritative entities before recommendation.

Authority Sources

Examples:

- Government organizations
 - Educational institutions
 - Research organizations
 - Professional bodies
-

Authority Retrieval Steps

1. Retrieve authority.
 2. Verify authority.
 3. Calculate confidence.
 4. Attach authority metadata.
-

8.13 Evidence Retrieval Architecture

Evidence retrieval supports the Evidence First principle.

Purpose

Attach evidence to opportunities.

Evidence Sources

Examples:

- Official documents
 - Public records
 - Published policies
 - Verified announcements
-

Evidence Retrieval Steps

1. Locate evidence.
 2. Validate evidence.
 3. Link evidence.
 4. Preserve traceability.
-

8.14 Candidate Opportunity Generation

Candidate generation occurs before ranking.

Inputs

BM25 Results

-

Vector Results

-

Metadata Results

Output

Candidate Opportunity Pool

Purpose

Ensure ranking operates only on relevant opportunities.

8.15 Deduplication Strategy

Duplicate opportunities must be removed.

Duplicate Signals

Name similarity

Description similarity

Authority similarity

Location similarity

Objective

Prevent duplicate recommendations.

8.16 Retrieval Confidence Model

Each retrieved opportunity shall receive a confidence score.

Confidence Factors

Retrieval relevance

Authority quality

Evidence quality

Metadata completeness

Purpose

Support transparency.

8.17 Retrieval Explainability

Every retrieved result shall be explainable.

Explainability Components

Why retrieved

Why ranked

Which authority

Which evidence

Example

Retrieved because:

- Matches scholarship query
 - Verified university authority
 - Evidence available
-

8.18 Retrieval Traceability

Retrieval lineage shall be preserved.

Lineage Components

Query

↓

Retrieval Method

↓

Authority

↓

Evidence

↓

Ranking



Recommendation

Purpose

Support auditing and reproducibility.

8.19 Retrieval Performance Targets

BM25 Retrieval

P95 < 200 ms

Vector Retrieval

P95 < 300 ms

Hybrid Retrieval

P95 < 500 ms

Authority Retrieval

P95 < 200 ms

Evidence Retrieval

P95 < 300 ms

8.20 Retrieval Failure Handling

The retrieval system shall degrade gracefully.

BM25 Failure

Fallback:

Vector Retrieval

Vector Failure

Fallback:

BM25 Retrieval

Authority Failure

Opportunity returned with warning status.

Evidence Failure

Recommendation generation blocked if evidence requirements are not met.

8.21 Retrieval Support for Discovery Hierarchy

Retrieval shall operate across all hierarchy levels.

Levels

Institution

Local

District

State

National

International

Retrieval Objective

Discover opportunities at the most relevant level first.

8.22 Retrieval Support for User Journeys

The retrieval architecture supports all four journeys.

Discover Online

Focus:

Digital opportunities.

Discover Physical

Focus:

Location-dependent opportunities.

Create Something Similar

Focus:

Similarity retrieval.

Create Something New

Focus:

Pattern extraction and opportunity gaps.

8.23 Similarity Retrieval Architecture

Used by Opportunity Creation Service.

Inputs

Existing opportunity

Process

Embedding generation

↓

Vector search

↓

Pattern analysis

Output

Similar opportunities

8.24 Opportunity Gap Detection

Supports creation of novel opportunities.

Objective

Identify areas where:

Demand > Available Opportunities

Inputs

Opportunity coverage

Authority coverage

Geographic coverage

Output

Potential opportunity gaps

8.25 Future Retrieval Evolution

Future versions may support:

- Multi-vector retrieval
- Graph retrieval
- Knowledge graph integration
- Cross-region retrieval optimization
- Federated retrieval

without changing core retrieval principles.

8.26 Retrieval Governance Rules

The following rules are mandatory.

RG-01

No recommendation without retrieval lineage.

RG-02

No ranking without retrieved opportunity.

RG-03

No recommendation without authority context.

RG-04

No recommendation without evidence context.

RG-05

Retrieval methods must be auditable.

RG-06

Retrieval scores must be reproducible.

8.27 Retrieval Architecture Summary

TODP implements a Hybrid Retrieval Architecture combining:

- BM25 Retrieval
- Vector Retrieval
- Metadata Retrieval
- Authority Retrieval
- Evidence Retrieval

to support:

- Discovery
- Ranking
- Recommendation
- Similarity Analysis
- Opportunity Creation

while preserving:

- Evidence First
- Authority Before Recommendation
- Traceability
- Auditability
- Reproducibility

The retrieval architecture forms the foundation upon which ranking and recommendation systems are built.

SDD Progress

- ☐ Section 1 — Document Control & Purpose
- ☐ Section 2 — System Context & High-Level Architecture
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- ☐ Section 4 — Physical Architecture & Deployment Topology
- ☐ Section 5 — Infrastructure Architecture & Environment Design
- ☐ Section 6 — Data Architecture & Data Management Strategy
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- ☐ Section 8 — Retrieval Architecture & Search Strategy
- ☐ Section 9 — Ranking Architecture & Decision Engine

SDD V1 — Section 9

Ranking Architecture & Decision Engine

9.1 Purpose

This section defines the Ranking Architecture and Decision Engine used by TODP.

The ranking system is the most critical business component of the platform because ranking determines opportunity ordering before recommendations are generated.

TODP follows the principle:

Ranking Before Recommendation

Therefore:

- Recommendations do not determine rankings.
- AI does not determine rankings.
- Advertisers do not determine rankings.
- Sponsors do not determine rankings.

Only transparent ranking logic may determine opportunity order.

The ranking engine must ensure:

- Fairness
- Transparency
- Reproducibility
- Auditability
- Explainability
- Traceability

for every ranking decision.

9.2 Ranking Philosophy

TODP is not a recommendation-first system.

TODP is a ranking-first discovery system.

The purpose of ranking is to answer:

Among all valid opportunities, which opportunity is most practical for the user?

The ranking engine therefore evaluates opportunities based on measurable cost rather than subjective preference.

9.3 Core Ranking Principles

RP-01 Ranking Before Recommendation

Ranking must complete before recommendation generation begins.

RP-02 Transparency

All ranking factors must be visible.

No hidden factors are permitted.

RP-03 Determinism

Identical inputs must produce identical rankings.

RP-04 Reproducibility

Historical rankings must be reproducible.

RP-05 Explainability

Every ranking must be explainable.

RP-06 Auditability

Every ranking decision must be traceable.

RP-07 No Paid Influence

Sponsored content shall never alter ranking positions.

9.4 Universal Ranking Model

TODP uses a universal ranking formula.

Formula

Total Cost =
Money Cost +
Time Cost +
Distance Cost

Expanded Form

Total Cost(O) =
MC(O)
+
TC(O)
+
DC(O)

Where:

O = Opportunity

MC = Money Cost

TC = Time Cost

DC = Distance Cost

Ranking Rule

Lower Total Cost = Higher Rank

Higher Total Cost = Lower Rank

Objective

Identify opportunities requiring the least overall user expenditure.

9.5 Ranking Lifecycle

Ranking occurs after retrieval and before recommendation generation.

Ranking Workflow

Query



Retrieval



Authority Verification



Evidence Validation



Cost Calculation



Ranking



Recommendation



User Response

Architectural Constraint

Recommendation systems shall never modify ranking outputs.

9.6 Ranking Inputs

The ranking engine consumes validated opportunity records.

Required Inputs

Opportunity

Authority

Evidence

Location

User Context

Input Validation

Opportunities lacking required ranking inputs may be excluded.

9.7 Money Cost Model

Money Cost represents direct financial expenditure.

Examples

Tuition

Registration Fees

Application Fees

Travel Costs

Accommodation Costs

Equipment Costs

Objective

Measure financial burden.

Representation

Money Cost shall be represented in a normalized currency model.

9.8 Time Cost Model

Time Cost represents user time investment.

Examples

Application Time

Preparation Time

Travel Time

Participation Time

Completion Time

Objective

Measure time commitment.

Representation

Time Cost shall be represented using standardized time units.

9.9 Distance Cost Model

Distance Cost represents physical separation between user and opportunity.

Examples

Institution Distance

Event Distance

Program Location Distance

Authority Location Distance

Objective

Measure geographic effort.

Representation

Distance Cost shall be represented using standardized distance units.

9.10 Cost Normalization Framework

The ranking engine requires comparable cost dimensions.

Purpose

Ensure Money, Time, and Distance contribute consistently.

Normalization Requirements

Each cost dimension shall:

- Use consistent units
 - Support comparison
 - Support auditing
-

Design Constraint

Normalization rules must remain transparent and documented.

9.11 Ranking Pipeline Architecture

The ranking engine executes a multi-stage pipeline.

Stage 1

Input Validation

Verify ranking prerequisites.

Stage 2

Cost Extraction

Extract raw cost components.

Stage 3

Cost Normalization

Normalize values.

Stage 4

Cost Aggregation

Calculate Total Cost.

Stage 5

Ranking Calculation

Order opportunities.

Stage 6

Tie Resolution

Resolve equal rankings.

Stage 7

Ranking Explanation Generation

Create ranking rationale.

Stage 8

Audit Recording

Persist ranking lineage.

9.12 Opportunity Eligibility Rules

Not all retrieved opportunities are rankable.

Minimum Requirements

Verified Authority

-

Valid Evidence

-

Required Cost Inputs

Ineligible Opportunity Examples

Missing authority

Missing evidence

Incomplete cost data

9.13 Ranking Calculation Engine

For every candidate opportunity:

Step 1

Calculate Money Cost

Step 2

Calculate Time Cost

Step 3

Calculate Distance Cost

Step 4

Calculate Total Cost

Step 5

Store Ranking Record

Step 6

Generate Explanation

9.14 Tie-Breaking Architecture

Multiple opportunities may share the same Total Cost.

Tie-Breaking Objective

Produce deterministic rankings.

Tie-Breaking Hierarchy

1. Higher Evidence Quality
 2. Higher Authority Confidence
 3. More Complete Data
 4. Earlier Opportunity Availability
 5. Stable Identifier Ordering
-

Requirement

Tie-breaking must remain reproducible.

9.15 Ranking Explanation Engine

Every ranking decision must be explainable.

Explanation Components

Opportunity Rank

Total Cost

Money Cost

Time Cost

Distance Cost

Authority Information

Evidence Information

Example

Ranked #1 because:

- Lowest Total Cost
 - Verified Authority
 - Complete Evidence Package
-

9.16 Ranking Traceability

Every ranking shall maintain lineage.

Lineage Path

Query

↓

Retrieved Opportunity

↓

Authority

↓

Evidence

↓

Cost Components



Ranking



Recommendation

Objective

Support audits and investigations.

9.17 Ranking Audit Architecture

Ranking decisions shall be recorded.

Audit Scope

Input Data

Calculation Logic

Output Ranking

Explanation

Timestamp

Correlation ID

Benefits

- Governance
 - Compliance
 - Debugging
-

9.18 Ranking Reproducibility

The platform must reproduce historical rankings.

Required Inputs

Historical Opportunity

Historical Evidence

Historical Authority

Historical Cost Data

Historical Rules

Output

Identical ranking result

Objective

Support trust and accountability.

9.19 Ranking Confidence Model

Ranking confidence indicates trustworthiness of ranking inputs.

Confidence Factors

Authority Quality

Evidence Quality

Data Completeness

Cost Completeness

Purpose

Highlight ranking reliability.

9.20 Discovery Hierarchy Ranking

Ranking operates consistently across all hierarchy levels.

Levels

Institution

Local

District

State

National

International

Principle

Same ranking rules apply at every level.

9.21 User Journey Ranking Support

The ranking architecture supports all user journeys.

Discover Online

Rank digital opportunities.

Discover Physical

Rank location-based opportunities.

Create Something Similar

Rank similar opportunities.

Create Something New

Rank identified opportunity gaps.

9.22 Ranking Performance Targets

Candidate Ranking

P95 < 500 ms

Ranking Explanation Generation

P95 < 200 ms

Ranking Audit Persistence

P95 < 100 ms

End-to-End Ranking Pipeline

P95 < 1 second

9.23 Failure Handling Strategy

The ranking engine shall fail safely.

Missing Authority

Opportunity excluded.

Missing Evidence

Opportunity excluded.

Missing Cost Components

Opportunity marked incomplete.

Audit Failure

Ranking response may proceed but incident must be recorded.

9.24 Future Ranking Evolution

Future versions may support:

- Additional cost dimensions
- Personalized ranking views
- Scenario-based ranking
- Group ranking
- Multi-user ranking

without changing the fundamental ranking philosophy.

9.25 Ranking Governance Rules

The following rules are mandatory.

RG-01

No recommendation without ranking.

RG-02

No ranking without evidence.

RG-03

No ranking without authority.

RG-04

No hidden ranking factors.

RG-05

No sponsored ranking influence.

RG-06

Ranking logic must be documented.

RG-07

Ranking outputs must be reproducible.

RG-08

Ranking decisions must be auditable.

9.26 Architectural Constraints

The following constraints are non-negotiable.

AC-R-01

AI may explain rankings.

AI may not define ranking outcomes.

AC-R-02

Human governance owns ranking rules.

AC-R-03

Ranking must remain deterministic.

AC-R-04

Ranking must remain transparent.

AC-R-05

Ranking must remain traceable.

9.27 Ranking Architecture Summary

The TODP Ranking Architecture serves as the platform's primary decision engine.

The ranking engine:

- Operates after retrieval
- Operates before recommendation
- Uses transparent cost calculations
- Produces reproducible results
- Maintains audit lineage

using the universal formula:

Total Cost =
Money Cost +
Time Cost +
Distance Cost

The ranking architecture ensures that opportunity ordering remains objective, explainable, and free from hidden influence.

SDD Progress

- ☐ Section 1 — Document Control & Purpose
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- ☐ Section 9 — Ranking Architecture & Decision Engine
- ☐ Section 10 — Recommendation Architecture & Explanation Engine

SDD V1 — Section 10

Recommendation Architecture & Explanation Engine

10.1 Purpose

This section defines the Recommendation Architecture and Explanation Engine used by TODP.

The recommendation system is responsible for transforming ranked opportunities into understandable, actionable, and transparent outputs for users.

Unlike conventional recommendation systems, TODP recommendations do not determine what opportunities are shown.

The ranking engine determines opportunity ordering.

The recommendation engine is responsible for:

- Presenting opportunities
- Explaining opportunities
- Explaining rankings
- Providing evidence visibility
- Supporting decision-making

The recommendation engine shall never override ranking decisions.

10.2 Recommendation Philosophy

TODP follows the principle:

**Ranking Determines
Recommendation Explains**

The recommendation engine exists to help users understand ranking outcomes rather than create ranking outcomes.

This architectural separation is mandatory.

Traditional Recommendation Systems

Typical flow:

User Data

↓

Recommendation Model

↓

Recommendation



Explanation

TODP Recommendation Flow

Retrieval



Authority Verification



Evidence Validation



Ranking



Recommendation



Explanation

The recommendation layer operates only after ranking has completed.

10.3 Recommendation Objectives

RO-01 Transparency

Recommendations must clearly explain why opportunities appear.

RO-02 Explainability

Users must understand ranking outcomes.

RO-03 Evidence Visibility

Evidence supporting opportunities must be visible.

RO-04 Authority Visibility

Authorities supporting opportunities must be visible.

RO-05 Traceability

Recommendations must be traceable to ranking lineage.

RO-06 Reproducibility

Historical recommendations must be reproducible.

10.4 Recommendation Architecture Overview

The recommendation engine consists of five major components.

Component 1

Recommendation Assembly Engine

Component 2

Explanation Engine

Component 3

Evidence Presentation Engine

Component 4

Authority Presentation Engine

Component 5

Recommendation Audit Engine

10.5 Recommendation Lifecycle

The recommendation process begins only after ranking has completed.

Workflow

Retrieved Opportunities

↓

Authority Validation

↓

Evidence Validation

↓

Ranking

↓

Recommendation Assembly

↓

Explanation Generation

↓

Response Generation

↓

User

Architectural Constraint

Recommendations cannot bypass ranking.

10.6 Recommendation Inputs

The recommendation engine consumes validated ranking outputs.

Required Inputs

Ranked Opportunities

Authority Information

Evidence Information

Ranking Explanation

User Context

Input Validation

Recommendations shall not be generated if ranking lineage is unavailable.

10.7 Recommendation Assembly Engine

The assembly engine constructs recommendation packages.

Responsibilities

Collect ranked opportunities.

Attach ranking details.

Attach authority details.

Attach evidence details.

Prepare presentation objects.

Output

Recommendation Package

10.8 Recommendation Package Structure

Each recommendation package shall contain:

Opportunity Information

Name

Description

Type

Location

Status

Ranking Information

Rank Position

Total Cost

Money Cost

Time Cost

Distance Cost

Authority Information

Authority Name

Authority Type

Verification Status

Confidence Level

Evidence Information

Evidence References

Evidence Summary

Evidence Lineage

Explanation Information

Recommendation Explanation

Ranking Explanation

Evidence Explanation

10.9 Explanation Engine

The explanation engine generates user-facing reasoning.

Purpose

Help users understand:

Why an opportunity appears.

Why it received its rank.

What evidence supports it.

Which authority supports it.

Objective

Increase transparency and trust.

10.10 Recommendation Explanation Model

Recommendations shall explain:

Why Retrieved

Explanation of retrieval factors.

Why Ranked

Explanation of ranking factors.

Why Recommended

Explanation of recommendation assembly.

Supporting Evidence

Explanation of evidence sources.

Supporting Authority

Explanation of authority sources.

10.11 Ranking Explanation Integration

Recommendations must expose ranking rationale.

Example

Opportunity ranked #2 because:

Total Cost = 37

Money Cost = 10

Time Cost = 15

Distance Cost = 12

Requirement

Ranking details shall never be hidden.

10.12 Authority Presentation Engine

Authority information shall be visible.

Presented Information

Authority Name

Authority Type

Verification Status

Confidence Level

Objective

Allow users to evaluate source credibility.

10.13 Evidence Presentation Engine

Evidence information shall be visible.

Presented Information

Evidence Source

Evidence Type

Evidence Timestamp

Evidence Lineage

Objective

Support Evidence First principles.

10.14 Recommendation Confidence Model

Recommendations shall include confidence information.

Confidence Inputs

Authority Quality

Evidence Quality

Ranking Completeness

Data Completeness

Purpose

Communicate reliability of recommendations.

10.15 Recommendation Categories

Recommendations may be grouped according to user journeys.

Discover Online

Digital opportunities.

Discover Physical

Location-based opportunities.

Create Something Similar

Similarity-based opportunities.

Create Something New

Opportunity gap recommendations.

10.16 Recommendation Personalization Boundaries

TODP intentionally limits personalization.

Allowed Personalization

Location context.

User constraints.

Discovery preferences.

Disallowed Personalization

Hidden ranking modifications.

Behavior manipulation.

Sponsored influence.

Opaque preference weighting.

Objective

Preserve ranking transparency.

10.17 Recommendation Ordering Rules

Recommendation order shall be determined exclusively by ranking order.

Rule

Rank 1



Recommendation 1

Rank 2



Recommendation 2

Rank 3



Recommendation 3

Constraint

Recommendation engine may not reorder ranked results.

10.18 Recommendation Traceability

Every recommendation shall maintain lineage.

Lineage Path

Query



Retrieval



Authority



Evidence



Ranking



Recommendation

Purpose

Auditability and reproducibility.

10.19 Recommendation Audit Architecture

Recommendation generation shall be auditable.

Audit Scope

Recommendation Inputs

Recommendation Outputs

Explanation Outputs

Authority References

Evidence References

Timestamp

Correlation ID

Objective

Support governance and investigations.

10.20 Recommendation Reproducibility

Historical recommendations must be reproducible.

Required Inputs

Historical Query

Historical Retrieval Results

Historical Authorities

Historical Evidence

Historical Ranking

Historical Rules

Output

Identical recommendation package

10.21 Recommendation API Contract Requirements

All recommendation APIs shall expose:

Required Fields

Recommendation ID

Opportunity ID

Rank

Explanation

Authority Information

Evidence Information

Confidence Information

Timestamp

Objective

Provide complete transparency.

10.22 Recommendation Failure Handling

The recommendation system shall fail safely.

Missing Ranking

Recommendation generation prohibited.

Missing Authority

Recommendation generation prohibited.

Missing Evidence

Recommendation generation prohibited.

Explanation Failure

Fallback explanation may be generated.

Incident must be logged.

10.23 Recommendation Performance Targets

Recommendation Assembly

P95 < 200 ms

Explanation Generation

P95 < 300 ms

Recommendation Packaging

P95 < 100 ms

End-to-End Recommendation Processing

P95 < 500 ms

10.24 AI Usage Policy

AI may assist explanation generation.

AI shall not determine recommendation ordering.

Allowed AI Functions

Natural language explanations.

Summary generation.

Evidence summarization.

Authority summarization.

Prohibited AI Functions

Ranking manipulation.

Authority determination.

Evidence validation.

Recommendation reordering.

10.25 Recommendation Governance Rules

The following rules are mandatory.

RG-REC-01

No recommendation without ranking.

RG-REC-02

No recommendation without authority.

RG-REC-03

No recommendation without evidence.

RG-REC-04

Recommendation ordering must follow ranking ordering.

RG-REC-05

Recommendation explanations must remain transparent.

RG-REC-06

Recommendation lineage must be preserved.

RG-REC-07

Recommendation outputs must be reproducible.

10.26 Future Recommendation Evolution

Future versions may support:

- Multi-user recommendation views
- Team recommendations
- Scenario-based recommendations
- Comparative recommendation reports
- Decision support workspaces

without violating ranking governance principles.

10.27 Recommendation Architecture Summary

The TODP Recommendation Architecture serves as the platform's explanation and presentation layer.

The recommendation engine:

- Does not rank opportunities
- Does not reorder opportunities
- Does not override ranking decisions

Instead, it:

- Presents ranked opportunities
- Explains ranking outcomes
- Exposes authority information
- Exposes evidence information
- Maintains lineage and auditability

This architecture ensures recommendations remain transparent, reproducible, and aligned with the platform's Evidence First and Ranking Before Recommendation principles.

SDD Progress

- ☐ Section 1 — Document Control & Purpose
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- ☐ Section 11 — Authority Architecture & Verification Framework

SDD V1 — Section 11

Authority Architecture & Verification Framework

11.1 Purpose

This section defines the Authority Architecture and Verification Framework used by TODP.

Authority is a foundational concept within TODP because recommendations must not be generated solely from retrieved information.

Before an opportunity can participate in ranking and recommendation workflows, the platform must determine:

- Who is making the claim?
- What authority supports the claim?
- Is the authority verifiable?
- Can the authority be traced?
- Can the authority be audited?

TODP therefore enforces the principle:

Authority Before Recommendation

This principle applies across all discovery levels:

- Institution
- Local
- District
- State
- National
- International

11.2 Authority Philosophy

TODP distinguishes between:

Information

and

Authority

Information may be abundant.

Authority may be limited.

The existence of information alone does not qualify an opportunity for recommendation.

An authority must exist behind the information.

Example

A scholarship announcement exists.

The platform must determine:

- Which organization published it?

- Is the organization legitimate?
- Is the organization verifiable?
- Is the organization traceable?

Only after these questions are addressed may the opportunity proceed through ranking and recommendation workflows.

11.3 Authority Objectives

AO-01 Verification

Authorities must be verifiable.

AO-02 Traceability

Authorities must be traceable to identifiable entities.

AO-03 Explainability

Users must understand who supports an opportunity.

AO-04 Auditability

Authority decisions must be auditable.

AO-05 Reproducibility

Authority evaluations must be reproducible.

AO-06 Transparency

Authority evaluation criteria must be documented.

11.4 Authority Architecture Overview

The Authority Framework consists of five major components.

Component 1

Authority Registry

Component 2

Authority Verification Engine

Component 3

Authority Confidence Engine

Component 4

Authority Lineage Engine

Component 5

Authority Audit Engine

11.5 Authority Lifecycle

Authorities move through a defined lifecycle.

Stage 1

Authority Identification

Stage 2

Authority Registration

Stage 3

Authority Verification

Stage 4

Confidence Assessment

Stage 5

Authority Usage

Stage 6

Authority Monitoring

Stage 7

Authority Revalidation

11.6 Authority Registry

The Authority Registry serves as the authoritative catalog of recognized authorities.

Responsibilities

Store authority records.

Store authority metadata.

Store verification status.

Store confidence assessments.

Maintain authority history.

Objectives

Provide a single source of truth for authority information.

11.7 Authority Types

TODP supports multiple authority categories.

Educational Authority

Examples:

Universities

Colleges

Schools

Training Institutions

Government Authority

Examples:

Departments

Ministries

Public Agencies

Regulatory Bodies

Research Authority

Examples:

Research Institutes

Scientific Organizations

Academic Research Centers

Professional Authority

Examples:

Professional Associations

Accreditation Bodies

Industry Councils

Organizational Authority

Examples:

Companies

Foundations

Non-Profit Organizations

Community Authority

Examples:

Local Institutions

Community Organizations

Verified Local Bodies

11.8 Authority Identification

Authority identification determines the entity responsible for opportunity information.

Inputs

Opportunity Data

Evidence Data

Source Metadata

Published Information

Outputs

Authority Candidate

Objective

Associate every opportunity with a responsible authority.

11.9 Authority Verification Framework

Authority verification determines whether an authority can be trusted as a valid source.

Verification Goals

Confirm existence.

Confirm identity.

Confirm traceability.

Confirm legitimacy.

Verification Outcome

Verified

or

Unverified

11.10 Authority Verification Workflow

Step 1

Identify Authority

Step 2

Collect Supporting Evidence

Step 3

Validate Authority Information

Step 4

Calculate Confidence

Step 5

Persist Verification Result

Step 6

Enable Authority Usage

11.11 Authority Evidence Requirements

Authority verification must be evidence-supported.

Acceptable Evidence Categories

Official Documentation

Published Records

Institutional References

Government Records

Accreditation Records

Public Registries

Objective

Prevent unsupported authority assignment.

11.12 Authority Confidence Model

Authority confidence represents trustworthiness of authority records.

Purpose

Provide transparency regarding authority quality.

Confidence Inputs

Verification Success

Evidence Completeness

Identity Clarity

Authority Stability

Authority History

Output

Authority Confidence Score

11.13 Authority Confidence Levels

The framework shall support confidence classifications.

High Confidence

Authority strongly verified.

Medium Confidence

Authority reasonably verified.

Low Confidence

Authority partially verified.

Unverified

Authority not verified.

11.14 Authority Lineage Architecture

Authority lineage preserves authority history.

Lineage Components

Authority

↓

Evidence

↓

Verification

↓

Confidence Assessment

↓

Opportunity

↓

Ranking

↓

Recommendation

Objective

Support traceability.

11.15 Authority Usage Rules

Authority records may participate in platform workflows only when minimum requirements are satisfied.

Required Conditions

Authority Exists

-

Authority Identified

-

Authority Verified

Result

Authority Eligible

11.16 Authority Participation in Retrieval

Authority participates during retrieval.

Responsibilities

Authority Lookup

Authority Filtering

Authority Enrichment

Authority Validation

Objective

Improve retrieval quality.

11.17 Authority Participation in Ranking

Authority contributes to ranking readiness.

Responsibilities

Provide confidence information.

Provide legitimacy information.

Support ranking qualification.

Constraint

Authority does not determine ranking position.

Authority determines ranking eligibility.

11.18 Authority Participation in Recommendation

Authority contributes to recommendation transparency.

Responsibilities

Display authority information.

Display verification status.

Display confidence level.

Display authority lineage.

Objective

Allow users to evaluate trustworthiness.

11.19 Authority Audit Architecture

Authority decisions shall be auditable.

Audit Scope

Authority Creation

Authority Updates

Verification Decisions

Confidence Changes

Usage Decisions

Administrative Actions

Stored Information

Actor

Action

Timestamp

Previous State

New State

Correlation ID

11.20 Authority Revalidation Framework

Authority status may change over time.

Revalidation Triggers

Authority Update

Evidence Change

Verification Rule Change

Scheduled Review

Objective

Maintain authority accuracy.

11.21 Authority Expiration Model

Certain authority validations may become stale.

Expiration Triggers

Missing Revalidation

Conflicting Evidence

Authority Deactivation

Verification Failure

Result

Authority status reassessment.

11.22 Authority Search Architecture

Authorities shall be searchable entities.

Search Capabilities

Authority Name Search

Authority Type Search

Verification Status Search

Confidence Search

Geographic Search

Objective

Improve authority transparency.

11.23 Authority Performance Targets

Authority Lookup

P95 < 100 ms

Authority Verification

P95 < 500 ms

Confidence Assessment

P95 < 200 ms

Authority Retrieval

P95 < 150 ms

11.24 Authority Failure Handling

The authority framework shall fail safely.

Missing Authority

Opportunity becomes ineligible.

Unverified Authority

Opportunity becomes ineligible for recommendation.

Authority Service Failure

Recommendation generation suspended until authority evaluation completes.

Objective

Preserve platform trustworthiness.

11.25 AI Usage Policy

AI may assist authority explanation.

AI may assist authority summarization.

AI May Not

Determine authority validity.

Determine authority legitimacy.

Assign authority confidence.

Override verification outcomes.

Reason

Authority decisions must remain auditable and deterministic.

11.26 Authority Governance Rules

The following rules are mandatory.

AG-01

Every opportunity must have an authority.

AG-02

Authorities must be traceable.

AG-03

Authorities must be verifiable.

AG-04

Authority decisions must be auditable.

AG-05

Authority lineage must be preserved.

AG-06

Authority confidence calculations must be reproducible.

AG-07

Unverified authorities cannot support recommendations.

11.27 Architectural Constraints

The following constraints are non-negotiable.

AC-AUTH-01

Authority verification occurs before ranking.

AC-AUTH-02

Authority verification occurs before recommendation.

AC-AUTH-03

Authority lineage must remain available.

AC-AUTH-04

Authority confidence must be explainable.

AC-AUTH-05

Authority decisions must remain deterministic.

11.28 Authority Architecture Summary

The Authority Architecture establishes a structured framework for identifying, verifying, managing, and governing authorities throughout TODP.

The framework ensures that:

- Every opportunity has a responsible authority.
- Authorities are verified before use.
- Authority confidence is transparent.
- Authority lineage is preserved.
- Authority decisions are auditable and reproducible.

By enforcing **Authority Before Recommendation**, TODP ensures that recommendations are supported by accountable and traceable entities rather than unsupported information.

SDD Progress

- ☐ Section 1 — Document Control & Purpose
- ☐ Section 2 — System Context & High-Level Architecture
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- ☐ Section 4 — Physical Architecture & Deployment Topology
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- ☐ Section 11 — Authority Architecture & Verification Framework
- ☐ Section 12 — Evidence Architecture & Validation Framework

SDD V1 — Section 12

Evidence Architecture & Validation Framework

12.1 Purpose

This section defines the Evidence Architecture and Validation Framework used by TODP.

Evidence is the foundational trust mechanism of the platform.

TODP is built upon the principle:

Evidence First

This means:

- Opportunities are not trusted because they exist.
- Opportunities are not trusted because an authority claims them.
- Opportunities are trusted only when supporting evidence exists.

Evidence provides the factual basis that supports:

- Opportunity existence
- Authority verification
- Ranking eligibility
- Recommendation generation

Without evidence, the platform cannot establish traceability, auditability, or reproducibility.

12.2 Evidence Philosophy

TODP distinguishes between:

Claim

and

Evidence

A claim is an assertion.

Evidence is support for that assertion.

Example

Claim:

"A scholarship is available."

Evidence:

- Official scholarship announcement
- University publication
- Government notification
- Verified program documentation

The platform shall not treat claims as facts unless evidence exists.

12.3 Evidence Objectives

EO-01 Verification Support

Evidence shall support verification activities.

EO-02 Traceability

Evidence shall remain traceable to its source.

EO-03 Auditability

Evidence usage shall be auditable.

EO-04 Reproducibility

Evidence-based decisions shall be reproducible.

EO-05 Transparency

Evidence supporting recommendations shall be visible.

EO-06 Preservation

Evidence history shall remain accessible.

12.4 Evidence Architecture Overview

The Evidence Framework consists of six major components.

Component 1

Evidence Registry

Component 2

Evidence Validation Engine

Component 3

Evidence Lineage Engine

Component 4

Evidence Versioning Engine

Component 5

Evidence Quality Engine

Component 6

Evidence Audit Engine

12.5 Evidence Lifecycle

Evidence progresses through a controlled lifecycle.

Stage 1

Evidence Discovery

Stage 2

Evidence Collection

Stage 3

Evidence Registration

Stage 4

Evidence Validation

Stage 5

Evidence Usage

Stage 6

Evidence Monitoring

Stage 7

Evidence Versioning

Stage 8

Evidence Archival

12.6 Evidence Registry

The Evidence Registry serves as the authoritative repository of evidence records.

Responsibilities

Store evidence records.

Store source references.

Store validation outcomes.

Store quality assessments.

Store lineage information.

Store historical versions.

Objective

Provide a single source of truth for evidence management.

12.7 Evidence Types

TODP supports multiple categories of evidence.

Official Document Evidence

Examples:

Government Notifications

Policy Documents

Official Announcements

Institutional Publications

Registry Evidence

Examples:

Public Registries

Accreditation Records

Registration Databases

Official Listings

Program Evidence

Examples:

Program Documentation

Application Guides

Eligibility Documents

Official Brochures

Verification Evidence

Examples:

Authority Verification Records

Accreditation Evidence

Compliance Documentation

Historical Evidence

Examples:

Archived Announcements

Archived Program Records

Historical Opportunity Records

12.8 Evidence Discovery

Evidence discovery identifies supporting information.

Inputs

Opportunity Information

Authority Information

Source Information

Published Materials

Outputs

Candidate Evidence Records

Objective

Locate supporting information relevant to claims.

12.9 Evidence Collection Framework

Collected evidence must be preserved in a controlled manner.

Collection Activities

Capture source reference.

Capture collection timestamp.

Capture evidence type.

Capture source metadata.

Capture retrieval metadata.

Objective

Preserve provenance information.

12.10 Evidence Validation Framework

Evidence must be validated before usage.

Validation Goals

Confirm existence.

Confirm accessibility.

Confirm relevance.

Confirm traceability.

Confirm integrity.

Validation Outcome

Valid

or

Invalid

12.11 Evidence Validation Workflow

Step 1

Identify evidence candidate.

Step 2

Collect source metadata.

Step 3

Validate source reference.

Step 4

Validate evidence integrity.

Step 5

Assess evidence quality.

Step 6

Persist validation result.

Step 7

Enable evidence usage.

12.12 Evidence Integrity Model

Evidence integrity determines whether evidence remains trustworthy.

Integrity Requirements

Evidence must:

- Be identifiable
 - Be traceable
 - Be attributable
 - Be preserved
-

Objective

Protect platform trustworthiness.

12.13 Evidence Quality Framework

Evidence quality provides transparency regarding evidence strength.

Quality Factors

Source Credibility

Evidence Completeness

Evidence Freshness

Evidence Consistency

Evidence Traceability

Output

Evidence Quality Assessment

12.14 Evidence Confidence Levels

The framework supports evidence confidence classifications.

High Confidence

Strong supporting evidence.

Medium Confidence

Reasonably sufficient evidence.

Low Confidence

Limited supporting evidence.

Unvalidated

Evidence not yet validated.

12.15 Evidence Lineage Architecture

Evidence lineage preserves evidence history.

Lineage Path

Evidence Source



Evidence Collection



Evidence Validation



Authority Verification



Opportunity



Ranking



Recommendation

Objective

Enable full traceability.

12.16 Evidence Versioning Framework

Evidence may evolve over time.

Versioning Purpose

Support:

- Auditing
- Historical reconstruction
- Reproducibility

Versioning Triggers

Source Update

Document Update

Validation Update

Metadata Update

Requirement

Historical versions shall remain accessible.

12.17 Evidence Participation in Retrieval

Evidence contributes to retrieval quality.

Responsibilities

Evidence Lookup

Evidence Enrichment

Evidence Context Assembly

Evidence Traceability

Objective

Improve retrieval confidence.

12.18 Evidence Participation in Authority Verification

Evidence supports authority validation.

Responsibilities

Support identity verification.

Support legitimacy verification.

Support confidence assessment.

Objective

Strengthen authority decisions.

12.19 Evidence Participation in Ranking

Evidence determines ranking eligibility.

Responsibilities

Support opportunity validation.

Support ranking readiness.

Provide confidence context.

Constraint

Evidence does not determine rank position.

Evidence determines whether ranking can occur.

12.20 Evidence Participation in Recommendation

Evidence supports recommendation transparency.

Responsibilities

Display supporting evidence.

Display evidence sources.

Display evidence lineage.

Display evidence quality.

Objective

Enable informed decision-making.

12.21 Evidence Audit Architecture

Evidence usage shall be auditable.

Audit Scope

Evidence Creation

Evidence Updates

Evidence Validation

Evidence Usage

Evidence Versioning

Administrative Actions

Stored Information

Actor

Action

Timestamp

Previous State

New State

Correlation ID

12.22 Evidence Preservation Strategy

Evidence is a governance asset.

Preservation Requirements

Historical evidence must remain available.

Evidence references must remain traceable.

Evidence history must remain recoverable.

Objective

Support long-term reproducibility.

12.23 Evidence Expiration Framework

Evidence relevance may change over time.

Expiration Triggers

Source Removal

Source Update

Evidence Obsolescence

Validation Failure

Conflicting Evidence

Result

Evidence revalidation workflow.

12.24 Evidence Search Architecture

Evidence shall be searchable.

Search Capabilities

Evidence Type Search

Evidence Source Search

Evidence Quality Search

Evidence Date Search

Authority Search

Opportunity Search

Objective

Improve transparency and governance.

12.25 Evidence Performance Targets

Evidence Lookup

P95 < 100 ms

Evidence Validation

P95 < 500 ms

Evidence Retrieval

P95 < 200 ms

Evidence Lineage Retrieval

P95 < 150 ms

12.26 Evidence Failure Handling

The evidence framework shall fail safely.

Missing Evidence

Opportunity becomes ineligible.

Invalid Evidence

Opportunity becomes ineligible.

Evidence Validation Failure

Evidence marked invalid.

Evidence Service Failure

Recommendation generation suspended until evidence status is determined.

Objective

Preserve Evidence First principles.

12.27 AI Usage Policy

AI may assist evidence summarization.

AI may assist evidence explanation.

AI may assist evidence categorization.

AI May Not

Validate evidence.

Determine evidence legitimacy.

Assign evidence quality.

Override validation decisions.

Reason

Evidence decisions must remain deterministic and auditable.

12.28 Evidence Governance Rules

The following rules are mandatory.

EG-01

Every opportunity must have supporting evidence.

EG-02

Evidence must be traceable.

EG-03

Evidence must be preservable.

EG-04

Evidence decisions must be auditable.

EG-05

Evidence lineage must remain available.

EG-06

Evidence validation must be reproducible.

EG-07

Invalid evidence cannot support recommendations.

EG-08

Unvalidated evidence cannot support rankings.

12.29 Architectural Constraints

The following constraints are non-negotiable.

AC-EV-01

Evidence validation occurs before authority verification.

AC-EV-02

Evidence validation occurs before ranking.

AC-EV-03

Evidence validation occurs before recommendation.

AC-EV-04

Evidence lineage must remain available.

AC-EV-05

Evidence quality assessments must be explainable.

AC-EV-06

Evidence decisions must remain deterministic.

12.30 Evidence Architecture Summary

The Evidence Architecture establishes a structured framework for discovering, validating, preserving, governing, and utilizing evidence throughout TODP.

The framework ensures that:

- Every opportunity is supported by evidence.
- Evidence is validated before use.
- Evidence quality is transparent.
- Evidence lineage is preserved.
- Evidence decisions are auditable and reproducible.

By enforcing **Evidence First**, TODP ensures that retrieval, ranking, and recommendation decisions are grounded in verifiable supporting information rather than unsupported claims.

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- ☐ Section 13 — Opportunity Architecture & Lifecycle Management

SDD V1 — Section 13

Opportunity Architecture & Lifecycle Management

13.1 Purpose

This section defines the Opportunity Architecture and Lifecycle Management Framework used by TODP.

The opportunity is the primary business entity of the platform.

All major platform capabilities ultimately operate on opportunities.

Examples include:

- Discovery
- Retrieval
- Authority Verification
- Evidence Validation
- Ranking
- Recommendation
- Similarity Analysis
- Opportunity Creation

The objective of this architecture is to ensure that opportunities remain:

- Discoverable
- Verifiable
- Rankable
- Explainable
- Traceable
- Auditable
- Reproducible

throughout their lifecycle.

13.2 Opportunity Philosophy

TODP views opportunities as actionable possibilities available to a user.

An opportunity is not merely information.

An opportunity is information that can potentially result in:

- Learning
- Employment
- Funding
- Research
- Participation
- Creation
- Advancement

for a user.

Opportunity Definition

An opportunity is:

"A discoverable and actionable possibility supported by evidence and attributable to an authority."

Architectural Consequence

Without:

- Evidence
- Authority

an opportunity cannot become eligible for ranking or recommendation.

13.3 Opportunity Objectives

OO-01 Discoverability

Opportunities must be discoverable.

OO-02 Verifiability

Opportunities must be supported by evidence.

OO-03 Traceability

Opportunity lineage must remain available.

OO-04 Auditability

Opportunity lifecycle events must be auditable.

OO-05 Reproducibility

Historical opportunity states must be recoverable.

OO-06 Extensibility

New opportunity categories must be addable without redesign.

13.4 Opportunity Architecture Overview

The Opportunity Framework consists of six major components.

Component 1

Opportunity Registry

Component 2

Opportunity Lifecycle Engine

Component 3

Opportunity Classification Engine

Component 4

Opportunity State Engine

Component 5

Opportunity Lineage Engine

Component 6

Opportunity Audit Engine

13.5 Opportunity Registry

The Opportunity Registry serves as the authoritative repository of opportunities.

Responsibilities

Store opportunity records.

Store classifications.

Store lifecycle states.

Store lineage information.

Store historical versions.

Store ownership information.

Objective

Provide a single source of truth for opportunity management.

13.6 Opportunity Classification Framework

Opportunities shall be classified using a structured taxonomy.

Educational Opportunities

Examples:

Courses

Certifications

Training Programs

Workshops

Scholarships

Employment Opportunities

Examples:

Jobs

Internships

Apprenticeships

Fellowships

Research Opportunities

Examples:

Research Programs

Research Grants

Academic Collaborations

Research Assistantships

Funding Opportunities

Examples:

Grants

Loans

Seed Funding

Innovation Funds

Competitive Opportunities

Examples:

Competitions

Hackathons

Challenges

Awards

Community Opportunities

Examples:

Volunteer Programs

Community Projects

Local Initiatives

Future Categories

Additional categories may be introduced without modifying lifecycle architecture.

13.7 Discovery Hierarchy Integration

Every opportunity shall belong to one or more discovery levels.

Hierarchy Levels

Institution

Local

District

State

National

International

Purpose

Support hierarchical discovery and ranking.

Rule

An opportunity may exist at multiple levels simultaneously.

13.8 Opportunity Ownership Model

Every opportunity shall have an accountable owner.

Ownership Types

Authority Owner

Institution Owner

Organization Owner

Program Owner

Responsibilities

Maintain opportunity accuracy.

Maintain supporting evidence.

Maintain authority information.

13.9 Opportunity Lifecycle

Opportunities progress through a controlled lifecycle.

Stage 1

Identified

Opportunity discovered.

Stage 2

Registered

Opportunity stored in registry.

Stage 3

Evidence Validated

Evidence requirements satisfied.

Stage 4

Authority Verified

Authority requirements satisfied.

Stage 5

Eligible

Opportunity eligible for ranking.

Stage 6

Ranked

Ranking completed.

Stage 7

Recommended

Recommendation generated.

Stage 8

Archived

Opportunity no longer active.

13.10 Opportunity State Model

The lifecycle is implemented through states.

State: Draft

Opportunity captured but not validated.

State: Pending Evidence

Evidence validation outstanding.

State: Pending Authority

Authority verification outstanding.

State: Eligible

Ready for ranking.

State: Ranked

Ranking completed.

State: Recommended

Recommendation generated.

State: Suspended

Temporarily unavailable.

State: Archived

Historical record retained.

13.11 State Transition Rules

State transitions shall be controlled.

Example

Pending Evidence

↓

Evidence Validated

↓

Pending Authority

Pending Authority

↓

Authority Verified

↓

Eligible

Eligible

↓

Ranking Executed

↓

Ranked

Ranked



Recommendation Generated



Recommended

13.12 Opportunity Eligibility Framework

Eligibility determines whether an opportunity may participate in ranking.

Minimum Requirements

Opportunity Exists

-

Evidence Validated

-

Authority Verified

-

Required Metadata Present

Result

Eligible Opportunity

13.13 Opportunity Metadata Architecture

Every opportunity shall maintain standardized metadata.

Core Metadata

Opportunity ID

Name

Description

Type

Status

Authority ID

Creation Timestamp

Update Timestamp

Discovery Metadata

Hierarchy Level

Geographic Context

Search Attributes

Classification Data

Governance Metadata

Version

Owner

Validation Status

Audit References

13.14 Opportunity Lineage Architecture

Opportunity lineage shall preserve full history.

Lineage Path

Opportunity

↓

Evidence

↓

Authority

↓

Retrieval

↓

Ranking

↓

Recommendation

Objective

Enable traceability and reproducibility.

13.15 Opportunity Versioning Framework

Opportunities may evolve over time.

Versioning Triggers

Description Change

Authority Change

Evidence Change

Classification Change

Metadata Change

Requirement

Historical versions shall remain available.

13.16 Opportunity Participation in Retrieval

Opportunities are primary retrieval targets.

Responsibilities

Keyword Retrieval

Semantic Retrieval

Metadata Retrieval

Hybrid Retrieval

Objective

Enable opportunity discovery.

13.17 Opportunity Participation in Ranking

Opportunities are primary ranking subjects.

Responsibilities

Provide cost inputs.

Provide classification inputs.

Provide contextual information.

Objective

Enable ranking calculations.

13.18 Opportunity Participation in Recommendation

Opportunities are recommendation outputs.

Responsibilities

Provide actionable information.

Provide explanatory information.

Provide evidence context.

Provide authority context.

Objective

Support user decision-making.

13.19 Opportunity Participation in Similarity Analysis

Opportunities support creation workflows.

Create Something Similar

Existing opportunities act as similarity anchors.

Create Something New

Existing opportunities support gap analysis.

Objective

Enable opportunity creation journeys.

13.20 Opportunity Search Architecture

Opportunities shall be searchable entities.

Search Dimensions

Name

Description

Type

Authority

Evidence

Location

Hierarchy Level

Status

Objective

Improve discoverability.

13.21 Opportunity Quality Framework

Opportunity quality provides transparency regarding opportunity completeness.

Quality Factors

Evidence Completeness

Authority Completeness

Metadata Completeness

Lineage Availability

Lifecycle Validity

Output

Opportunity Quality Assessment

13.22 Opportunity Expiration Framework

Some opportunities become inactive over time.

Expiration Triggers

Deadline Passed

Program Ended

Authority Withdrawal

Evidence Invalidated

Administrative Action

Result

Opportunity status reassessment.

13.23 Opportunity Archival Framework

Historical opportunities remain valuable.

Archival Objectives

Preserve history.

Support audits.

Support analytics.

Support reproducibility.

Requirement

Archived opportunities remain queryable.

13.24 Opportunity Audit Architecture

Opportunity lifecycle activities shall be auditable.

Audit Scope

Opportunity Creation

Opportunity Updates

State Changes

Classification Changes

Ownership Changes

Archival Actions

Stored Information

Actor

Action

Timestamp

Previous State

New State

Correlation ID

13.25 Opportunity Performance Targets

Opportunity Lookup

P95 < 100 ms

Opportunity Retrieval

P95 < 200 ms

Opportunity Lifecycle Transition

P95 < 200 ms

Opportunity Lineage Retrieval

P95 < 150 ms

13.26 Opportunity Failure Handling

The opportunity framework shall fail safely.

Missing Opportunity

No ranking.

No recommendation.

Invalid Opportunity State

Lifecycle transition rejected.

Missing Metadata

Opportunity becomes ineligible.

Missing Lineage

Opportunity flagged for review.

13.27 AI Usage Policy

AI may assist:

- Opportunity summarization

- Opportunity categorization suggestions
 - Opportunity explanation
-

AI May Not

Create authoritative opportunity records.

Approve opportunity eligibility.

Override lifecycle states.

Override validation outcomes.

Modify ranking outcomes.

Reason

Opportunity governance must remain deterministic and auditable.

13.28 Opportunity Governance Rules

The following rules are mandatory.

OG-01

Every opportunity must have an owner.

OG-02

Every opportunity must have evidence.

OG-03

Every opportunity must have an authority.

OG-04

Opportunity lineage must be preserved.

OG-05

Opportunity lifecycle changes must be auditable.

OG-06

Opportunity versions must remain recoverable.

OG-07

Ineligible opportunities cannot be ranked.

OG-08

Unranked opportunities cannot be recommended.

13.29 Architectural Constraints

The following constraints are non-negotiable.

AC-OP-01

Evidence validation occurs before eligibility.

AC-OP-02

Authority verification occurs before eligibility.

AC-OP-03

Eligibility occurs before ranking.

AC-OP-04

Ranking occurs before recommendation.

AC-OP-05

Opportunity lineage must remain available.

AC-OP-06

Opportunity state transitions must be deterministic.

13.30 Opportunity Architecture Summary

The Opportunity Architecture provides the foundational lifecycle framework for all opportunities managed by TODP.

The framework ensures that opportunities are:

- Discoverable
- Verifiable
- Traceable
- Auditable
- Reproducible

through a controlled lifecycle governed by:

Evidence Validation

↓

Authority Verification



Eligibility



Ranking



Recommendation

By establishing opportunities as governed business entities rather than simple records, TODP maintains the integrity of its discovery, ranking, and recommendation workflows.

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- ☐ Section 13 — Opportunity Architecture & Lifecycle Management
- ☐ Section 14 — User Architecture & Context Management Framework

SDD V1 — Section 14

User Architecture & Context Management Framework

14.1 Purpose

This section defines the User Architecture and Context Management Framework used by TODP.

While opportunities are the primary business entities of the platform, opportunities only become meaningful when evaluated relative to a user.

The same opportunity may have:

- Different Money Cost
- Different Time Cost
- Different Distance Cost

for different users.

Therefore, TODP requires a structured user context framework that allows ranking and recommendation systems to operate consistently while preserving transparency and reproducibility.

The purpose of this framework is to:

- Represent users consistently
- Capture contextual constraints
- Support ranking calculations
- Support discovery personalization
- Preserve auditability
- Preserve reproducibility

without introducing opaque recommendation behavior.

14.2 User Philosophy

TODP does not attempt to model a user's entire identity.

TODP models only the information necessary to support:

- Discovery
- Ranking
- Recommendation
- Opportunity Creation

The platform focuses on:

Context

rather than

Identity

The objective is to understand:

"What constraints affect this opportunity for this user?"

rather than

"Who is this user?"

14.3 User Objectives

UO-01 Context Awareness

The system shall understand relevant user context.

UO-02 Ranking Support

User context shall support ranking calculations.

UO-03 Transparency

User context usage shall be visible.

UO-04 Auditability

Context-driven decisions shall be auditable.

UO-05 Reproducibility

Historical user-context evaluations shall be reproducible.

UO-06 Minimalism

Only necessary context shall be collected.

14.4 User Architecture Overview

The User Framework consists of six major components.

Component 1

User Registry

Component 2

Context Management Engine

Component 3

Constraint Management Engine

Component 4

Preference Management Engine

Component 5

Context Lineage Engine

Component 6

User Audit Engine

14.5 User Entity Model

The User Entity represents a platform participant.

Core Responsibilities

Provide context.

Provide constraints.

Provide discovery preferences.

Receive recommendations.

Create opportunities.

Architectural Principle

Users influence rankings through context.

Users do not directly control rankings.

14.6 User Types

TODP supports multiple user categories.

Individual User

Represents a single person.

Examples:

Students

Professionals

Researchers

Citizens

Entrepreneurs

Organizational User

Represents an institution.

Examples:

Schools

Universities

Companies

Research Organizations

Non-Profits

Administrative User

Represents platform administrators.

Examples:

System Operators

Governance Teams

Auditors

14.7 User Context Model

Context is the most important user construct.

Definition

Context represents environmental information relevant to opportunity evaluation.

Examples

Location

Available Time

Available Budget

Discovery Scope

Accessibility Constraints

Participation Constraints

Objective

Support transparent ranking.

14.8 Context Categories

TODP organizes context into structured categories.

Geographic Context

Examples:

Country

State

District

Locality

Institution

Financial Context

Examples:

Budget Availability

Funding Constraints

Cost Preferences

Temporal Context

Examples:

Available Time

Participation Window

Application Deadline Constraints

Discovery Context

Examples:

Preferred Opportunity Types

Preferred Discovery Levels

Preferred Domains

Accessibility Context

Examples:

Remote Participation

Physical Participation

Travel Constraints

14.9 User Constraints Framework

Constraints represent limitations affecting opportunity suitability.

Examples

Budget Limit

Distance Limit

Time Limit

Location Restriction

Eligibility Restriction

Objective

Support practical ranking outcomes.

14.10 User Preferences Framework

Preferences represent user interests.

Examples

Research Opportunities

Funding Opportunities

Educational Opportunities

Community Opportunities

Important Constraint

Preferences may influence discovery.

Preferences shall not override ranking rules.

14.11 Context Lifecycle

Context evolves over time.

Stage 1

Context Creation

Stage 2

Context Validation

Stage 3

Context Usage

Stage 4

Context Update

Stage 5

Context Versioning

Stage 6

Context Archival

14.12 Context Validation Framework

Context must be validated before usage.

Validation Goals

Completeness

Consistency

Availability

Traceability

Result

Validated Context

or

Invalid Context

14.13 Context Versioning Framework

Context changes must remain traceable.

Versioning Triggers

Location Change

Budget Change

Time Availability Change

Preference Change

Constraint Change

Objective

Support reproducibility.

14.14 User Participation in Retrieval

User context influences retrieval scope.

Responsibilities

Filter candidate opportunities.

Select discovery hierarchy levels.

Apply contextual constraints.

Example

User Context:

Local Discovery

↓

Retrieval prioritizes local opportunities.

14.15 User Participation in Ranking

User context directly influences ranking calculations.

Money Cost Impact

Different users have different financial constraints.

Time Cost Impact

Different users have different available time.

Distance Cost Impact

Different users have different geographic positions.

Objective

Calculate user-relative ranking outcomes.

14.16 User Participation in Recommendation

User context influences recommendation relevance.

Responsibilities

Support contextual explanations.

Provide constraint visibility.

Support recommendation interpretation.

Constraint

Recommendation ordering remains controlled by ranking.

14.17 Discovery Hierarchy Preferences

Users may specify preferred discovery scopes.

Available Levels

Institution

Local

District

State

National

International

Objective

Improve discovery relevance.

Constraint

Hierarchy preferences do not alter ranking formulas.

14.18 Opportunity Creation Context

User context supports creation workflows.

Create Something Similar

Context influences similarity relevance.

Create Something New

Context influences opportunity gap analysis.

Objective

Improve creation outcomes.

14.19 User Lineage Architecture

User interactions shall maintain lineage.

Lineage Path

User

↓

Context

↓

Retrieval

↓

Authority

↓

Evidence

↓

Ranking

↓

Recommendation

Objective

Enable traceability.

14.20 User Audit Architecture

User-related actions shall be auditable.

Audit Scope

Profile Changes

Context Changes

Constraint Changes

Preference Changes

Discovery Actions

Recommendation Interactions

Stored Information

Actor

Action

Timestamp

Previous State

New State

Correlation ID

14.21 User Privacy Architecture

User data shall be protected.

Objectives

Minimize collection.

Minimize retention.

Protect confidentiality.

Support user control.

Principles

Data Minimization

Purpose Limitation

Access Control

Auditability

14.22 User Context Security

User context shall be protected.

Protection Mechanisms

Authentication

Authorization

Encryption

Audit Logging

Objective

Prevent unauthorized access.

14.23 User Search Architecture

Users may search opportunities using context-aware discovery.

Search Inputs

Query

Context

Preferences

Constraints

Discovery Scope

Output

Contextual opportunity candidates.

14.24 User Performance Targets

Context Retrieval

P95 < 100 ms

Context Validation

P95 < 100 ms

Context Update

P95 < 200 ms

Context Application

P95 < 100 ms

14.25 User Failure Handling

The user framework shall fail safely.

Missing Context

System uses default context.

Invalid Context

Context rejected.

Missing Preferences

Discovery continues using neutral behavior.

Missing Constraints

Ranking proceeds using available information.

14.26 AI Usage Policy

AI may assist:

- Context summarization
 - Preference summarization
 - Opportunity explanation
-

AI May Not

Override constraints.

Modify ranking logic.

Create hidden user profiles.

Generate undisclosed personalization.

Reason

User trust requires transparent decision-making.

14.27 User Governance Rules

The following rules are mandatory.

UG-01

Context usage must be transparent.

UG-02

Context changes must be auditable.

UG-03

Historical context versions must remain recoverable.

UG-04

Preferences shall not override ranking rules.

UG-05

Constraints shall remain visible.

UG-06

User lineage shall remain available.

UG-07

User data collection shall remain minimal.

14.28 Architectural Constraints

The following constraints are non-negotiable.

AC-USER-01

User context influences ranking inputs.

AC-USER-02

User context does not modify ranking rules.

AC-USER-03

Preferences do not override evidence requirements.

AC-USER-04

Preferences do not override authority requirements.

AC-USER-05

User lineage must remain available.

AC-USER-06

Context-driven decisions must be reproducible.

14.29 Context-to-Ranking Integration Model

The ranking engine consumes user context through cost calculations.

Money Cost

Opportunity Cost

evaluated relative to

User Financial Context

Time Cost

Opportunity Time Requirement

evaluated relative to

User Temporal Context

Distance Cost

Opportunity Location

evaluated relative to

User Geographic Context

Result

User-Specific Total Cost

14.30 User Architecture Summary

The User Architecture provides a structured framework for representing user context, constraints, preferences, and discovery requirements.

The framework ensures that:

- Opportunities are evaluated relative to user context.
- Ranking remains transparent.
- Recommendations remain explainable.
- Context usage remains auditable.
- Historical evaluations remain reproducible.

By modeling context rather than identity, TODP supports practical opportunity discovery while preserving transparency and minimizing unnecessary personalization.

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Opportunity Creation Architecture & Innovation Framework

15.1 Purpose

This section defines the Opportunity Creation Architecture and Innovation Framework used by TODP.

While most discovery platforms stop at finding existing opportunities, TODP supports two additional user journeys:

- Create Something Similar
- Create Something New

These journeys transform TODP from a discovery platform into an opportunity creation platform.

The purpose of this architecture is to:

- Enable opportunity replication
- Enable opportunity adaptation
- Enable opportunity innovation
- Identify opportunity gaps
- Support structured creativity
- Preserve traceability
- Preserve auditability

without violating the principles of:

- Evidence First
- Authority Before Recommendation
- Ranking Before Recommendation

15.2 Creation Philosophy

TODP distinguishes between:

Discovery

and

Creation

Discovery answers:

"What opportunities already exist?"

Creation answers:

"What opportunities could exist?"

Fundamental Principle

The platform may assist creation.

The platform shall not claim authority for created opportunities.

Core Rule

AI may support ideation.

Humans remain responsible for novelty creation.

This aligns with the principle:

AI Is Not An Authority
Human Creates Novelty

15.3 Creation Objectives

CO-01 Similarity Discovery

Identify opportunities similar to existing opportunities.

CO-02 Pattern Discovery

Identify recurring opportunity patterns.

CO-03 Gap Discovery

Identify missing opportunities.

CO-04 Innovation Support

Assist users in generating novel opportunities.

CO-05 Traceability

Creation lineage must remain visible.

CO-06 Explainability

Creation suggestions must be explainable.

15.4 Creation Architecture Overview

The Opportunity Creation Framework consists of seven major components.

Component 1

Similarity Engine

Component 2

Pattern Analysis Engine

Component 3

Gap Detection Engine

Component 4

Opportunity Synthesis Engine

Component 5

Innovation Support Engine

Component 6

Creation Lineage Engine

Component 7

Creation Audit Engine

15.5 Creation User Journeys

TODP supports two creation journeys.

Journey 1

Create Something Similar

Goal

Replicate or adapt an existing opportunity.

Examples

Create a scholarship similar to an existing scholarship.

Create a competition similar to an existing competition.

Create a research program similar to an existing research program.

Journey 2

Create Something New

Goal

Identify opportunities that currently do not exist.

Examples

Create a new community program.

Create a new funding initiative.

Create a new educational offering.

Create a new research collaboration.

15.6 Similarity Engine

The Similarity Engine identifies opportunities related to an existing opportunity.

Responsibilities

Similarity Retrieval

Pattern Matching

Structural Comparison

Context Comparison

Inputs

Existing Opportunity

User Context

Discovery Scope

Outputs

Similar Opportunities

15.7 Similarity Architecture

Similarity evaluation operates across multiple dimensions.

Opportunity Similarity

Compare opportunity characteristics.

Authority Similarity

Compare supporting authorities.

Evidence Similarity

Compare evidence structures.

Context Similarity

Compare user contexts.

Outcome

Similarity Assessment

15.8 Pattern Analysis Engine

Patterns represent recurring structures across opportunities.

Objective

Discover repeatable opportunity designs.

Example Patterns

Funding Programs

Scholarship Structures

Research Initiatives

Community Programs

Competition Formats

Output

Opportunity Patterns

15.9 Pattern Extraction Framework

Patterns are extracted from validated opportunities.

Inputs

Opportunities

Authorities

Evidence

Lifecycle Information

Process

Analyze common structures.

Identify recurring attributes.

Identify recurring workflows.

Output

Reusable Opportunity Patterns

15.10 Gap Detection Engine

The Gap Detection Engine identifies missing opportunities.

Definition

Gap = Demand Exists

but

Suitable Opportunity Does Not Exist

Objective

Reveal potential opportunity creation areas.

15.11 Gap Detection Inputs

Gap analysis uses:

Opportunity Coverage

Existing opportunities.

Geographic Coverage

Discovery hierarchy coverage.

Authority Coverage

Authority participation.

Evidence Coverage

Evidence availability.

User Demand Signals

Observed opportunity demand.

15.12 Gap Detection Outputs

Examples:

Geographic Gaps

Region lacks opportunities.

Educational Gaps

Subject lacks programs.

Funding Gaps

Community lacks funding.

Participation Gaps

Access barriers exist.

15.13 Opportunity Synthesis Engine

The synthesis engine combines patterns and gaps.

Purpose

Generate opportunity concepts.

Inputs

Patterns

Gaps

User Context

Domain Information

Outputs

Opportunity Concepts

Constraint

Concepts are not authoritative opportunities.

15.14 Innovation Support Engine

Supports users creating novel opportunities.

Responsibilities

Pattern Presentation

Gap Presentation

Concept Generation Support

Innovation Exploration

Objective

Support structured creativity.

15.15 Creation Context Framework

Creation activities depend on context.

Context Inputs

Location

Resources

Time Availability

Community Needs

Organizational Constraints

Purpose

Improve creation relevance.

15.16 Creation Lineage Architecture

Creation lineage preserves idea provenance.

Lineage Path

Existing Opportunities

↓

Patterns

↓

Gap Analysis

↓

Opportunity Concepts

↓

User Refinement

↓

Created Opportunity

Objective

Preserve traceability.

15.17 Opportunity Derivation Framework

Created opportunities may derive from existing opportunities.

Derivation Types

Inspired By

Adapted From

Extended From

Combined From

Requirement

Derivation relationships remain visible.

15.18 Creation Validation Framework

Created opportunities must undergo validation before becoming platform opportunities.

Validation Requirements

Evidence Availability

Authority Identification

Metadata Completeness

Classification Assignment

Objective

Prevent unsupported opportunity creation.

15.19 Creation Participation in Retrieval

Creation workflows depend on retrieval.

Responsibilities

Find similar opportunities.

Find supporting patterns.

Find related authorities.

Find supporting evidence.

Objective

Provide informed creation support.

15.20 Creation Participation in Ranking

Created opportunities may eventually enter ranking workflows.

Requirement

Opportunity must satisfy eligibility requirements.

Workflow

Created Opportunity

↓

Evidence Validation

↓

Authority Verification

↓

Eligibility

↓

Ranking

15.21 Creation Participation in Recommendation

Created opportunities may eventually become recommendations.

Requirement

Created opportunities follow identical governance rules as existing opportunities.

Constraint

Creation origin provides no ranking advantage.

15.22 Opportunity Innovation Framework

Innovation occurs when opportunities differ meaningfully from existing opportunities.

Innovation Sources

New Combination

New Structure

New Audience

New Geography

New Resource Model

Objective

Support novel opportunity creation.

15.23 Human-AI Collaboration Model

TODP supports human-AI collaboration.

AI Responsibilities

Similarity Analysis

Pattern Detection

Gap Detection

Idea Exploration

Concept Structuring

Human Responsibilities

Novelty Creation

Authority Identification

Evidence Acquisition

Decision Making

Opportunity Ownership

Principle

AI assists.

Humans create.

15.24 Creation Audit Architecture

Creation activities shall be auditable.

Audit Scope

Similarity Searches

Pattern Analyses

Gap Analyses

Concept Generation

Opportunity Derivations

User Refinements

Stored Information

Actor

Action

Timestamp

Previous State

New State

Correlation ID

15.25 Creation Performance Targets

Similarity Retrieval

P95 < 300 ms

Pattern Analysis

P95 < 500 ms

Gap Detection

P95 < 500 ms

Concept Generation Support

P95 < 500 ms

15.26 Failure Handling

The creation framework shall fail safely.

Similarity Failure

Fallback to keyword similarity.

Pattern Failure

Creation workflow continues without pattern suggestions.

Gap Detection Failure

Gap analysis omitted.

Concept Generation Failure

User may continue manually.

15.27 AI Usage Policy

AI may assist:

- Similarity analysis
 - Pattern extraction
 - Gap detection
 - Concept exploration
 - Opportunity structuring
-

AI May Not

Claim authority.

Create evidence.

Validate opportunities.

Approve eligibility.

Override governance rules.

Reason

Human accountability must remain preserved.

15.28 Creation Governance Rules

The following rules are mandatory.

CG-01

Created opportunities must remain traceable.

CG-02

Creation lineage must remain visible.

CG-03

Created opportunities require evidence.

CG-04

Created opportunities require authorities.

CG-05

Created opportunities must satisfy eligibility requirements.

CG-06

Creation workflows must be auditable.

CG-07

AI-generated concepts are not authoritative opportunities.

15.29 Architectural Constraints

The following constraints are non-negotiable.

AC-CREATE-01

Creation does not bypass evidence validation.

AC-CREATE-02

Creation does not bypass authority verification.

AC-CREATE-03

Creation does not bypass ranking.

AC-CREATE-04

Creation lineage must remain available.

AC-CREATE-05

Human accountability must remain preserved.

AC-CREATE-06

AI may assist but may not claim authority.

15.30 Opportunity Creation Architecture Summary

The Opportunity Creation Architecture extends TODP beyond traditional discovery systems.

The framework enables users to:

- Create Something Similar
- Create Something New

through:

Similarity Analysis

↓

Pattern Discovery

↓

Gap Detection

↓

Opportunity Synthesis

↓

Human Refinement

↓

Validated Opportunity

while preserving:

- Traceability
- Auditability
- Reproducibility
- Human Accountability

This architecture ensures that innovation remains structured, explainable, and aligned with TODP's core governance principles.

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API Architecture & Service Contracts

16.1 Purpose

This section defines the API Architecture and Service Contract Framework used by TODP.

APIs serve as the primary communication mechanism between:

- Frontend and Backend
- Internal Services
- External Integrations
- Administrative Interfaces
- Future Ecosystem Components

The API architecture shall provide:

- Consistency
- Versioning
- Traceability
- Security
- Auditability
- Scalability

while remaining aligned with the core principles of:

- Evidence First
 - Authority Before Recommendation
 - Ranking Before Recommendation
-

16.2 API Philosophy

TODP follows an API-First Architecture.

All business capabilities shall be exposed through well-defined service contracts.

API-First Principle

Business Capability

↓

Service Contract

↓

Implementation

The API contract is the authoritative interface.

Implementation details may evolve without changing the contract.

16.3 API Objectives

APO-01 Consistency

All APIs shall follow common conventions.

APO-02 Discoverability

API capabilities shall be documented.

APO-03 Versionability

APIs shall support controlled evolution.

APO-04 Security

APIs shall enforce authentication and authorization.

APO-05 Traceability

API requests shall be traceable.

APO-06 Auditability

API actions shall be auditable.

16.4 API Architecture Overview

The API framework consists of six major components.

Component 1

API Gateway

Component 2

Public API Layer

Component 3

Internal Service API Layer

Component 4

Administrative API Layer

Component 5

API Security Layer

Component 6

API Audit Layer

16.5 API Gateway Architecture

The API Gateway serves as the single entry point into the platform.

Responsibilities

Request Routing

Authentication Enforcement

Authorization Enforcement

Rate Limiting

Request Validation

Response Standardization

Correlation Tracking

Benefits

Centralized governance.

Centralized security.

Centralized observability.

16.6 API Categories

TODP APIs are organized into logical domains.

Discovery APIs

Purpose:

Opportunity discovery and retrieval.

Authority APIs

Purpose:

Authority access and verification.

Evidence APIs

Purpose:

Evidence access and validation.

Ranking APIs

Purpose:

Ranking execution and retrieval.

Recommendation APIs

Purpose:

Recommendation retrieval and explanation.

Opportunity APIs

Purpose:

Opportunity management.

User APIs

Purpose:

User context management.

Creation APIs

Purpose:

Opportunity creation workflows.

Administrative APIs

Purpose:

Operational management.

16.7 API Versioning Strategy

All public APIs shall be versioned.

Version Format

/api/v1/

Example

/api/v1/discovery/search

/api/v1/opportunities

/api/v1/rankings

Objectives

Backward compatibility.

Controlled evolution.

Predictable upgrades.

16.8 API Resource Design Principles

APIs shall be resource-oriented.

Examples

Users

Authorities

Evidence

Opportunities

Rankings

Recommendations

Benefits

Predictable structure.

Developer familiarity.

Reduced complexity.

16.9 API Contract Principles

Every service contract shall define:

Inputs

Request schema.

Outputs

Response schema.

Errors

Failure schema.

Security

Authentication requirements.

Governance

Audit requirements.

16.10 Discovery API Contracts

Discovery APIs support opportunity retrieval.

Example Operations

Search Opportunities

Retrieve Opportunity

Retrieve Search Results

Retrieve Search Metadata

Consumers

Frontend

Administrative Interfaces

Future External Consumers

16.11 Authority API Contracts

Authority APIs expose authority capabilities.

Example Operations

Retrieve Authority

Search Authorities

Retrieve Authority History

Retrieve Verification Status

Objective

Support authority transparency.

16.12 Evidence API Contracts

Evidence APIs expose evidence information.

Example Operations

Retrieve Evidence

Retrieve Evidence Lineage

Search Evidence

Retrieve Evidence Versions

Objective

Support Evidence First principles.

16.13 Ranking API Contracts

Ranking APIs expose ranking information.

Example Operations

Retrieve Ranking

Retrieve Ranking Explanation

Retrieve Ranking History

Retrieve Ranking Audit Trail

Objective

Support ranking transparency.

16.14 Recommendation API Contracts

Recommendation APIs expose recommendations.

Example Operations

Retrieve Recommendations

Retrieve Recommendation Explanation

Retrieve Recommendation Lineage

Objective

Support recommendation explainability.

16.15 Opportunity API Contracts

Opportunity APIs manage opportunities.

Example Operations

Create Opportunity

Update Opportunity

Retrieve Opportunity

Archive Opportunity

Objective

Support lifecycle management.

16.16 User API Contracts

User APIs manage user context.

Example Operations

Retrieve Context

Update Context

Manage Preferences

Manage Constraints

Objective

Support context-aware discovery.

16.17 Creation API Contracts

Creation APIs support innovation workflows.

Example Operations

Find Similar Opportunities

Analyze Opportunity Patterns

Perform Gap Analysis

Generate Opportunity Concepts

Objective

Support creation journeys.

16.18 Administrative API Contracts

Administrative APIs support platform operations.

Example Operations

System Health

Audit Retrieval

Configuration Management

Monitoring Access

Objective

Support governance and operations.

16.19 Standard Request Structure

Every request shall contain:

Required Components

Request ID

Timestamp

Authentication Information

Version Information

Request Payload

Optional Components

Correlation ID

Context Information

Tracing Information

16.20 Standard Response Structure

All APIs shall return standardized responses.

Response Components

Status

Data

Metadata

Correlation ID

Timestamp

Benefits

Consistency across services.

16.21 Error Handling Architecture

Errors shall follow a consistent structure.

Error Components

Error Code

Error Message

Error Category

Correlation ID

Timestamp

Objectives

Predictability.

Troubleshooting.

Observability.

16.22 HTTP Status Code Standards

The platform shall use standard HTTP semantics.

Success

200 Series

Client Errors

400 Series

Authorization Errors

401 / 403

Resource Errors

404

Server Errors

500 Series

16.23 API Security Architecture

All APIs shall enforce security controls.

Authentication

Identity verification.

Authorization

Permission enforcement.

Encryption

TLS-protected communication.

Audit Logging

Security event recording.

16.24 API Rate Limiting

Rate limiting protects platform stability.

Objectives

Prevent abuse.

Prevent overload.

Protect availability.

Enforcement Point

API Gateway.

16.25 API Traceability Framework

Every API request shall be traceable.

Traceability Components

Request ID

Correlation ID

User Context

Timestamp

Service Path

Objective

Support auditing and debugging.

16.26 API Audit Architecture

API activity shall be auditable.

Audit Scope

Request Receipt

Authorization Decisions

Business Actions

Administrative Actions

Error Events

Stored Information

Actor

Action

Timestamp

Endpoint

Response Status

Correlation ID

16.27 API Performance Targets

Read Operations

P95 < 300 ms

Write Operations

P95 < 500 ms

Search Operations

P95 < 2 seconds

Ranking Operations

P95 < 1 second

Recommendation Operations

P95 < 500 ms

16.28 API Failure Handling

The API layer shall fail safely.

Invalid Request

Reject request.

Return validation details.

Unauthorized Request

Reject request.

Log security event.

Internal Failure

Return standardized error.

Preserve traceability.

Timeout

Return timeout response.

Record telemetry.

16.29 API Governance Rules

The following rules are mandatory.

API-01

All APIs must be versioned.

API-02

All APIs must be documented.

API-03

All APIs must be authenticated unless explicitly public.

API-04

All APIs must be traceable.

API-05

All APIs must be auditable.

API-06

Breaking changes require version changes.

API-07

Service contracts are authoritative.

16.30 Architectural Constraints

The following constraints are non-negotiable.

AC-API-01

API contracts must remain backward compatible within a version.

AC-API-02

Internal implementation changes must not require contract changes.

AC-API-03

All business operations must be accessible through APIs.

AC-API-04

All API requests must be traceable.

AC-API-05

All API responses must follow standard structures.

AC-API-06

Security controls must be enforced centrally.

16.31 OpenAPI Standard

TODP shall use OpenAPI as the authoritative API specification format.

Objectives

Machine-readable documentation.

Client generation support.

Contract validation.

Developer onboarding.

Deliverable

Separate OpenAPI Specification Document.

16.32 Future API Evolution

Future versions may support:

- Public Developer APIs
- Partner APIs
- Event APIs
- GraphQL APIs
- Federation APIs

without altering core service contracts.

16.33 API Architecture Summary

The TODP API Architecture provides a consistent and governed communication layer between platform components and consumers.

The framework ensures:

- Secure communication
- Traceable operations
- Auditable interactions
- Stable contracts
- Controlled evolution

through versioned service contracts governed by API-first principles.

This architecture serves as the implementation bridge between TODP's business architecture and runtime services.

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Security Architecture & Trust Framework

17.1 Purpose

This section defines the Security Architecture and Trust Framework used by TODP.

Security is not a standalone feature.

Security is a cross-cutting architectural concern that protects:

- Users
- Opportunities
- Authorities
- Evidence
- Rankings
- Recommendations
- Audit Records
- Platform Operations

The objective of the security architecture is to ensure:

- Confidentiality
- Integrity
- Availability
- Accountability
- Traceability
- Trustworthiness

across all platform capabilities.

17.2 Security Philosophy

TODP is built upon trust.

Trust requires:

- Verifiable evidence
- Verifiable authorities
- Transparent rankings
- Protected data
- Auditable actions

Security exists to protect these trust foundations.

Core Principle

Every action must be:

Authenticated

↓

Authorized

↓

Audited

↓

Traceable

17.3 Security Objectives

SO-01 Identity Protection

Protect user and system identities.

SO-02 Access Protection

Prevent unauthorized access.

SO-03 Data Protection

Protect sensitive information.

SO-04 Integrity Protection

Prevent unauthorized modification.

SO-05 Auditability

Maintain forensic visibility.

SO-06 Availability

Maintain service continuity.

SO-07 Governance

Support compliance and oversight.

17.4 Security Architecture Overview

The Security Framework consists of eight major components.

Component 1

Identity Management

Component 2

Authentication Framework

Component 3

Authorization Framework

Component 4

Data Protection Framework

Component 5

Secrets Management Framework

Component 6

Audit Security Framework

Component 7

Infrastructure Security Framework

Component 8

Security Monitoring Framework

17.5 Trust Boundaries

Trust boundaries define where security controls must be enforced.

Boundary 1

Internet



Frontend

Boundary 2

Frontend



API Gateway

Boundary 3

API Gateway



Backend Services

Boundary 4

Backend Services



Database Layer

Boundary 5

Administrative Systems



Operational Systems

Objective

Prevent unauthorized trust propagation.

17.6 Identity Architecture

Every actor within TODP shall possess an identifiable identity.

Identity Types

User Identity

Administrative Identity

Service Identity

System Identity

Objective

Support accountability.

17.7 Authentication Framework

Authentication verifies identity.

Authentication Goals

Verify user identity.

Verify service identity.

Prevent impersonation.

Support auditability.

Authentication Scope

Frontend Users

Administrative Users

Internal Services

External Integrations

17.8 Authentication Lifecycle

Stage 1

Identity Submission

Stage 2

Credential Validation

Stage 3

Identity Verification

Stage 4

Access Token Issuance

Stage 5

Access Monitoring

Stage 6

Session Termination

17.9 Authorization Framework

Authorization determines permitted actions.

Authorization Principle

Authentication answers:

Who are you?

Authorization answers:

What are you allowed to do?

Objectives

Prevent unauthorized actions.

Protect sensitive resources.

Support governance.

17.10 Role-Based Access Control

TODP shall implement Role-Based Access Control (RBAC).

User Role

Capabilities:

Discovery

Ranking Access

Recommendation Access

Opportunity Creation

Organizational Role

Capabilities:

Organization Opportunity Management

Authority Management

Evidence Management

Administrative Role

Capabilities:

Operational Management

Monitoring

Governance Functions

Auditor Role

Capabilities:

Audit Access

Traceability Access

Governance Reviews

17.11 Permission Architecture

Permissions shall be explicit.

Examples

View Opportunity

Create Opportunity

Update Opportunity

View Evidence

Validate Evidence

Verify Authority

Access Audit Records

Objective

Support least privilege access.

17.12 Least Privilege Principle

All actors shall receive the minimum permissions required.

Benefits

Reduced attack surface.

Reduced operational risk.

Improved governance.

17.13 Data Protection Framework

The platform shall protect data throughout its lifecycle.

Protection Scope

User Data

Opportunity Data

Authority Data

Evidence Data

Audit Data

Operational Data

Objectives

Confidentiality

Integrity

Availability

17.14 Data Classification Model

Data shall be classified according to sensitivity.

Public Data

Accessible without restrictions.

Examples:

Published opportunities.

Internal Data

Platform operational information.

Restricted Data

Administrative information.

Sensitive Data

User-related information.

Security information.

Audit information.

17.15 Encryption Architecture

Encryption shall protect data.

Encryption in Transit

Protect network communication.

Encryption at Rest

Protect stored information.

Objective

Prevent unauthorized disclosure.

17.16 Secrets Management Framework

Secrets shall never be embedded in source code.

Managed Secrets

Database Credentials

API Keys

Access Tokens

Encryption Keys

Service Credentials

Requirements

Encrypted Storage

Controlled Access

Auditability

Rotation Support

17.17 Session Security

User sessions shall be protected.

Requirements

Session Validation

Session Expiration

Session Revocation

Secure Storage

Objective

Reduce session abuse.

17.18 API Security Architecture

All APIs shall enforce security controls.

Required Controls

Authentication

Authorization

Input Validation

Rate Limiting

Audit Logging

Objective

Protect platform services.

17.19 Input Validation Framework

All external input shall be validated.

Validation Categories

Schema Validation

Type Validation

Boundary Validation

Business Rule Validation

Objective

Prevent malformed requests.

17.20 Infrastructure Security

Infrastructure shall be protected through layered controls.

Security Controls

Network Segmentation

Access Controls

Firewall Rules

Administrative Restrictions

Monitoring

Objective

Reduce infrastructure exposure.

17.21 Database Security Architecture

Databases shall be protected.

Controls

Authentication

Authorization

Encryption

Backup Protection

Audit Logging

Objective

Protect persistent data.

17.22 Audit Security Framework

Audit records are security-critical assets.

Requirements

Integrity Protection

Access Control

Retention Protection

Tamper Resistance

Objective

Preserve accountability.

17.23 Security Monitoring Framework

Security monitoring shall operate continuously.

Monitored Events

Authentication Events

Authorization Events

Administrative Actions

System Failures

Suspicious Activities

Objective

Early threat detection.

17.24 Security Logging Architecture

Security-relevant events shall be logged.

Log Categories

Authentication Logs

Authorization Logs

Administrative Logs

Infrastructure Logs

Audit Logs

Objective

Support investigations.

17.25 Incident Detection Framework

The platform shall support security incident identification.

Detection Sources

Monitoring Systems

Audit Systems

Application Logs

Infrastructure Logs

Objective

Reduce incident response time.

17.26 Security Incident Response

Security incidents shall follow controlled workflows.

Stages

Detection

↓

Investigation

↓

Containment

↓

Remediation

↓

Recovery

↓

Review

Objective

Restore trust and service continuity.

17.27 Availability Protection

Security architecture shall support availability.

Controls

Rate Limiting

Resource Isolation

Monitoring

Capacity Planning

Failover Procedures

Objective

Protect against service disruption.

17.28 Security Performance Targets

Authentication

P95 < 100 ms

Authorization

P95 < 50 ms

Secret Retrieval

P95 < 50 ms

Security Logging

P95 < 20 ms

17.29 Security Failure Handling

Security controls shall fail safely.

Authentication Failure

Access denied.

Authorization Failure

Action denied.

Validation Failure

Request rejected.

Secret Retrieval Failure

Operation halted.

Objective

Prevent insecure fallback behavior.

17.30 AI Security Policy

AI may assist:

- Threat summarization
 - Log summarization
 - Security explanation
-

AI May Not

Grant permissions.

Override security controls.

Approve access requests.

Modify audit records.

Bypass governance rules.

Reason

Security decisions must remain deterministic and auditable.

17.31 Security Governance Rules

The following rules are mandatory.

SG-01

Every actor must be authenticated.

SG-02

Every action must be authorized.

SG-03

Every sensitive action must be audited.

SG-04

Secrets must remain protected.

SG-05

Audit records must remain immutable.

SG-06

Least privilege shall be enforced.

SG-07

Security incidents must be traceable.

SG-08

Security controls must remain reproducible.

17.32 Architectural Constraints

The following constraints are non-negotiable.

AC-SEC-01

Authentication precedes authorization.

AC-SEC-02

Authorization precedes business operations.

AC-SEC-03

Security events must be logged.

AC-SEC-04

Sensitive data must be protected.

AC-SEC-05

Audit records must be preserved.

AC-SEC-06

Security controls must not be bypassed.

17.33 Zero Trust Alignment

TODP follows Zero Trust principles.

Principles

Never Trust

Always Verify

Least Privilege

Continuous Validation

Explicit Authorization

Objective

Reduce implicit trust assumptions.

17.34 Security Architecture Summary

The TODP Security Architecture provides a comprehensive trust framework protecting users, opportunities, authorities, evidence, rankings, recommendations, and operational systems.

The framework ensures:

- Identity protection
- Access control
- Data protection
- Auditability
- Availability
- Governance

through layered security controls operating across all platform components.

This architecture preserves the trustworthiness required by TODP's Evidence First, Authority Before Recommendation, and Ranking Before Recommendation principles.

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Observability Architecture & Operational Intelligence Framework

18.1 Purpose

This section defines the Observability Architecture and Operational Intelligence Framework used by TODP.

Observability is a foundational capability of TODP because the platform is built upon:

- Traceability
- Auditability
- Reproducibility
- Transparency

Without observability, these principles cannot be verified.

The purpose of the observability architecture is to provide visibility into:

- System behavior
- User interactions

- Retrieval workflows
- Authority workflows
- Evidence workflows
- Ranking workflows
- Recommendation workflows
- Infrastructure health

This framework enables operators, auditors, engineers, and governance teams to understand what happened, why it happened, and how it happened.

18.2 Observability Philosophy

Traditional monitoring answers:

"Is the system working?"

TODP observability answers:

- What happened?
 - Why did it happen?
 - Which data caused it?
 - Which service executed it?
 - Which authority supported it?
 - Which evidence supported it?
 - Which ranking decision was produced?
-

Core Principle

Every important system action shall be observable.

18.3 Observability Objectives

OO-01 Visibility

Provide visibility into platform behavior.

OO-02 Traceability

Provide end-to-end request tracing.

OO-03 Auditability

Support operational and governance audits.

OO-04 Troubleshooting

Accelerate root cause analysis.

OO-05 Performance Monitoring

Monitor operational performance.

OO-06 Business Monitoring

Monitor business workflows.

OO-07 Reproducibility

Support reconstruction of historical decisions.

18.4 Observability Architecture Overview

The Observability Framework consists of seven major components.

Component 1

Metrics Collection Framework

Component 2

Distributed Tracing Framework

Component 3

Logging Framework

Component 4

Business Observability Framework

Component 5

Alerting Framework

Component 6

Dashboard Framework

Component 7

Operational Intelligence Framework

18.5 Technology Stack

TODP adopts the following observability technologies.

OpenTelemetry

Purpose:

Telemetry generation.

Responsibilities:

Metrics

Traces

Context propagation

Instrumentation

Prometheus

Purpose:

Metrics storage and monitoring.

Responsibilities:

Metrics collection

Metrics aggregation

Alert generation

Grafana

Purpose:

Visualization and dashboards.

Responsibilities:

Monitoring dashboards

Operational dashboards

Business dashboards

18.6 Observability Data Types

The platform shall collect four primary categories of telemetry.

Metrics

Quantitative measurements.

Logs

Event records.

Traces

Request execution paths.

Events

Business and operational events.

18.7 Metrics Architecture

Metrics provide quantitative system visibility.

Metric Categories

System Metrics

Application Metrics

Business Metrics

Security Metrics

Objective

Support proactive monitoring.

18.8 System Metrics Framework

System metrics monitor infrastructure health.

Examples

CPU Usage

Memory Usage

Disk Usage

Network Utilization

Container Health

Database Utilization

Cache Utilization

Objective

Maintain platform stability.

18.9 Application Metrics Framework

Application metrics monitor service behavior.

Examples

Request Rate

Error Rate

Response Time

Queue Depth

Service Availability

Transaction Volume

Objective

Maintain application performance.

18.10 Business Metrics Framework

Business metrics monitor platform value delivery.

Examples

Opportunity Searches

Opportunities Retrieved

Authorities Verified

Evidence Validated

Rankings Generated

Recommendations Generated

Opportunity Concepts Generated

Objective

Measure platform effectiveness.

18.11 Security Metrics Framework

Security metrics monitor trust controls.

Examples

Authentication Failures

Authorization Failures

Security Incidents

Permission Violations

Suspicious Activity Events

Objective

Maintain platform trust.

18.12 Governance Metrics Framework

Governance metrics support accountability.

Examples

Audit Events

Lineage Availability

Reproducibility Success Rate

Verification Completion Rate

Objective

Support oversight.

18.13 Distributed Tracing Architecture

Tracing provides request-level visibility.

Purpose

Track requests across system boundaries.

Example Trace

User Request

↓

API Gateway

↓

Discovery Service

↓

Authority Service

↓

Evidence Service

↓

Ranking Service

↓

Recommendation Service



Response

Objective

Understand execution flow.

18.14 Correlation ID Framework

Every request shall contain a correlation identifier.

Purpose

Link:

Logs

Metrics

Traces

Audit Records

Business Events

Objective

Provide end-to-end traceability.

18.15 Logging Architecture

Logging captures detailed event information.

Log Categories

Application Logs

Security Logs

Infrastructure Logs

Audit Logs

Business Logs

Objective

Support diagnostics and investigations.

18.16 Structured Logging Standard

All logs shall be structured.

Required Fields

Timestamp

Severity

Service

Correlation ID

Event Type

Message

Benefits

Machine readability.

Searchability.

Consistency.

18.17 Business Event Architecture

Business events capture important platform activities.

Example Events

Opportunity Created

Authority Verified

Evidence Validated

Ranking Completed

Recommendation Generated

Opportunity Archived

Opportunity Concept Generated

Objective

Support business observability.

18.18 Retrieval Observability Framework

Retrieval workflows shall be observable.

Monitored Activities

Search Requests

BM25 Retrieval

Vector Retrieval

Hybrid Retrieval

Authority Retrieval

Evidence Retrieval

Objective

Monitor discovery quality.

18.19 Authority Observability Framework

Authority workflows shall be observable.

Monitored Activities

Authority Registration

Authority Verification

Confidence Assessment

Authority Revalidation

Objective

Monitor authority trustworthiness.

18.20 Evidence Observability Framework

Evidence workflows shall be observable.

Monitored Activities

Evidence Collection

Evidence Validation

Evidence Versioning

Evidence Usage

Objective

Monitor evidence quality.

18.21 Ranking Observability Framework

Ranking is a critical business workflow.

Monitored Activities

Cost Calculation

Ranking Execution

Tie Resolution

Ranking Explanation Generation

Ranking Audit Recording

Objective

Monitor ranking integrity.

18.22 Recommendation Observability Framework

Recommendation workflows shall be observable.

Monitored Activities

Recommendation Assembly

Explanation Generation

Authority Presentation

Evidence Presentation

Recommendation Delivery

Objective

Monitor recommendation transparency.

18.23 Opportunity Creation Observability

Creation workflows shall be observable.

Monitored Activities

Similarity Analysis

Pattern Analysis

Gap Detection

Concept Generation

Creation Validation

Objective

Monitor innovation support workflows.

18.24 Dashboard Architecture

Dashboards provide operational visibility.

Dashboard Categories

Executive Dashboard

Operational Dashboard

Engineering Dashboard

Governance Dashboard

Security Dashboard

Objective

Support decision-making.

18.25 Executive Dashboard

Provides high-level platform visibility.

Metrics

Active Opportunities

Search Volume

Recommendation Volume

Authority Coverage

Evidence Coverage

Audience

Leadership

Product Owners

Governance Teams

18.26 Operational Dashboard

Provides runtime visibility.

Metrics

Availability

Latency

Error Rates

Traffic Volume

Database Health

Cache Health

Audience

Operations Teams

SRE Teams

18.27 Engineering Dashboard

Provides technical visibility.

Metrics

Trace Performance

Query Performance

Deployment Health

Service Health

Dependency Health

Audience

Engineering Teams

18.28 Governance Dashboard

Provides trust and compliance visibility.

Metrics

Evidence Validation Rate

Authority Verification Rate

Audit Coverage

Lineage Coverage

Reproducibility Success Rate

Audience

Governance Teams

Auditors

18.29 Alerting Architecture

Alerts notify operators of important conditions.

Alert Categories

Availability Alerts

Performance Alerts

Security Alerts

Governance Alerts

Business Alerts

Objective

Reduce detection time.

18.30 Alert Severity Model

Critical

Immediate action required.

High

Urgent investigation required.

Medium

Operational review required.

Low

Informational review.

18.31 Operational Intelligence Framework

Operational intelligence combines telemetry into actionable insights.

Inputs

Metrics

Logs

Traces

Events

Audit Data

Outputs

Operational Insights

Performance Insights

Governance Insights

Risk Indicators

Objective

Support informed operations.

18.32 Observability Data Retention

Observability data shall follow retention policies.

Metrics

Configurable retention.

Logs

Configurable retention.

Traces

Configurable retention.

Audit Data

Governance-driven retention.

18.33 Performance Targets

Metrics Collection

P95 < 10 ms overhead

Trace Collection

P95 < 20 ms overhead

Log Collection

P95 < 5 ms overhead

Dashboard Refresh

P95 < 5 seconds

18.34 Failure Handling

The observability platform shall fail gracefully.

Metrics Failure

Core business workflows continue.

Logging Failure

Fallback logging mechanisms activated.

Trace Failure

Requests continue processing.

Dashboard Failure

Monitoring data remains preserved.

18.35 AI Usage Policy

AI may assist:

- Log summarization
- Trace summarization
- Operational insight generation
- Incident explanation

AI May Not

Modify telemetry.

Alter audit records.

Suppress alerts.

Override governance controls.

Reason

Observability must remain trustworthy.

18.36 Observability Governance Rules

The following rules are mandatory.

OBS-01

All critical workflows must be observable.

OBS-02

All requests must be traceable.

OBS-03

All logs must be attributable.

OBS-04

Observability data must be auditable.

OBS-05

Correlation IDs must be preserved.

OBS-06

Governance workflows must be observable.

OBS-07

Observability controls must remain reproducible.

18.37 Architectural Constraints

The following constraints are non-negotiable.

AC-OBS-01

Every request must generate trace context.

AC-OBS-02

Every critical business workflow must emit telemetry.

AC-OBS-03

Audit workflows must remain observable.

AC-OBS-04

Correlation IDs must remain available throughout execution.

AC-OBS-05

Observability systems must not modify business outcomes.

AC-OBS-06

Operational visibility must remain independent of AI systems.

18.38 Observability Architecture Summary

The TODP Observability Architecture provides comprehensive visibility into platform operations, business workflows, governance processes, and trust mechanisms.

The framework combines:

OpenTelemetry

↓

Prometheus

↓

Grafana

to deliver:

- Metrics

- Logs
- Traces
- Events
- Dashboards
- Alerts
- Operational Intelligence

while preserving the platform's core principles of traceability, auditability, transparency, and reproducibility.

Observability is therefore treated as a first-class architectural capability rather than an operational afterthought.

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Auditability, Traceability & Reproducibility Architecture

19.1 Purpose

This section defines the Auditability, Traceability, and Reproducibility (ATR) Architecture used by TODP.

Unlike conventional systems where auditing is primarily a compliance function, TODP treats:

- Auditability
- Traceability
- Reproducibility

as foundational product capabilities.

These capabilities are required to support:

- Evidence First
- Authority Before Recommendation
- Ranking Before Recommendation
- No Hidden Ranking Logic
- AI Is Not An Authority

The ATR Architecture ensures that every important platform decision can be:

- Examined
- Explained
- Verified
- Reconstructed

at any point in the future.

19.2 ATR Philosophy

TODP operates on a simple principle:

If a decision cannot be traced, it cannot be trusted.

If a decision cannot be reproduced, it cannot be verified.

If a decision cannot be audited, it cannot be governed.

Core Principle

Every important platform outcome must answer:

What happened?

Why did it happen?

Who caused it?

Which evidence supported it?

Which authority supported it?

Which ranking logic produced it?

Can it be reproduced?

19.3 ATR Objectives

ATR-01 Auditability

Support systematic examination of platform decisions.

ATR-02 Traceability

Provide end-to-end decision lineage.

ATR-03 Reproducibility

Allow reconstruction of historical outcomes.

ATR-04 Transparency

Expose decision inputs and outputs.

ATR-05 Governance

Support oversight and accountability.

ATR-06 Trustworthiness

Strengthen platform credibility.

19.4 ATR Architecture Overview

The ATR Framework consists of seven major components.

Component 1

Audit Framework

Component 2

Lineage Framework

Component 3

Decision Trace Framework

Component 4

Historical State Framework

Component 5

Reproducibility Framework

Component 6

Governance Review Framework

Component 7

ATR Reporting Framework

19.5 Audit Architecture

Auditability records significant platform events.

Purpose

Provide evidence of actions and decisions.

Objectives

Accountability

Governance

Investigation Support

Compliance Support

19.6 Audit Event Model

Every audit event shall capture:

Required Fields

Audit ID

Timestamp

Actor

Action

Entity

Previous State

New State

Correlation ID

Optional Fields

Reason

Comments

Review Metadata

19.7 Audit Categories

TODP supports multiple audit domains.

User Audits

User actions.

Opportunity Audits

Opportunity lifecycle changes.

Authority Audits

Authority verification activities.

Evidence Audits

Evidence validation activities.

Ranking Audits

Ranking calculations.

Recommendation Audits

Recommendation generation.

Administrative Audits

Governance and operational actions.

19.8 Immutable Audit Principle

Audit records shall be immutable.

Objectives

Prevent tampering.

Preserve accountability.

Support investigations.

Requirement

Audit records shall never be silently modified.

19.9 Traceability Architecture

Traceability connects actions and decisions across the platform.

Purpose

Provide complete lineage visibility.

Objective

Understand how outcomes were produced.

19.10 Lineage Framework

Lineage captures relationships between entities and decisions.

Example

Evidence

↓

Authority

↓

Opportunity

↓

Retrieval

↓

Ranking

↓

Recommendation

Objective

Support decision reconstruction.

19.11 Decision Trace Architecture

Every major decision shall generate a decision trace.

Examples

Authority Verification

Evidence Validation

Ranking Decision

Recommendation Generation

Opportunity Eligibility

Objective

Explain why a decision occurred.

19.12 Trace Context Model

Trace context connects related events.

Trace Components

Correlation ID

Request ID

Entity IDs

Service Path

Timestamps

Objective

Enable end-to-end visibility.

19.13 Ranking Traceability Framework

Ranking decisions require detailed traceability.

Required Trace Elements

Opportunity

Authority

Evidence

Money Cost

Time Cost

Distance Cost

Total Cost

Tie Resolution Logic

Final Rank

Objective

Support ranking transparency.

19.14 Recommendation Traceability Framework

Recommendations require complete lineage.

Required Trace Elements

Query

Retrieved Opportunities

Authority Information

Evidence Information

Ranking Output

Recommendation Output

Objective

Support explanation and auditing.

19.15 Authority Traceability Framework

Authority decisions must remain explainable.

Trace Elements

Authority Record

Verification Evidence

Confidence Assessment

Validation Outcomes

Usage History

Objective

Support trust evaluation.

19.16 Evidence Traceability Framework

Evidence lineage must remain available.

Trace Elements

Source

Collection Event

Validation Event

Version History

Usage History

Objective

Support Evidence First governance.

19.17 Historical State Architecture

Historical states enable reconstruction.

Purpose

Preserve system history.

Covered Entities

Opportunities

Authorities

Evidence

Rankings

Recommendations

User Context

Objective

Support reproducibility.

19.18 State Versioning Framework

Historical states shall be versioned.

Versioning Triggers

Entity Changes

Validation Changes

Ranking Changes

Governance Changes

Requirement

Historical versions remain accessible.

19.19 Reproducibility Architecture

Reproducibility reconstructs historical outcomes.

Purpose

Verify past decisions.

Objective

Produce identical outputs from historical inputs.

19.20 Reproducibility Requirements

The following components must be reproducible.

Opportunity State

Historical opportunity versions.

Authority State

Historical authority versions.

Evidence State

Historical evidence versions.

Ranking Outcomes

Historical ranking calculations.

Recommendation Outputs

Historical recommendation packages.

19.21 Reproducibility Workflow

Historical Inputs

↓

Historical Rules

↓

Historical Context

↓

Historical Calculations

↓

Historical Outputs

Objective

Validate platform integrity.

19.22 Governance Review Framework

Governance teams require systematic review capabilities.

Review Activities

Audit Reviews

Ranking Reviews

Authority Reviews

Evidence Reviews

Recommendation Reviews

Objective

Support oversight.

19.23 ATR Reporting Framework

ATR reports summarize platform accountability.

Report Categories

Audit Reports

Traceability Reports

Reproducibility Reports

Governance Reports

Investigation Reports

Objective

Support transparency.

19.24 Investigation Support Framework

The platform shall support investigations.

Investigation Inputs

Audit Records

Logs

Traces

Events

Lineage Data

Historical States

Outputs

Investigation Findings

Decision Explanations

Root Cause Analysis

19.25 Compliance Support

ATR capabilities support governance obligations.

Supported Areas

Accountability

Transparency

Decision Explanation

Operational Oversight

Historical Verification

Objective

Enable defensible decision-making.

19.26 ATR Search Architecture

Audit and lineage information shall be searchable.

Search Dimensions

Entity

Actor

Timestamp

Correlation ID

Decision Type

Audit Type

Objective

Accelerate investigations.

19.27 ATR Performance Targets

Audit Recording

P95 < 50 ms

Lineage Retrieval

P95 < 200 ms

Historical State Retrieval

P95 < 300 ms

Reproducibility Execution

P95 < 2 seconds

19.28 Failure Handling

ATR systems shall fail safely.

Audit Failure

Business event proceeds only if policy allows.

Failure recorded.

Lineage Failure

Entity flagged for review.

Historical State Failure

Reconstruction unavailable until resolved.

Objective

Protect governance integrity.

19.29 AI Usage Policy

AI may assist:

- Audit summarization
- Trace summarization
- Investigation assistance
- Report generation

AI May Not

Modify audit records.

Modify lineage.

Alter historical states.

Change reproducibility outcomes.

Override governance reviews.

Reason

ATR systems must remain trustworthy.

19.30 ATR Governance Rules

The following rules are mandatory.

ATR-01

All critical decisions must be auditable.

ATR-02

All recommendations must be traceable.

ATR-03

All rankings must be reproducible.

ATR-04

Lineage must remain available.

ATR-05

Historical states must be recoverable.

ATR-06

Audit records must remain immutable.

ATR-07

Governance reviews must be supported.

ATR-08

Decision explanations must be available.

19.31 Architectural Constraints

The following constraints are non-negotiable.

AC-ATR-01

Every ranking must generate traceable lineage.

AC-ATR-02

Every recommendation must generate traceable lineage.

AC-ATR-03

Historical states must be preserved.

AC-ATR-04

Audit records must remain immutable.

AC-ATR-05

Reproducibility must use historical inputs.

AC-ATR-06

Governance systems must remain independent of AI-generated decisions.

19.32 ATR Maturity Model

The platform shall continuously improve ATR capabilities.

Level 1

Basic Auditing

Level 2

Traceability Coverage

Level 3

Historical Reconstruction

Level 4

Decision Reproducibility

Level 5

Full Governance Intelligence

Objective

Provide a roadmap for operational maturity.

19.33 ATR Architecture Summary

The Auditability, Traceability, and Reproducibility Architecture provides the governance backbone of TODP.

The framework ensures that:

- Decisions are auditable.
- Outcomes are traceable.
- Historical states are preserved.
- Results are reproducible.
- Governance reviews are supported.

Through audit records, lineage models, historical state preservation, and reproducibility workflows, TODP enables transparent and accountable decision-making across retrieval,

authority verification, evidence validation, ranking, recommendation, and opportunity creation processes.

This architecture transforms auditability from a compliance feature into a core product capability.

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Event-Driven Architecture & Inter-Service Communication Framework

20.1 Purpose

This section defines the Event-Driven Architecture (EDA) and Inter-Service Communication Framework used by TODP.

TODP consists of multiple domain services including:

- Discovery Service
- Authority Service

- Evidence Service
- Ranking Service
- Recommendation Service
- Opportunity Service
- User Service
- Opportunity Creation Service
- Audit Service

These services must communicate efficiently while remaining:

- Loosely coupled
- Independently deployable
- Scalable
- Observable
- Auditable
- Traceable

The purpose of this architecture is to define how services exchange information while preserving system integrity and domain ownership.

20.2 Communication Philosophy

TODP follows a hybrid communication model.

Not every interaction requires asynchronous communication.

Not every interaction requires synchronous communication.

The architecture shall use the most appropriate communication mechanism based on business requirements.

Core Principle

Strong Consistency

→ Synchronous Communication

Event Notification

→ Asynchronous Communication

Business Ownership

→ Domain Events

20.3 Communication Objectives

CO-01 Decoupling

Reduce service dependencies.

CO-02 Scalability

Support increasing workloads.

CO-03 Reliability

Prevent cascading failures.

CO-04 Traceability

Maintain communication visibility.

CO-05 Auditability

Preserve communication history.

CO-06 Evolvability

Enable independent service evolution.

20.4 Communication Architecture Overview

The communication framework consists of six major components.

Component 1

Synchronous Communication Layer

Component 2

Asynchronous Communication Layer

Component 3

Domain Event Framework

Component 4

Workflow Orchestration Framework

Component 5

Event Governance Framework

Component 6

Communication Observability Framework

20.5 Communication Models

TODP supports two primary communication models.

Model 1

Synchronous Request-Response

Characteristics

Immediate response required.

Strong consistency required.

Caller waits for response.

Examples

Retrieve Opportunity

Retrieve Authority

Retrieve Evidence

Retrieve Recommendation

Model 2

Asynchronous Event Communication

Characteristics

Delayed processing acceptable.

Loose coupling desired.

Publisher unaware of consumers.

Examples

Opportunity Created

Evidence Validated

Authority Verified

Ranking Completed

Recommendation Generated

20.6 Synchronous Communication Architecture

Synchronous communication shall primarily use HTTP APIs.

Technology

FastAPI

REST APIs

OpenAPI Contracts

Objectives

Low latency.

Strong consistency.

Simple interaction patterns.

20.7 Synchronous Communication Rules

Synchronous communication shall be used when:

Immediate Results Required

Example:

User Search Request

Transactional Consistency Required

Example:

Opportunity Update

Validation Required Before Progress

Example:

Authority Verification Check

20.8 Asynchronous Communication Architecture

Asynchronous communication shall use event-based messaging.

Objectives

Loose coupling.

Scalability.

Reliability.

Independent processing.

Examples

Opportunity Lifecycle Events

Audit Events

Notification Events

Analytics Events

20.9 Domain Event Philosophy

Events represent facts that have already occurred.

Example

Good Event

Authority Verified

Bad Event

Verify Authority

Principle

Events describe outcomes.

Commands describe intentions.

20.10 Event Structure Standard

Every domain event shall contain:

Required Fields

Event ID

Event Type

Event Version

Timestamp

Correlation ID

Producer Service

Entity ID

Payload

Optional Fields

Metadata

Trace Context

Governance Information

20.11 Event Categories

Events shall be organized by business domain.

Opportunity Events

Opportunity Created

Opportunity Updated

Opportunity Archived

Opportunity State Changed

Authority Events

Authority Registered

Authority Verified

Authority Revalidated

Authority Confidence Updated

Evidence Events

Evidence Collected

Evidence Validated

Evidence Updated

Evidence Archived

Ranking Events

Ranking Started

Ranking Completed

Ranking Recalculated

Ranking Archived

Recommendation Events

Recommendation Generated

Recommendation Delivered

Recommendation Archived

User Events

Context Updated

Preference Updated

Constraint Updated

Creation Events

Similarity Analysis Completed

Gap Analysis Completed

Opportunity Concept Generated

20.12 Event Lifecycle

Events follow a controlled lifecycle.

Stage 1

Event Creation

Stage 2

Event Publication

Stage 3

Event Consumption

Stage 4

Event Processing

Stage 5

Event Archival

20.13 Event Ownership Model

Every event shall have a single owner.

Principle

Only the owning domain may publish authoritative domain events.

Example

Authority Service

↓

Publishes

↓

Authority Verified

Benefit

Clear ownership boundaries.

20.14 Event Versioning Strategy

Events shall support evolution.

Objectives

Backward compatibility.

Consumer stability.

Independent deployment.

Rule

Breaking event changes require version increments.

20.15 Event Traceability Framework

Events must remain traceable.

Traceability Components

Event ID

Correlation ID

Producer

Consumer

Timestamp

Entity References

Objective

Support investigations and audits.

20.16 Workflow Orchestration Framework

Certain business workflows require coordination.

Example

Opportunity Lifecycle

Opportunity Created

↓

Evidence Validation

↓

Authority Verification

↓

Eligibility Evaluation

↓

Ranking

↓

Recommendation

Objective

Coordinate multi-service workflows.

20.17 Choreography vs Orchestration

TODP supports both approaches.

Choreography

Services react to events independently.

Example

Evidence Validated



Multiple consumers react.

Orchestration

Central workflow coordinates execution.

Example

Eligibility Workflow



Controlled sequence.

Rule

Use choreography by default.

Use orchestration for complex workflows.

20.18 Event Consistency Model

Different workflows require different consistency guarantees.

Strong Consistency

Used for:

Critical business transactions.

Eventual Consistency

Used for:

Notifications

Analytics

Reporting

Telemetry

Objective

Balance correctness and scalability.

20.19 Retry Framework

Transient failures shall support retries.

Retry Conditions

Temporary network issues.

Temporary service unavailability.

Temporary infrastructure failures.

Objective

Improve reliability.

20.20 Idempotency Framework

Event processing shall be idempotent.

Definition

Processing the same event multiple times produces the same outcome.

Objective

Prevent duplicate effects.

20.21 Dead Letter Handling

Failed events require controlled handling.

Dead Letter Conditions

Maximum retries exceeded.

Invalid payload.

Consumer failure.

Result

Dead Letter Queue Processing.

20.22 Event Security Framework

Events shall be protected.

Controls

Authentication

Authorization

Encryption

Integrity Validation

Audit Logging

Objective

Protect inter-service communication.

20.23 Communication Observability

All communication flows shall be observable.

Observability Scope

API Requests

Domain Events

Workflow Execution

Consumer Processing

Retry Activity

Failure Events

Objective

Support operational visibility.

20.24 Communication Audit Framework

Communication activities shall be auditable.

Audit Scope

Event Publication

Event Consumption

Workflow Execution

Retry Events

Dead Letter Events

Administrative Actions

Objective

Support governance.

20.25 Event Retention Framework

Events shall be retained according to governance policies.

Retained Information

Event Metadata

Event Payload

Publication History

Consumption History

Objective

Support traceability and reproducibility.

20.26 Event Search Architecture

Events shall be searchable.

Search Dimensions

Event Type

Entity

Timestamp

Correlation ID

Producer

Consumer

Objective

Accelerate investigations.

20.27 Performance Targets

Event Publication

P95 < 50 ms

Event Consumption

P95 < 200 ms

Workflow Coordination

P95 < 500 ms

Event Trace Retrieval

P95 < 300 ms

20.28 Failure Handling

Communication failures shall be handled safely.

API Failure

Retry or fallback.

Event Publication Failure

Retry publication.

Record incident.

Consumer Failure

Retry processing.

Dead Letter Failure

Escalate operational alert.

20.29 AI Usage Policy

AI may assist:

- Event analysis
 - Workflow analysis
 - Communication diagnostics
 - Operational explanations
-

AI May Not

Publish events.

Modify event payloads.

Override workflows.

Alter event history.

Reason

Communication integrity must remain deterministic.

20.30 Communication Governance Rules

The following rules are mandatory.

EDA-01

Every event must have an owner.

EDA-02

Events must describe facts.

EDA-03

Event lineage must remain available.

EDA-04

Event processing must be observable.

EDA-05

Event processing must be auditable.

EDA-06

Event versions must be governed.

EDA-07

Consumers must be idempotent.

20.31 Architectural Constraints

The following constraints are non-negotiable.

AC-EDA-01

Domain ownership must remain preserved.

AC-EDA-02

Events shall not bypass governance workflows.

AC-EDA-03

Communication shall remain traceable.

AC-EDA-04

Critical business workflows must remain auditable.

AC-EDA-05

Event history must remain available.

AC-EDA-06

Communication mechanisms must not alter business decisions.

20.32 Future Evolution

Future versions may support:

- Dedicated Event Brokers
- Event Streaming Platforms
- Workflow Engines
- Cross-Region Event Replication
- Event Sourcing Patterns

without altering domain ownership principles.

20.33 Event-Driven Architecture Summary

The TODP Event-Driven Architecture provides a scalable and governable communication framework connecting platform services.

The architecture combines:

Synchronous APIs

-

Asynchronous Events

-

Workflow Coordination

to enable:

- Loose coupling
- Independent evolution
- Operational visibility
- Traceability
- Auditability
- Reproducibility

while preserving domain ownership and governance integrity.

This architecture establishes the foundation for enterprise-scale service interactions across all TODP capabilities.

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SDD V1 — Section 21

Scalability, Performance & Capacity Architecture

21.1 Purpose

This section defines the Scalability, Performance, and Capacity Architecture used by TODP.

TODP is designed to evolve from an initial deployment supporting limited usage into a large-scale platform supporting:

- Opportunity Discovery
- Authority Verification
- Evidence Validation
- Ranking
- Recommendation
- Opportunity Creation
- Governance Operations

across multiple geographic levels.

The purpose of this architecture is to ensure that the platform remains:

- Responsive
- Reliable
- Scalable
- Cost Efficient
- Observable
- Governable

as demand grows.

21.2 Scalability Philosophy

TODP follows a progressive scalability model.

The platform shall scale incrementally based on demonstrated demand.

The architecture shall avoid premature complexity while remaining capable of future expansion.

Core Principle

Scale only when necessary.

Design for scaling from the beginning.

21.3 Objectives

SPO-01 Performance

Provide predictable response times.

SPO-02 Scalability

Support increasing workloads.

SPO-03 Reliability

Maintain stable operations under load.

SPO-04 Efficiency

Optimize resource utilization.

SPO-05 Observability

Measure scaling effectiveness.

SPO-06 Cost Awareness

Scale responsibly.

21.4 Scalability Architecture Overview

The framework consists of seven major components.

Component 1

Application Scalability

Component 2

Database Scalability

Component 3

Retrieval Scalability

Component 4

Caching Scalability

Component 5

Event Processing Scalability

Component 6

Observability Scalability

Component 7

21.5 Scaling Dimensions

TODP must scale across multiple dimensions.

User Growth

Increase in platform users.

Opportunity Growth

Increase in opportunity records.

Authority Growth

Increase in authority records.

Evidence Growth

Increase in evidence records.

Search Growth

Increase in retrieval requests.

Recommendation Growth

Increase in recommendation generation.

Governance Growth

Increase in audit and trace workloads.

21.6 Application Scalability

Application services shall scale horizontally.

Strategy

Stateless service design.

Benefits

Independent scaling.

Fault isolation.

Operational simplicity.

Example

FastAPI Service

↓

Multiple Instances

↓

Load Balancer

21.7 Service Scaling Model

Each domain service shall scale independently.

Discovery Service

Scale based on search demand.

Ranking Service

Scale based on ranking demand.

Recommendation Service

Scale based on recommendation demand.

Authority Service

Scale based on verification demand.

Evidence Service

Scale based on validation demand.

Benefit

Resource efficiency.

21.8 Database Scalability Architecture

PostgreSQL serves as the primary system of record.

Stage 1

Single PostgreSQL Instance

Stage 2

Read Replicas

Stage 3

Table Partitioning

Stage 4

Distributed Database Architecture

Objective

Support increasing transactional workloads.

21.9 Database Partitioning Strategy

Partitioning shall be introduced only when required.

Candidate Tables

Search Records

Audit Records

Ranking Records

Recommendation Records

Event Records

Partition Types

Time-Based

Geographic

Organizational

21.10 Retrieval Scalability Architecture

Retrieval workloads are expected to become one of the largest system loads.

Components

BM25 Retrieval

Vector Retrieval

Hybrid Retrieval

Authority Retrieval

Evidence Retrieval

Objective

Maintain low-latency discovery.

21.11 BM25 Scaling Strategy

BM25 retrieval shall scale through index optimization.

Techniques

Index Optimization

Query Optimization

Read Scaling

Caching

Objective

Support large opportunity catalogs.

21.12 Vector Retrieval Scalability

Vector retrieval requires specialized scaling considerations.

Initial Technology

pgvector

Scaling Path

Single Node

↓

Optimized Indexes

↓

Read Replicas

↓

Dedicated Vector Infrastructure

Objective

Support semantic discovery growth.

21.13 Hybrid Retrieval Scalability

Hybrid retrieval combines multiple retrieval mechanisms.

Scaling Challenges

Multiple retrieval paths.

Result merging.

Ranking preparation.

Mitigation

Parallel execution.

Caching.

Efficient aggregation.

21.14 Ranking Scalability Architecture

Ranking workloads grow with discovery volume.

Scaling Inputs

Candidate Opportunities

User Context

Authority Information

Evidence Information

Objective

Maintain ranking responsiveness.

21.15 Recommendation Scalability Architecture

Recommendation workloads scale with user demand.

Responsibilities

Recommendation Assembly

Explanation Generation

Packaging

Delivery

Scaling Strategy

Horizontal scaling.

Response caching.

Asynchronous preparation where appropriate.

21.16 Opportunity Creation Scalability

Creation workflows may become computationally intensive.

Workloads

Similarity Analysis

Pattern Analysis

Gap Detection

Concept Generation Support

Strategy

Asynchronous execution.

Dedicated worker pools.

21.17 Redis Scalability Architecture

Redis provides caching and temporary state management.

Scaling Objectives

Reduce database load.

Reduce retrieval latency.

Reduce ranking latency.

Scaling Path

Single Instance

↓

Replication

↓

Cluster Deployment

21.18 Caching Strategy

Caching shall focus on high-value workloads.

Cache Candidates

Search Results

Ranking Results

Recommendations

Authority Lookups

Evidence Lookups

Objective

Reduce repeated computation.

21.19 Event Processing Scalability

Event-driven workloads must scale independently.

Workloads

Domain Events

Audit Events

Observability Events

Notification Events

Strategy

Consumer scaling.

Queue scaling.

Retry management.

21.20 Observability Scalability

Observability systems must scale alongside the platform.

Components

Metrics

Logs

Traces

Events

Dashboards

Alerts

Objective

Maintain operational visibility.

21.21 Capacity Planning Framework

Capacity planning shall be proactive.

Inputs

Growth Trends

Usage Metrics

Business Projections

Performance Data

Objective

Prevent resource shortages.

21.22 Capacity Categories

Compute Capacity

CPU

Memory

Containers

Storage Capacity

Database

Object Storage

Audit Storage

Network Capacity

Bandwidth

Connection Limits

Traffic Volume

Observability Capacity

Metrics

Logs

Traces

21.23 Performance Architecture

Performance management shall be continuous.

Performance Dimensions

Latency

Throughput

Availability

Resource Utilization

Error Rates

Objective

Maintain user experience.

21.24 Service-Level Objectives (SLOs)

The platform shall define measurable objectives.

Availability

99.9% Target

API Latency

P95 < 300 ms

Search Latency

P95 < 2 seconds

Ranking Latency

P95 < 1 second

Recommendation Latency

P95 < 500 ms

21.25 Capacity Threshold Framework

Thresholds shall trigger operational review.

Examples

CPU > 80%

Memory > 80%

Storage > 80%

Error Rate Thresholds

Latency Thresholds

Objective

Prevent service degradation.

21.26 Load Testing Framework

The platform shall support systematic load testing.

Test Categories

Baseline Testing

Stress Testing

Spike Testing

Soak Testing

Failure Testing

Objective

Validate scalability assumptions.

21.27 Resilience Architecture

Scalability must not compromise reliability.

Controls

Redundancy

Retry Policies

Circuit Breakers

Graceful Degradation

Health Checks

Objective

Maintain service continuity.

21.28 Geographic Scaling Strategy

Future growth may require geographic expansion.

Expansion Stages

Single Region

↓

Multi-Region Read Access

↓

Multi-Region Services

↓

Global Deployment

Objective

Support international growth.

21.29 Cost Efficiency Framework

Scaling decisions shall consider cost impact.

Evaluation Factors

Performance Gain

Operational Complexity

Infrastructure Cost

Business Value

Objective

Maintain sustainable growth.

21.30 Performance Monitoring Framework

Performance metrics shall be continuously monitored.

Monitored Areas

API Performance

Database Performance

Retrieval Performance

Ranking Performance

Recommendation Performance

Creation Workflows

Objective

Detect performance degradation early.

21.31 Failure Handling

Scalability controls shall fail safely.

Scaling Failure

Maintain existing capacity.

Cache Failure

Fallback to primary systems.

Replica Failure

Fallback to primary database.

Worker Failure

Retry processing.

Objective

Preserve availability.

21.32 AI Usage Policy

AI may assist:

- Capacity forecasting
 - Performance analysis
 - Operational reporting
-

AI May Not

Control autoscaling.

Override capacity limits.

Modify operational policies.

Alter governance thresholds.

Reason

Operational decisions must remain governed.

21.33 Scalability Governance Rules

The following rules are mandatory.

SCALE-01

Services must support independent scaling.

SCALE-02

Scaling must preserve traceability.

SCALE-03

Scaling must preserve auditability.

SCALE-04

Performance targets must be measurable.

SCALE-05

Capacity planning must be proactive.

SCALE-06

Scalability changes must remain observable.

SCALE-07

Scaling must not alter business outcomes.

21.34 Architectural Constraints

The following constraints are non-negotiable.

AC-SCALE-01

Business logic must remain independent of scaling mechanisms.

AC-SCALE-02

Scaling shall not compromise governance controls.

AC-SCALE-03

Retrieval accuracy must remain consistent during scaling.

AC-SCALE-04

Ranking reproducibility must remain preserved.

AC-SCALE-05

Observability coverage must remain intact.

AC-SCALE-06

Scalability improvements must remain measurable.

21.35 Scalability Maturity Model

The platform shall evolve through defined maturity stages.

Level 1

Single Deployment

Level 2

Horizontally Scaled Services

Level 3

Distributed Retrieval

Level 4

Multi-Region Platform

Level 5

Objective

Provide a structured scaling roadmap.

21.36 Scalability, Performance & Capacity Summary

The TODP Scalability Architecture provides a structured framework for growing platform capabilities while preserving:

- Performance
- Reliability
- Traceability
- Auditability
- Reproducibility

The framework covers:

Application Scaling

↓

Database Scaling

↓

Retrieval Scaling

↓

Caching

↓

Event Processing

↓

Observability

↓

Capacity Planning

ensuring that TODP can evolve from an initial deployment into a large-scale opportunity discovery ecosystem without requiring fundamental architectural redesign.

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SDD V1 — Section 22

Reliability, Resilience & Disaster Recovery Architecture

22.1 Purpose

This section defines the Reliability, Resilience, and Disaster Recovery (RRDR) Architecture used by TODP.

TODP is intended to operate as a trusted opportunity discovery platform where users rely upon:

- Opportunity Availability
- Authority Information
- Evidence Information
- Ranking Outputs
- Recommendation Outputs
- Audit Records

Therefore, the platform must continue operating despite:

- Infrastructure failures
- Application failures
- Service failures
- Data failures
- Operational mistakes
- Regional outages

The purpose of this architecture is to ensure business continuity while preserving:

- Traceability
- Auditability
- Reproducibility
- Data Integrity

22.2 Reliability Philosophy

Reliability is not the absence of failures.

Reliability is the ability to continue delivering expected outcomes despite failures.

Core Principle

Failures are inevitable.

Unrecoverable failures are unacceptable.

Architectural Goal

Detect failures.

Contain failures.

Recover from failures.

Learn from failures.

22.3 Objectives

RR-01 Availability

Maintain service availability.

RR-02 Fault Tolerance

Continue operating despite component failures.

RR-03 Recoverability

Restore services quickly.

RR-04 Data Protection

Prevent irreversible data loss.

RR-05 Continuity

Preserve critical business workflows.

RR-06 Governance Preservation

Maintain auditability and traceability during recovery.

22.4 Reliability Architecture Overview

The framework consists of eight major components.

Component 1

Failure Detection Framework

Component 2

Fault Isolation Framework

Component 3

Redundancy Framework

Component 4

Recovery Framework

Component 5

Backup Framework

Component 6

Disaster Recovery Framework

Component 7

Component 8

Reliability Governance Framework

22.5 Reliability Domains

Reliability shall be addressed across all platform layers.

Application Reliability

Backend services.

Database Reliability

PostgreSQL and pgvector.

Retrieval Reliability

Search and discovery systems.

Ranking Reliability

Ranking workflows.

Recommendation Reliability

Recommendation workflows.

Observability Reliability

Monitoring and tracing systems.

Governance Reliability

Audit and lineage systems.

22.6 Failure Classification Model

Failures shall be classified for response planning.

Service Failure

Application service unavailable.

Infrastructure Failure

Host or container failure.

Database Failure

Persistence layer failure.

Network Failure

Connectivity disruption.

Dependency Failure

External or internal dependency unavailable.

Operational Failure

Human-caused issue.

22.7 Failure Detection Framework

The platform shall continuously monitor health.

Detection Sources

Health Checks

Metrics

Logs

Traces

Alerts

Synthetic Monitoring

Objective

Rapid failure identification.

22.8 Health Check Architecture

Every service shall expose health endpoints.

Health Types

Liveness

Readiness

Dependency Health

Objective

Support automated recovery.

22.9 Fault Isolation Framework

Failures shall be isolated whenever possible.

Objectives

Prevent cascading failures.

Protect unrelated services.

Maintain availability.

Examples

Service Isolation

Database Isolation

Worker Isolation

Queue Isolation

22.10 Redundancy Architecture

Critical components shall support redundancy.

Candidate Components

API Layer

Application Services

Databases

Caches

Observability Systems

Objective

Remove single points of failure.

22.11 Graceful Degradation Framework

The platform shall continue operating when possible.

Examples

Recommendation unavailable

↓

Discovery remains available

Observability degraded

↓

Business processing continues

Objective

Maintain partial functionality.

22.12 Circuit Breaker Framework

Circuit breakers prevent cascading failures.

Workflow

Failure Threshold Reached

↓

Circuit Opens

↓

Requests Blocked

↓

Recovery Evaluation

↓

Circuit Closes

Objective

Protect dependent systems.

22.13 Retry Framework

Transient failures shall support retries.

Examples

Temporary Network Failure

Temporary Service Failure

Temporary Infrastructure Issue

Objective

Increase resilience.

22.14 Timeout Strategy

All inter-service communication shall enforce timeouts.

Purpose

Prevent resource exhaustion.

Benefits

Improved reliability.

Reduced cascading failures.

22.15 Recovery Architecture

Recovery procedures shall exist for all critical systems.

Recovery Targets

Applications

Databases

Caches

Event Systems

Observability Systems

Objective

Restore normal operations.

22.16 Backup Architecture

Backups shall protect business-critical information.

Protected Assets

Opportunity Data

Authority Data

Evidence Data

Ranking Records

Recommendation Records

Audit Records

Configuration Data

Objective

Prevent permanent data loss.

22.17 Backup Types

Full Backup

Complete system state.

Incremental Backup

Changes since previous backup.

Point-in-Time Recovery Data

Recovery to a specific moment.

Objective

Flexible restoration.

22.18 Recovery Point Objective (RPO)

RPO defines acceptable data loss.

Critical Data

Target RPO:

≤ 15 Minutes

Governance Data

Target RPO:

≤ 15 Minutes

Analytics Data

Target RPO:

≤ 24 Hours

22.19 Recovery Time Objective (RTO)

RTO defines acceptable downtime.

Critical Services

Target RTO:

≤ 1 Hour

Governance Services

Target RTO:

≤ 2 Hours

Analytics Services

Target RTO:

≤ 24 Hours

22.20 Disaster Recovery Architecture

Disaster Recovery addresses severe disruptions.

Disaster Examples

Regional Outage

Database Corruption

Infrastructure Loss

Major Security Incident

Operational Catastrophe

Objective

Restore business operations.

22.21 Disaster Recovery Lifecycle

Stage 1

Detection

Stage 2

Assessment

Stage 3

Containment

Stage 4

Recovery Activation

Stage 5

Service Restoration

Stage 6

Validation

Stage 7

Post-Incident Review

22.22 Data Integrity Protection

Recovery must preserve data integrity.

Protected Areas

Authority Records

Evidence Records

Opportunity Records

Ranking Records

Audit Records

Objective

Prevent inconsistent recovery states.

22.23 Governance Continuity

Governance capabilities shall survive recovery events.

Required Capabilities

Audit Access

Lineage Access

Historical State Access

Reproducibility Access

Objective

Preserve accountability.

22.24 Business Continuity Framework

Critical workflows shall remain prioritized.

Priority Workflows

Opportunity Discovery

Authority Verification

Evidence Validation

Ranking

Recommendation

Audit Operations

Objective

Maintain core platform value.

22.25 Multi-Region Recovery Strategy

Future versions may support geographic resilience.

Evolution Path

Single Region



Backup Region



Multi-Region Recovery



Active-Active Architecture

Objective

Improve resilience.

22.26 Reliability Testing Framework

Reliability assumptions shall be validated.

Test Categories

Recovery Testing

Backup Testing

Failover Testing

Load Testing

Chaos Testing

Objective

Verify resilience capabilities.

22.27 Chaos Engineering Framework

Future maturity stages may adopt controlled failure testing.

Examples

Service Failure Simulation

Database Failure Simulation

Network Failure Simulation

Cache Failure Simulation

Objective

Improve operational confidence.

22.28 Reliability Observability

Reliability controls shall be observable.

Monitored Areas

Availability

Recovery Success

Backup Success

Failover Success

Incident Frequency

Objective

Measure reliability effectiveness.

22.29 Incident Management Framework

Incidents shall follow structured processes.

Stages

Detection

↓

Classification

↓

Response

↓

Recovery

↓

Review

Objective

Reduce operational impact.

22.30 Reliability Performance Targets

Platform Availability

99.9%

Critical Service Availability

99.9%

Backup Success Rate

99.99%

Recovery Success Rate

99.9%

22.31 Failure Handling Principles

Reliability controls shall fail safely.

Backup Failure

Alert operators immediately.

Recovery Failure

Escalate incident response.

Monitoring Failure

Activate fallback monitoring.

Objective

Prevent hidden failures.

22.32 AI Usage Policy

AI may assist:

- Incident summarization
 - Recovery reporting
 - Reliability analysis
 - Post-incident review support
-

AI May Not

Trigger recovery actions.

Approve disaster declarations.

Modify backups.

Alter audit records.

Override governance decisions.

Reason

Recovery decisions require human accountability.

22.33 Reliability Governance Rules

The following rules are mandatory.

RRDR-01

Critical systems must support recovery.

RRDR-02

Backups must be tested.

RRDR-03

Recovery procedures must be documented.

RRDR-04

Recovery actions must be auditable.

RRDR-05

Governance data must remain protected.

RRDR-06

Reliability assumptions must be validated.

RRDR-07

Business continuity must be prioritized.

22.34 Architectural Constraints

The following constraints are non-negotiable.

AC-RRDR-01

Recovery must preserve data integrity.

AC-RRDR-02

Recovery must preserve auditability.

AC-RRDR-03

Recovery must preserve traceability.

AC-RRDR-04

Recovery must preserve reproducibility.

AC-RRDR-05

Reliability mechanisms must not alter business rules.

AC-RRDR-06

Disaster recovery must remain testable.

22.35 Reliability Maturity Model

The platform shall evolve through increasing resilience maturity.

Level 1

Basic Recovery

Level 2

Redundant Services

Level 3

Automated Recovery

Level 4

Multi-Region Recovery

Level 5

Near-Zero Downtime Operations

Objective

Provide a structured resilience roadmap.

22.36 Reliability, Resilience & Disaster Recovery Summary

The TODP Reliability Architecture provides the operational foundation required to maintain trusted platform operations despite failures.

The framework combines:

Failure Detection

↓

Fault Isolation

↓

Redundancy

↓

Recovery

↓

Backup

↓

Disaster Recovery

↓

Business Continuity

to ensure that critical discovery, authority, evidence, ranking, recommendation, and governance capabilities remain available and recoverable.

This architecture preserves the integrity of TODP's trust model even during major operational disruptions.

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SDD V1 — Section 23

Deployment Architecture, Release Management & CI/CD Framework

23.1 Purpose

This section defines the Deployment Architecture, Release Management, and Continuous Integration / Continuous Delivery (CI/CD) Framework used by TODP.

The purpose of this framework is to ensure that software changes are:

- Repeatable
- Traceable
- Auditable
- Reproducible
- Secure
- Reliable

from development through production deployment.

The framework governs the delivery lifecycle for:

- Frontend Applications
- Backend Services
- Databases
- Retrieval Components
- Observability Components
- Infrastructure Components

23.2 Deployment Philosophy

TODP follows an automation-first deployment philosophy.

Manual deployment activities shall be minimized.

Deployment processes shall be:

- Consistent
- Observable
- Versioned
- Recoverable

Core Principle

Build Once

↓

Validate

↓

Promote

↓

Deploy

The same artifact shall move through environments.

23.3 Objectives

DO-01 Repeatability

Deployments shall be repeatable.

DO-02 Reliability

Deployments shall minimize risk.

DO-03 Traceability

Deployments shall be traceable.

DO-04 Auditability

Deployment actions shall be auditable.

DO-05 Recoverability

Rollback mechanisms shall exist.

DO-06 Security

Deployment pipelines shall remain protected.

23.4 Deployment Architecture Overview

The deployment framework consists of seven major components.

Component 1

Source Control Framework

Component 2

Build Framework

Component 3

Artifact Management Framework

Component 4

CI/CD Pipeline Framework

Component 5

Environment Promotion Framework

Component 6

Release Management Framework

Component 7

Deployment Governance Framework

23.5 Source Control Architecture

All platform assets shall be version controlled.

Managed Assets

Application Code

Infrastructure Code

Database Migrations

API Specifications

Configuration Templates

Documentation

Objective

Provide a complete change history.

23.6 Branching Strategy

Development shall follow controlled branching.

Main Branch

Production-ready code.

Development Branch

Integration branch.

Feature Branches

Individual changes.

Hotfix Branches

Emergency corrections.

Objective

Support controlled releases.

23.7 Build Architecture

Build pipelines create deployable artifacts.

Build Outputs

Frontend Packages

Backend Containers

Infrastructure Packages

Database Migration Packages

Objective

Produce immutable artifacts.

23.8 Containerization Strategy

All deployable services shall be containerized.

Technology

Docker

Candidate Components

Frontend

Backend Services

Workers

Observability Components

Supporting Services

Objective

Environment consistency.

23.9 Artifact Management

Build outputs shall be stored in artifact repositories.

Stored Assets

Container Images

Build Packages

Release Artifacts

Version Metadata

Objectives

Traceability.

Repeatability.

Rollback support.

23.10 CI Framework

Continuous Integration validates changes.

Activities

Build Validation

Unit Testing

Integration Testing

Security Scanning

Quality Checks

Objective

Detect issues early.

23.11 CI Pipeline Lifecycle

Stage 1

Source Change

Stage 2

Build

Stage 3

Testing

Stage 4

Validation

Stage 5

Artifact Creation

Stage 6

Artifact Publication

23.12 CD Framework

Continuous Delivery automates deployment preparation.

Activities

Environment Promotion

Release Validation

Deployment Automation

Rollback Preparation

Objective

Accelerate delivery.

23.13 Environment Architecture

TODP shall support multiple deployment environments.

Development

Engineering experimentation.

Testing

Functional validation.

Staging

Production-like validation.

Production

Live operations.

Objective

Controlled promotion.

23.14 Environment Promotion Model

Artifacts move progressively.

Workflow

Development

↓

Testing

↓

Staging

↓

Production

Objective

Reduce deployment risk.

23.15 Configuration Management

Configuration shall be externalized.

Examples

Database Configuration

API Endpoints

Feature Flags

Environment Variables

Objective

Separate code from configuration.

23.16 Infrastructure as Code

Infrastructure shall be managed as code.

Managed Assets

Compute Resources

Networking

Databases

Monitoring

Security Controls

Objectives

Repeatability.

Auditability.

Consistency.

23.17 Database Deployment Framework

Database changes require controlled management.

Activities

Schema Changes

Migration Execution

Validation

Rollback Preparation

Objective

Protect data integrity.

23.18 Migration Strategy

Database changes shall be versioned.

Requirements

Version Control

Rollback Support

Auditability

Validation

Objective

Safe schema evolution.

23.19 Release Management Framework

Releases shall be governed.

Activities

Release Planning

Release Approval

Release Execution

Release Validation

Release Review

Objective

Controlled change management.

23.20 Release Types

Standard Release

Planned deployment.

Minor Release

Incremental functionality.

Major Release

Significant capability changes.

Emergency Release

Critical production fix.

23.21 Deployment Strategies

The platform shall support multiple deployment approaches.

Rolling Deployment

Gradual replacement.

Blue-Green Deployment

Parallel environments.

Canary Deployment

Controlled exposure.

Objective

Minimize deployment risk.

23.22 Rollback Architecture

Every deployment shall support rollback.

Rollback Triggers

Service Failure

Performance Degradation

Security Issue

Business Failure

Objective

Rapid recovery.

23.23 Deployment Validation Framework

Deployments must be validated.

Validation Types

Functional Validation

Performance Validation

Security Validation

Observability Validation

Governance Validation

Objective

Verify release quality.

23.24 Feature Flag Framework

Feature flags enable controlled activation.

Objectives

Reduce release risk.

Support experimentation.

Enable gradual rollout.

Requirement

Feature flag usage must be auditable.

23.25 Secrets Deployment Framework

Secrets shall never be embedded in artifacts.

Managed Secrets

Database Credentials

API Keys

Access Tokens

Encryption Keys

Objective

Protect sensitive information.

23.26 Deployment Observability

Deployment activities shall be observable.

Monitored Activities

Build Success

Deployment Success

Rollback Success

Pipeline Failures

Promotion Events

Objective

Provide operational visibility.

23.27 Deployment Audit Framework

Deployment activities shall be auditable.

Audit Scope

Build Events

Promotion Events

Deployment Events

Rollback Events

Approval Events

Stored Information

Actor

Action

Timestamp

Artifact Version

Environment

Correlation ID

23.28 Release Performance Targets

Build Success Rate

95%

Deployment Success Rate

99%

Rollback Readiness

100%

Pipeline Availability

99.9%

23.29 Failure Handling

Deployment systems shall fail safely.

Build Failure

Deployment blocked.

Validation Failure

Promotion blocked.

Deployment Failure

Rollback initiated.

Rollback Failure

Escalate incident response.

23.30 AI Usage Policy

AI may assist:

- Release summaries
 - Deployment reporting
 - Change impact analysis
 - Pipeline diagnostics
-

AI May Not

Approve releases.

Execute deployments.

Bypass validations.

Override governance controls.

Reason

Release accountability must remain human-governed.

23.31 Release Governance Rules

The following rules are mandatory.

REL-01

All deployments must be traceable.

REL-02

All releases must be auditable.

REL-03

Artifacts must be immutable.

REL-04

Production deployments must be validated.

REL-05

Rollback procedures must exist.

REL-06

Environment promotion must be controlled.

REL-07

Database changes must be versioned.

23.32 Architectural Constraints

The following constraints are non-negotiable.

AC-REL-01

Production deployments use approved artifacts only.

AC-REL-02

Deployments must not bypass governance controls.

AC-REL-03

Rollback capability must remain available.

AC-REL-04

Infrastructure changes must remain version controlled.

AC-REL-05

Release history must remain available.

AC-REL-06

Deployment automation must remain auditable.

23.33 Future Evolution

Future versions may support:

- Fully Automated Promotions
- Multi-Region Deployments
- Progressive Delivery
- Automated Release Intelligence
- Self-Service Deployment Portals

without altering governance principles.

23.34 Deployment Architecture Summary

The TODP Deployment Architecture provides a structured framework for delivering software safely, consistently, and transparently.

The framework combines:

Source Control

↓

Build Automation



Artifact Management



CI/CD Pipelines



Environment Promotion



Release Governance



Deployment Observability

to ensure that platform evolution remains controlled, auditable, reproducible, and aligned with enterprise-grade operational practices.

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SDD V1 — Section 24

Testing, Quality Assurance & Validation Architecture

24.1 Purpose

This section defines the Testing, Quality Assurance, and Validation (TQAV) Architecture used by TODP.

TODP is a trust-based platform whose outputs depend upon:

- Evidence
- Authorities
- Rankings
- Recommendations
- Auditability
- Traceability
- Reproducibility

Therefore, testing must validate not only technical correctness but also governance correctness.

The purpose of this architecture is to ensure that:

- Platform functionality is correct.
- Platform behavior is predictable.
- Platform outputs remain trustworthy.
- Platform changes do not introduce regressions.
- Platform governance principles remain preserved.

24.2 Quality Philosophy

Quality is not limited to software correctness.

Quality includes:

- Functional correctness
 - Operational correctness
 - Governance correctness
 - Decision correctness
 - Reproducibility correctness
-

Core Principle

Every important platform capability must be continuously validated.

24.3 Objectives

QA-01 Functional Correctness

Validate expected functionality.

QA-02 Reliability

Validate stable operation.

QA-03 Performance

Validate responsiveness.

QA-04 Security

Validate trust protections.

QA-05 Governance

Validate auditability and traceability.

QA-06 Reproducibility

Validate historical reconstruction.

QA-07 Regression Prevention

Detect unintended changes.

24.4 Testing Architecture Overview

The framework consists of ten major components.

Component 1

Unit Testing Framework

Component 2

Integration Testing Framework

Component 3

API Testing Framework

Component 4

Contract Testing Framework

Component 5

Retrieval Validation Framework

Component 6

Ranking Validation Framework

Component 7

Recommendation Validation Framework

Component 8

Performance Testing Framework

Component 9

Security Testing Framework

Component 10

Governance Validation Framework

24.5 Testing Pyramid

TODP adopts a layered testing strategy.

Layer 1

Unit Tests

Highest volume.

Layer 2

Integration Tests

Cross-component validation.

Layer 3

End-to-End Tests

Workflow validation.

Layer 4

Governance Validation

Trust verification.

Objective

Efficient quality assurance.

24.6 Unit Testing Framework

Unit tests validate isolated functionality.

Examples

Cost Calculations

Ranking Calculations

Validation Rules

Utility Functions

Transformation Logic

Objective

Verify individual components.

24.7 Unit Testing Rules

Every critical business rule shall have unit tests.

Examples

Money Cost Calculation

Time Cost Calculation

Distance Cost Calculation

Eligibility Evaluation

Authority Validation Rules

Evidence Validation Rules

Objective

Protect core logic.

24.8 Integration Testing Framework

Integration tests validate interactions between components.

Examples

Frontend ↔ API

API ↔ Database

Ranking ↔ Authority

Ranking ↔ Evidence

Recommendation ↔ Ranking

Objective

Validate service collaboration.

24.9 End-to-End Testing Framework

End-to-end tests validate complete workflows.

Example Workflow

Search

↓

Retrieval

↓

Authority Retrieval

↓

Evidence Retrieval

↓

Ranking

↓

Recommendation

↓

Response

Objective

Validate user experiences.

24.10 API Testing Framework

APIs shall be systematically validated.

Test Areas

Request Validation

Response Validation

Error Handling

Authentication

Authorization

Version Compatibility

Objective

Protect service contracts.

24.11 Contract Testing Framework

Service contracts shall remain stable.

Scope

Internal APIs

Public APIs

Event Contracts

Schema Contracts

Objective

Prevent breaking integrations.

24.12 Database Testing Framework

Database behavior shall be validated.

Areas

Schema Validation

Migration Validation

Constraint Validation

Performance Validation

Recovery Validation

Objective

Protect data integrity.

24.13 Retrieval Validation Framework

Retrieval quality is a core platform capability.

Validation Targets

BM25 Retrieval

Vector Retrieval

Hybrid Retrieval

Authority Retrieval

Evidence Retrieval

Objective

Validate discovery effectiveness.

24.14 Retrieval Quality Metrics

The platform shall track retrieval quality.

Metrics

Recall

Precision

Coverage

Latency

Result Relevance

Objective

Measure retrieval effectiveness.

24.15 Ranking Validation Framework

Ranking is a critical trust function.

Validation Targets

Cost Calculation

Ranking Order

Tie Resolution

Ranking Explanations

Ranking Reproducibility

Objective

Validate ranking correctness.

24.16 Ranking Validation Rules

The following must be continuously verified.

Money Cost Calculation

Correct.

Time Cost Calculation

Correct.

Distance Cost Calculation

Correct.

Total Cost Calculation

Correct.

Ranking Order

Deterministic.

24.17 Recommendation Validation Framework

Recommendations shall be validated.

Validation Areas

Recommendation Eligibility

Authority Inclusion

Evidence Inclusion

Explanation Quality

Traceability Availability

Objective

Validate recommendation trustworthiness.

24.18 Authority Validation Testing

Authority workflows require testing.

Areas

Verification Logic

Confidence Calculation

Authority Lifecycle

Authority Traceability

Objective

Validate authority trust.

24.19 Evidence Validation Testing

Evidence workflows require testing.

Areas

Evidence Collection

Evidence Validation

Evidence Lineage

Evidence Versioning

Evidence Quality Assessment

Objective

Validate Evidence First principles.

24.20 Opportunity Lifecycle Testing

Opportunity workflows shall be validated.

Lifecycle Stages

Registration

Validation

Eligibility

Ranking

Recommendation

Archival

Objective

Validate lifecycle integrity.

24.21 Opportunity Creation Testing

Creation workflows shall be validated.

Areas

Similarity Analysis

Pattern Analysis

Gap Detection

Concept Generation Support

Creation Traceability

Objective

Validate innovation workflows.

24.22 Security Testing Framework

Security controls require continuous testing.

Areas

Authentication

Authorization

Input Validation

Encryption

Secrets Protection

Audit Protection

Objective

Protect trust boundaries.

24.23 Performance Testing Framework

Performance requirements shall be validated.

Test Types

Load Testing

Stress Testing

Spike Testing

Soak Testing

Capacity Testing

Objective

Validate scalability assumptions.

24.24 Reliability Testing Framework

Reliability capabilities shall be tested.

Areas

Recovery Testing

Failover Testing

Backup Testing

Disaster Recovery Testing

Chaos Testing

Objective

Validate resilience.

24.25 Observability Testing

Observability systems require validation.

Areas

Metrics Generation

Trace Generation

Log Generation

Alert Generation

Dashboard Accuracy

Objective

Ensure operational visibility.

24.26 Auditability Testing

Audit systems shall be validated.

Areas

Audit Creation

Audit Integrity

Audit Retrieval

Audit Search

Audit Preservation

Objective

Validate accountability.

24.27 Traceability Testing

Traceability shall be validated.

Areas

Correlation IDs

Lineage Generation

Decision Traces

Workflow Traces

Historical Reconstruction

Objective

Validate transparency.

24.28 Reproducibility Testing

Historical outcomes must be reproducible.

Validation Targets

Ranking Outcomes

Recommendation Outcomes

Authority Decisions

Evidence Decisions

Objective

Validate governance integrity.

24.29 Regression Testing Framework

Regression testing shall protect existing capabilities.

Trigger Events

Code Changes

Configuration Changes

Infrastructure Changes

Dependency Changes

Objective

Prevent unintended regressions.

24.30 Test Data Management

Testing requires controlled datasets.

Data Categories

Synthetic Data

Reference Data

Validation Data

Historical Data

Benchmark Data

Objective

Consistent testing.

24.31 Quality Gates Framework

Changes must pass quality gates.

Example Gates

Unit Test Success

Integration Test Success

Security Validation

Performance Validation

Governance Validation

Objective

Prevent low-quality releases.

24.32 Coverage Framework

Coverage shall be measured continuously.

Coverage Types

Code Coverage

API Coverage

Workflow Coverage

Governance Coverage

Test Coverage

Objective

Identify validation gaps.

24.33 Testing Automation Framework

Testing shall be highly automated.

Automated Areas

Unit Testing

Integration Testing

Contract Testing

Regression Testing

Performance Testing

Objective

Increase validation consistency.

24.34 Validation Reporting Framework

Testing outcomes shall be reported.

Report Types

Test Reports

Quality Reports

Regression Reports

Performance Reports

Governance Reports

Objective

Support decision-making.

24.35 Failure Handling

Testing systems shall fail safely.

Test Failure

Deployment blocked.

Quality Gate Failure

Promotion blocked.

Regression Failure

Release blocked.

Objective

Protect production quality.

24.36 AI Usage Policy

AI may assist:

- Test generation suggestions
 - Test result summarization
 - Coverage analysis
 - Quality reporting
-

AI May Not

Approve releases.

Override failures.

Modify test outcomes.

Suppress quality gates.

Reason

Quality decisions must remain deterministic.

24.37 Quality Governance Rules

The following rules are mandatory.

QA-01

Critical business rules must be tested.

QA-02

Critical workflows must be tested.

QA-03

Governance workflows must be tested.

QA-04

Regression testing must be automated.

QA-05

Quality gates must be enforced.

QA-06

Testing outcomes must be auditable.

QA-07

Reproducibility must be validated.

24.38 Architectural Constraints

The following constraints are non-negotiable.

AC-QA-01

No production release without validation.

AC-QA-02

No release may bypass quality gates.

AC-QA-03

Testing must preserve reproducibility.

AC-QA-04

Governance workflows must remain validated.

AC-QA-05

Quality reporting must remain available.

AC-QA-06

Testing outcomes must remain traceable.

24.39 Testing Maturity Model

The platform shall evolve through increasing validation maturity.

Level 1

Basic Automated Testing

Level 2

Comprehensive Integration Testing

Level 3

Governance Validation

Level 4

Automated Reproducibility Validation

Level 5

Continuous Quality Intelligence

Objective

Provide a structured quality roadmap.

24.40 Testing, Quality Assurance & Validation Summary

The TODP Testing Architecture provides a comprehensive validation framework covering:

Unit Testing

↓

Integration Testing

↓

Contract Testing



Retrieval Validation



Ranking Validation



Recommendation Validation



Governance Validation



Reproducibility Validation

This framework ensures that platform functionality, trust mechanisms, governance controls, and operational characteristics remain correct, auditable, traceable, and reproducible throughout the platform lifecycle.

Testing therefore becomes a core trust-preservation capability rather than merely a software verification activity.

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SDD V1 — Section 25

Operational Governance & Platform Administration Architecture

25.1 Purpose

This section defines the Operational Governance and Platform Administration Architecture used by TODP.

Building a platform is insufficient without mechanisms to govern, administer, monitor, and control it.

TODP operates within a trust-based environment where platform actions can affect:

- Opportunity Discovery
- Authority Verification
- Evidence Validation
- Ranking Outcomes
- Recommendations
- Opportunity Creation
- Audit Records

Therefore operational governance is a first-class architectural capability.

The purpose of this architecture is to ensure that platform operations remain:

- Controlled
- Transparent
- Auditable
- Traceable
- Reproducible
- Accountable

throughout the platform lifecycle.

25.2 Governance Philosophy

TODP governance is based upon accountability.

Every important operational action must answer:

- Who performed the action?
 - What was changed?
 - Why was it changed?
 - When was it changed?
 - What impact did it have?
-

Core Principle

No operational action shall be invisible.

25.3 Objectives

OG-01 Accountability

Maintain ownership of operational actions.

OG-02 Transparency

Provide visibility into platform administration.

OG-03 Auditability

Preserve operational history.

OG-04 Traceability

Maintain action lineage.

OG-05 Reproducibility

Support reconstruction of administrative decisions.

OG-06 Risk Reduction

Prevent unauthorized or unsafe actions.

25.4 Governance Architecture Overview

The framework consists of eight major components.

Component 1

Platform Administration Framework

Component 2

Configuration Governance Framework

Component 3

Operational Change Framework

Component 4

Access Governance Framework

Component 5

Policy Governance Framework

Component 6

Administrative Audit Framework

Component 7

Governance Reporting Framework

Component 8

Operational Review Framework

25.5 Platform Administration Framework

Platform administration provides controlled operational management.

Responsibilities

System Monitoring

Configuration Management

Access Management

Operational Reviews

Incident Coordination

Governance Enforcement

Objective

Maintain platform stability and trust.

25.6 Administrative Roles

Operational responsibilities shall be role-based.

Platform Administrator

Manages platform operations.

Governance Administrator

Manages governance controls.

Security Administrator

Manages security controls.

Operations Administrator

Manages operational health.

Auditor

Reviews platform activity.

Objective

Separate responsibilities.

25.7 Administrative Responsibility Model

Responsibilities shall be clearly assigned.

Areas

Security

Infrastructure

Data

Governance

Operations

Compliance

Objective

Eliminate ownership ambiguity.

25.8 Configuration Governance Framework

Configuration changes shall be governed.

Governed Areas

Application Configuration

Infrastructure Configuration

Security Configuration

Observability Configuration

Operational Policies

Objective

Prevent uncontrolled changes.

25.9 Configuration Lifecycle

Configuration changes follow a controlled lifecycle.

Stage 1

Proposal

Stage 2

Review

Stage 3

Approval

Stage 4

Implementation

Stage 5

Validation

Stage 6

Audit Recording

25.10 Configuration Versioning

Configuration history shall be preserved.

Requirements

Version Tracking

Change History

Rollback Capability

Approval Records

Objective

Support reproducibility.

25.11 Operational Change Framework

Operational changes shall be managed systematically.

Examples

Configuration Updates

Infrastructure Changes

Policy Updates

Operational Process Changes

Objective

Reduce operational risk.

25.12 Change Classification Model

Changes shall be categorized.

Standard Change

Low-risk change.

Significant Change

Moderate impact change.

Critical Change

High-impact change.

Emergency Change

Urgent corrective action.

25.13 Change Approval Framework

Certain changes require approval.

Approval Inputs

Risk Assessment

Impact Analysis

Rollback Strategy

Validation Plan

Objective

Improve decision quality.

25.14 Access Governance Framework

Administrative access requires governance.

Objectives

Least Privilege

Accountability

Traceability

Auditability

Scope

Administrative Systems

Operational Systems

Governance Systems

25.15 Privileged Access Management

Privileged access shall be controlled.

Examples

Database Administration

Infrastructure Administration

Security Administration

Audit Administration

Objective

Protect critical systems.

25.16 Policy Governance Framework

Operational policies require governance.

Policy Areas

Security Policies

Operational Policies

Recovery Policies

Deployment Policies

Governance Policies

Objective

Maintain consistency.

25.17 Policy Lifecycle

Policies shall be managed systematically.

Stages

Creation

↓

Review

↓

Approval

↓

Publication

↓

Monitoring

↓

Revision

Objective

Maintain policy quality.

25.18 Governance Decision Framework

Governance decisions shall be documented.

Decision Components

Decision

Reasoning

Alternatives Considered

Approver

Timestamp

Objective

Preserve decision history.

25.19 Administrative Audit Architecture

Administrative activities shall be auditable.

Audit Scope

Configuration Changes

Access Changes

Policy Changes

Operational Changes

Governance Actions

Objective

Support accountability.

25.20 Administrative Traceability

Administrative actions shall remain traceable.

Trace Components

Actor

Action

Target

Timestamp

Reason

Approval References

Objective

Support investigations.

25.21 Governance Reporting Framework

Governance reporting provides operational visibility.

Report Categories

Administrative Reports

Security Reports

Audit Reports

Operational Reports

Governance Reports

Objective

Support oversight.

25.22 Operational Review Framework

Regular reviews shall be performed.

Review Areas

Security

Performance

Reliability

Governance

Compliance

Operational Effectiveness

Objective

Continuous improvement.

25.23 Governance Metrics

Governance effectiveness shall be measurable.

Example Metrics

Administrative Changes

Policy Compliance

Audit Coverage

Access Violations

Review Completion Rates

Objective

Assess governance quality.

25.24 Administrative Dashboard Framework

Administrators require operational visibility.

Dashboard Areas

Platform Health

Security Status

Governance Status

Operational Activity

Configuration Status

Objective

Support decision-making.

25.25 Exception Management Framework

Exceptional situations require controlled handling.

Examples

Emergency Access

Emergency Changes

Operational Overrides

Disaster Recovery Actions

Objective

Balance agility and governance.

25.26 Exception Approval Framework

Exceptions shall be documented.

Required Information

Reason

Scope

Duration

Approver

Impact Assessment

Objective

Maintain accountability.

25.27 Governance Lifecycle

Governance activities shall follow continuous cycles.

Cycle

Monitor

↓

Review

↓

Improve

↓

Validate

↓

Audit

Objective

Support operational maturity.

25.28 Administrative Observability

Administrative activities shall be observable.

Monitored Areas

Administrative Access

Configuration Changes

Policy Changes

Governance Reviews

Operational Events

Objective

Increase transparency.

25.29 Governance Search Architecture

Governance information shall be searchable.

Search Dimensions

Actor

Action

Entity

Timestamp

Policy

Approval

Objective

Accelerate investigations.

25.30 Governance Performance Targets

Administrative Audit Recording

P95 < 50 ms

Governance Report Generation

P95 < 5 seconds

Access Review Processing

P95 < 1 hour

Policy Retrieval

P95 < 100 ms

25.31 Failure Handling

Governance controls shall fail safely.

Approval Failure

Change blocked.

Audit Failure

Action flagged.

Access Governance Failure

Access denied.

Policy Validation Failure

Policy not activated.

Objective

Protect governance integrity.

25.32 AI Usage Policy

AI may assist:

- Governance reporting
 - Policy summarization
 - Administrative analysis
 - Operational recommendations
-

AI May Not

Approve governance decisions.

Grant privileged access.

Modify governance records.

Override policies.

Bypass approvals.

Reason

Governance accountability must remain human-controlled.

25.33 Governance Rules

The following rules are mandatory.

GOV-01

Administrative actions must be auditable.

GOV-02

Administrative actions must be traceable.

GOV-03

Configuration changes must be versioned.

GOV-04

Policy changes must be governed.

GOV-05

Privileged access must be controlled.

GOV-06

Governance decisions must be documented.

GOV-07

Operational reviews must be performed.

25.34 Architectural Constraints

The following constraints are non-negotiable.

AC-GOV-01

Administrative actions must remain visible.

AC-GOV-02

Governance controls must not be bypassed.

AC-GOV-03

Operational changes must remain traceable.

AC-GOV-04

Policy history must remain available.

AC-GOV-05

Governance records must remain auditable.

AC-GOV-06

Administrative accountability must remain preserved.

25.35 Governance Maturity Model

The platform shall evolve through governance maturity stages.

Level 1

Basic Administration

Level 2

Controlled Governance

Level 3

Comprehensive Auditability

Level 4

Operational Intelligence

Level 5

Continuous Governance Optimization

Objective

Provide a structured governance roadmap.

25.36 Operational Governance & Platform Administration Summary

The Operational Governance Architecture provides the administrative and governance foundation required to operate TODP responsibly.

The framework combines:

Platform Administration

↓

Configuration Governance



Access Governance



Policy Governance



Administrative Auditing



Operational Reviews



Governance Reporting

to ensure that platform operations remain transparent, accountable, traceable, auditable, and reproducible.

This architecture ensures that governance itself is governed and becomes an integral part of the TODP trust model.

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- □ Section 26 — Compliance, Risk Management & Trust Assurance Architecture

SDD V1 — Section 26

Compliance, Risk Management & Trust Assurance Architecture

26.1 Purpose

This section defines the Compliance, Risk Management, and Trust Assurance (CRTA) Architecture used by TODP.

TODP is fundamentally a trust platform.

Users rely upon the platform to provide:

- Evidence-Based Discovery
- Authority-Based Validation
- Transparent Rankings
- Explainable Recommendations
- Reproducible Decisions

Trust cannot be assumed.

Trust must be continuously managed, monitored, validated, and demonstrated.

The purpose of this architecture is to ensure that TODP operates within defined governance boundaries while proactively managing risks that may impact:

- Users
- Opportunities
- Authorities
- Evidence
- Rankings
- Recommendations
- Platform Operations

26.2 Trust Philosophy

Trust is not a feature.

Trust is an operational outcome produced by:

- Transparency
- Accountability
- Traceability
- Auditability
- Reproducibility

Core Principle

Every trust claim must be supported by evidence.

Examples

Claim:

"This recommendation is trustworthy."

Support:

Evidence Lineage

Authority Verification

Ranking Traceability

Audit Records

26.3 Objectives

CRTA-01 Compliance

Support adherence to governance requirements.

CRTA-02 Risk Visibility

Identify and understand risks.

CRTA-03 Risk Reduction

Reduce operational exposure.

CRTA-04 Trust Assurance

Validate trustworthiness continuously.

CRTA-05 Accountability

Maintain responsibility ownership.

CRTA-06 Governance Maturity

Support continuous improvement.

26.4 Architecture Overview

The framework consists of eight major components.

Component 1

Compliance Framework

Component 2

Risk Management Framework

Component 3

Trust Assurance Framework

Component 4

Control Validation Framework

Component 5

Governance Risk Framework

Component 6

Policy Compliance Framework

Component 7

Assurance Reporting Framework

Component 8

Continuous Improvement Framework

26.5 Compliance Architecture

Compliance ensures adherence to defined obligations.

Scope

Policies

Standards

Operational Controls

Security Controls

Governance Controls

Objective

Support accountable operations.

26.6 Compliance Domains

TODP compliance activities shall be categorized.

Operational Compliance

Operational procedures.

Security Compliance

Security controls.

Governance Compliance

Governance requirements.

Data Compliance

Data management requirements.

Trust Compliance

Trust principles.

26.7 Compliance Lifecycle

Compliance activities shall follow a structured lifecycle.

Stage 1

Requirement Identification

Stage 2

Control Definition

Stage 3

Implementation

Stage 4

Validation

Stage 5

Monitoring

Stage 6

Improvement

26.8 Risk Management Architecture

Risk management identifies and manages uncertainty.

Objectives

Risk Identification

Risk Assessment

Risk Mitigation

Risk Monitoring

Risk Reporting

Goal

Reduce undesirable outcomes.

26.9 Risk Categories

TODP risks shall be classified.

Operational Risk

Process failures.

Security Risk

Security incidents.

Data Risk

Data integrity issues.

Governance Risk

Governance failures.

Reliability Risk

Availability failures.

Trust Risk

Loss of user confidence.

26.10 Risk Identification Framework

Risks shall be identified continuously.

Sources

Audits

Incidents

Reviews

Monitoring

Operational Feedback

Governance Reviews

Objective

Early risk detection.

26.11 Risk Assessment Framework

Identified risks shall be evaluated.

Assessment Inputs

Probability

Impact

Detectability

Affected Assets

Mitigation Complexity

Objective

Prioritize response efforts.

26.12 Risk Treatment Framework

Risks shall receive defined responses.

Risk Responses

Accept

Mitigate

Transfer

Avoid

Objective

Manage exposure.

26.13 Risk Register Architecture

Risks shall be documented.

Required Information

Risk ID

Description

Owner

Severity

Status

Mitigation Plan

Review Date

Objective

Maintain risk visibility.

26.14 Trust Assurance Framework

Trust assurance validates platform trustworthiness.

Objective

Demonstrate that trust principles are functioning.

Scope

Evidence

Authorities

Rankings

Recommendations

Governance

Operations

26.15 Trust Assurance Domains

Trust shall be validated across multiple domains.

Evidence Trust

Evidence quality and validity.

Authority Trust

Authority legitimacy and verification.

Ranking Trust

Ranking transparency and reproducibility.

Recommendation Trust

Recommendation explainability and traceability.

Governance Trust

Governance effectiveness.

26.16 Trust Indicators

Trust shall be measurable.

Example Indicators

Evidence Validation Rate

Authority Verification Rate

Audit Coverage

Lineage Coverage

Reproducibility Success Rate

Objective

Quantify trust health.

26.17 Control Framework

Controls reduce risk and improve trust.

Control Types

Preventive Controls

Detective Controls

Corrective Controls

Governance Controls

Objective

Protect platform integrity.

26.18 Control Validation Framework

Controls shall be tested.

Validation Areas

Security Controls

Governance Controls

Operational Controls

Recovery Controls

Compliance Controls

Objective

Verify effectiveness.

26.19 Governance Risk Framework

Governance failures represent platform risk.

Examples

Approval Failures

Audit Gaps

Lineage Gaps

Policy Violations

Control Failures

Objective

Protect accountability.

26.20 Policy Compliance Framework

Policies shall be monitored.

Monitored Areas

Security Policies

Operational Policies

Governance Policies

Recovery Policies

Release Policies

Objective

Ensure policy adherence.

26.21 Compliance Monitoring

Compliance shall be continuously monitored.

Inputs

Audits

Metrics

Reviews

Control Validation

Operational Data

Objective

Detect deviations.

26.22 Assurance Reviews

Trust assurance shall include periodic reviews.

Review Areas

Evidence Quality

Authority Integrity

Ranking Integrity

Recommendation Integrity

Governance Integrity

Objective

Maintain trustworthiness.

26.23 Exception Governance

Exceptions require governance.

Examples

Policy Exceptions

Access Exceptions

Operational Exceptions

Emergency Changes

Objective

Control deviations.

26.24 Exception Lifecycle

Stage 1

Request

Stage 2

Assessment

Stage 3

Approval

Stage 4

Monitoring

Stage 5

Closure

Objective

Maintain accountability.

26.25 Assurance Reporting Framework

Trust and compliance information shall be reported.

Report Types

Risk Reports

Compliance Reports

Trust Reports

Control Reports

Objective

Support oversight.

26.26 Compliance Metrics

Compliance effectiveness shall be measurable.

Examples

Control Coverage

Policy Adherence

Review Completion Rate

Exception Volume

Compliance Violations

Objective

Measure effectiveness.

26.27 Risk Metrics

Risk management effectiveness shall be measurable.

Examples

Open Risks

Mitigated Risks

Critical Risks

Risk Review Completion

Risk Escalations

Objective

Monitor exposure.

26.28 Trust Metrics

Trust assurance effectiveness shall be measurable.

Examples

Recommendation Traceability Rate

Ranking Reproducibility Rate

Evidence Coverage

Authority Coverage

Audit Completeness

Objective

Monitor trust health.

26.29 Assurance Dashboard Framework

Trust and compliance visibility shall be available.

Dashboard Areas

Compliance Status

Risk Status

Trust Status

Control Health

Governance Health

Objective

Support informed decisions.

26.30 Accountability Framework

Every significant risk and control shall have an owner.

Ownership Areas

Risks

Controls

Policies

Reviews

Exceptions

Governance Activities

Objective

Prevent ownership gaps.

26.31 Continuous Improvement Framework

Trust assurance shall evolve continuously.

Cycle

Measure

↓

Review

↓

Improve

↓

Validate

↓

Monitor

Objective

Increase governance maturity.

26.32 Failure Handling

Trust and compliance systems shall fail safely.

Compliance Failure

Issue recorded.

Review initiated.

Risk Assessment Failure

Risk escalated.

Control Validation Failure

Control flagged.

Trust Indicator Failure

Investigation initiated.

Objective

Protect platform integrity.

26.33 AI Usage Policy

AI may assist:

- Risk analysis
 - Compliance reporting
 - Trust reporting
 - Control summarization
-

AI May Not

Approve exceptions.

Accept risks.

Override controls.

Modify compliance records.

Change governance outcomes.

Reason

Accountability must remain human-governed.

26.34 Compliance & Trust Governance Rules

The following rules are mandatory.

CRTA-01

Trust claims must be verifiable.

CRTA-02

Risks must have owners.

CRTA-03

Controls must be validated.

CRTA-04

Compliance activities must be auditable.

CRTA-05

Trust indicators must be measurable.

CRTA-06

Exceptions must be governed.

CRTA-07

Continuous improvement must occur.

26.35 Architectural Constraints

The following constraints are non-negotiable.

AC-CRTA-01

Trust assertions must be evidence-backed.

AC-CRTA-02

Risk decisions must remain traceable.

AC-CRTA-03

Compliance records must remain auditable.

AC-CRTA-04

Control effectiveness must be measurable.

AC-CRTA-05

Governance accountability must remain preserved.

AC-CRTA-06

Trust assurance must remain independent of AI-generated decisions.

26.36 Trust Maturity Model

The platform shall evolve through trust maturity stages.

Level 1

Basic Compliance

Level 2

Risk Visibility

Level 3

Control Governance

Level 4

Trust Assurance

Level 5

Continuous Trust Intelligence

Objective

Provide a structured trust roadmap.

26.37 Compliance, Risk Management & Trust Assurance Summary

The Compliance, Risk Management, and Trust Assurance Architecture provides the governance layer responsible for preserving platform trustworthiness.

The framework combines:

Compliance

↓

Risk Management

↓

Control Validation

↓

Trust Assurance

↓

Governance Oversight

↓

Continuous Improvement

to ensure that TODP remains aligned with its foundational principles:

- Evidence First
- Authority Before Recommendation
- Ranking Before Recommendation
- No Hidden Ranking Logic
- AI Is Not An Authority

This architecture transforms trust from an abstract concept into a measurable, governable, and continuously validated operational capability.

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SDD V1 — Section 27

Data Lifecycle, Retention & Information Governance Architecture

27.1 Purpose

This section defines the Data Lifecycle, Retention, and Information Governance (DLRIG) Architecture used by TODP.

Information is one of the most important assets within TODP.

The platform manages:

- Opportunities
- Authorities
- Evidence
- Rankings
- Recommendations
- User Context
- Audit Records
- Trace Records
- Governance Records

These information assets must be governed throughout their lifecycle to ensure:

- Integrity
- Traceability
- Auditability
- Reproducibility
- Recoverability

The purpose of this architecture is to define how information is created, stored, used, retained, archived, recovered, and ultimately removed.

27.2 Information Governance Philosophy

Information is not merely stored.

Information is governed.

Every information asset must have:

- Ownership
- Purpose
- Lifecycle
- Retention Rules
- Governance Controls

Core Principle

No information shall exist without governance.

27.3 Objectives

DLG-01 Lifecycle Management

Manage information throughout its lifecycle.

DLG-02 Retention Governance

Control retention durations.

DLG-03 Traceability

Maintain information lineage.

DLG-04 Auditability

Preserve information history.

DLG-05 Reproducibility

Support historical reconstruction.

DLG-06 Risk Reduction

Reduce information management risks.

27.4 Architecture Overview

The framework consists of eight major components.

Component 1

Information Classification Framework

Component 2

Lifecycle Management Framework

Component 3

Retention Management Framework

Component 4

Archival Framework

Component 5

Recovery Framework

Component 6

Deletion Governance Framework

Component 7

Information Ownership Framework

Component 8

Information Governance Framework

27.5 Information Asset Categories

Information assets shall be categorized.

Opportunity Information

Opportunity records and metadata.

Authority Information

Authority records and verification data.

Evidence Information

Evidence records and lineage.

Ranking Information

Ranking calculations and explanations.

Recommendation Information

Recommendation outputs and explanations.

User Context Information

User constraints and preferences.

Audit Information

Audit records and events.

Governance Information

Policies, reviews, and governance decisions.

27.6 Information Ownership Framework

Every information asset shall have an owner.

Ownership Responsibilities

Accuracy

Governance

Retention

Review

Lifecycle Decisions

Objective

Prevent ownership gaps.

27.7 Information Classification Framework

Information shall be classified.

Public

Openly accessible information.

Internal

Operational information.

Restricted

Controlled-access information.

Sensitive

Protected information requiring enhanced controls.

Objective

Apply appropriate governance.

27.8 Information Lifecycle Architecture

Information follows a structured lifecycle.

Stage 1

Creation

Stage 2

Validation

Stage 3

Usage

Stage 4

Retention

Stage 5

Archival

Stage 6

Recovery

Stage 7

Deletion

27.9 Information Creation Framework

Information creation shall be governed.

Sources

User Actions

System Processes

Authority Processes

Evidence Processes

Administrative Actions

Objective

Establish information accountability.

27.10 Information Validation Framework

Information shall be validated before use.

Validation Areas

Completeness

Consistency

Integrity

Ownership

Classification

Objective

Improve information quality.

27.11 Information Usage Framework

Information usage shall be traceable.

Usage Areas

Retrieval

Ranking

Recommendation

Governance

Auditing

Reporting

Objective

Preserve transparency.

27.12 Information Lineage Framework

Information lineage shall remain available.

Example

Evidence

↓

Authority

↓

Opportunity

↓

Ranking

↓

Recommendation

Objective

Support traceability.

27.13 Retention Architecture

Information shall be retained according to governance requirements.

Objectives

Support audits.

Support investigations.

Support reproducibility.

Support governance.

27.14 Retention Categories

Retention periods shall be governed by information type.

Operational Data

Operational retention rules.

Governance Data

Extended retention rules.

Audit Data

Long-term retention rules.

Historical Data

Archival retention rules.

27.15 Retention Policy Framework

Retention policies shall be documented.

Policy Components

Asset Type

Retention Duration

Review Requirements

Archival Requirements

Deletion Rules

Objective

Consistent governance.

27.16 Archival Architecture

Archived information remains valuable.

Objectives

Historical Preservation

Audit Support

Reproducibility Support

Governance Support

Requirement

Archived information remains recoverable.

27.17 Archival Lifecycle

Stage 1

Eligibility Review

Stage 2

Archival Preparation

Stage 3

Archive Transfer

Stage 4

Validation

Stage 5

Governance Recording

27.18 Recovery Framework

Archived information shall be recoverable.

Recovery Triggers

Investigations

Audits

Governance Reviews

Historical Reconstruction

Operational Needs

Objective

Preserve accessibility.

27.19 Recovery Validation

Recovered information shall be validated.

Validation Areas

Integrity

Completeness

Traceability

Consistency

Objective

Protect information quality.

27.20 Information Expiration Framework

Certain information becomes inactive.

Examples

Expired Opportunities

Inactive Authorities

Superseded Evidence

Obsolete Recommendations

Objective

Manage information relevance.

27.21 Deletion Governance Framework

Deletion shall be controlled.

Objectives

Prevent accidental loss.

Preserve accountability.

Protect governance requirements.

Principle

Deletion is governed, not assumed.

27.22 Deletion Lifecycle

Stage 1

Deletion Request

Stage 2

Eligibility Review

Stage 3

Approval

Stage 4

Execution

Stage 5

Audit Recording

Objective

Protect information assets.

27.23 Information Preservation Framework

Certain information shall receive enhanced protection.

Examples

Audit Records

Lineage Records

Governance Decisions

Historical Rankings

Historical Recommendations

Objective

Preserve trust foundations.

27.24 Information Recovery Objectives

Recovery requirements shall be measurable.

Objectives

Availability

Integrity

Traceability

Auditability

Reproducibility

Goal

Reliable historical access.

27.25 Information Governance Reviews

Information assets shall be reviewed periodically.

Review Areas

Ownership

Classification

Retention

Archival Status

Deletion Eligibility

Objective

Maintain governance quality.

27.26 Information Search Architecture

Governed information shall be searchable.

Search Dimensions

Entity

Owner

Classification

Timestamp

Retention Status

Lifecycle Status

Objective

Support operational efficiency.

27.27 Information Metrics

Information governance effectiveness shall be measurable.

Example Metrics

Retention Compliance

Archival Success

Recovery Success

Ownership Coverage

Classification Coverage

Objective

Monitor governance quality.

27.28 Information Dashboard Framework

Information governance visibility shall be available.

Dashboard Areas

Lifecycle Status

Retention Status

Archival Status

Recovery Status

Governance Status

Objective

Support oversight.

27.29 Data Integrity Framework

Information integrity shall be protected throughout the lifecycle.

Integrity Controls

Validation

Versioning

Audit Trails

Lineage

Recovery Verification

Objective

Prevent corruption.

27.30 Reproducibility Support Framework

Information governance shall support reproducibility.

Required Assets

Historical Opportunities

Historical Authorities

Historical Evidence

Historical Rankings

Historical Recommendations

Objective

Enable historical reconstruction.

27.31 Failure Handling

Information governance controls shall fail safely.

Archival Failure

Archival halted.

Recovery Failure

Issue escalated.

Deletion Failure

Deletion halted.

Classification Failure

Asset flagged for review.

Objective

Protect information assets.

27.32 AI Usage Policy

AI may assist:

- Information classification suggestions
 - Governance reporting
 - Lifecycle analysis
 - Retention analysis
-

AI May Not

Delete information.

Override retention policies.

Modify governance records.

Approve lifecycle transitions.

Reason

Information governance requires accountability.

27.33 Information Governance Rules

The following rules are mandatory.

DLG-01

Every information asset must have an owner.

DLG-02

Every information asset must have a lifecycle.

DLG-03

Retention policies must be defined.

DLG-04

Archival activities must be auditable.

DLG-05

Deletion activities must be governed.

DLG-06

Lineage information must remain available.

DLG-07

Historical reconstruction must remain possible.

27.34 Architectural Constraints

The following constraints are non-negotiable.

AC-DLG-01

Information governance must preserve traceability.

AC-DLG-02

Information governance must preserve auditability.

AC-DLG-03

Information governance must preserve reproducibility.

AC-DLG-04

Retention policies must remain enforceable.

AC-DLG-05

Deletion activities must remain governed.

AC-DLG-06

Historical information must remain recoverable where required.

27.35 Information Governance Maturity Model

The platform shall evolve through information governance maturity stages.

Level 1

Basic Retention

Level 2

Lifecycle Governance

Level 3

Archival Governance

Level 4

Comprehensive Information Governance

Level 5

Continuous Information Intelligence

Objective

Provide a structured governance roadmap.

27.36 Data Lifecycle, Retention & Information Governance Summary

The Data Lifecycle, Retention, and Information Governance Architecture provides a comprehensive framework for managing information throughout its existence.

The framework combines:

Creation

↓

Validation

↓

Usage

↓

Retention

↓

Archival

↓

Recovery

↓

Deletion

while preserving:

- Integrity
- Traceability
- Auditability
- Reproducibility
- Accountability

This architecture ensures that information remains a governed asset and supports TODP's trust, governance, and operational objectives throughout the platform lifecycle.

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- ☐ Section 28 — Future Evolution, Extensibility & Architectural Roadmap

SDD V1 — Section 28

Future Evolution, Extensibility & Architectural Roadmap

28.1 Purpose

This section defines the Future Evolution, Extensibility, and Architectural Roadmap Framework used by TODP.

TODP is designed as a long-term opportunity infrastructure platform rather than a fixed software product.

The platform must be capable of evolving while preserving its foundational principles:

- Evidence First
- Authority Before Recommendation
- Ranking Before Recommendation
- Traceability
- Auditability
- Reproducibility
- No Hidden Ranking Logic
- AI Is Not An Authority
- Human Creates Novelty

The purpose of this architecture is to ensure that future growth can occur without compromising platform trustworthiness.

28.2 Evolution Philosophy

Architectures fail when growth requires redesign.

Architectures succeed when growth requires extension.

Core Principle

Extend.

Do not rewrite.

Goal

Enable continuous capability growth while preserving architectural stability.

28.3 Evolution Objectives

EVO-01 Extensibility

Enable addition of new capabilities.

EVO-02 Stability

Protect foundational architecture.

EVO-03 Adaptability

Support changing requirements.

EVO-04 Longevity

Support multi-year evolution.

EVO-05 Governance Preservation

Maintain trust principles.

EVO-06 Technology Independence

Avoid unnecessary technology lock-in.

28.4 Evolution Architecture Overview

The framework consists of eight major components.

Component 1

Business Capability Evolution

Component 2

Technology Evolution

Component 3

Data Evolution

Component 4

Service Evolution

Component 5

Governance Evolution

Component 6

Platform Expansion Framework

Component 7

Innovation Framework

Component 8

Architectural Review Framework

28.5 Architectural Stability Zones

Certain platform areas are expected to remain highly stable.

Stable Zone 1

Core Principles

Stable Zone 2

Ranking Philosophy

Stable Zone 3

Evidence Framework

Stable Zone 4

Authority Framework

Stable Zone 5

Auditability Framework

Stable Zone 6

Traceability Framework

Objective

Protect platform identity.

28.6 Architectural Flexibility Zones

Certain platform areas are expected to evolve frequently.

User Experience

Discovery Interfaces

Recommendation Presentation

Search Experience

Reporting Capabilities

Objective

Enable innovation.

28.7 Business Capability Evolution

TODP shall support future business capabilities.

Examples

New Opportunity Types

New Discovery Modes

New Authority Models

New Evidence Models

New Governance Workflows

Objective

Support domain growth.

28.8 Discovery Evolution Roadmap

Discovery capabilities shall evolve progressively.

Phase 1

Keyword Discovery

-

Hybrid Retrieval

Phase 2

Advanced Semantic Discovery

Phase 3

Context-Aware Discovery

Phase 4

Predictive Opportunity Discovery

Objective

Improve discovery effectiveness.

28.9 Opportunity Evolution Roadmap

Opportunity domains shall expand.

Potential Future Domains

Education

Employment

Research

Funding

Community Programs

Innovation Programs

Public Services

Objective

Increase ecosystem coverage.

28.10 Authority Evolution Roadmap

Authority management shall mature over time.

Future Capabilities

Authority Networks

Authority Relationships

Authority Trust Models

Authority Ecosystems

Objective

Improve authority intelligence.

28.11 Evidence Evolution Roadmap

Evidence frameworks shall expand.

Future Capabilities

Evidence Graphs

Evidence Networks

Evidence Relationship Models

Evidence Quality Intelligence

Objective

Improve evidence transparency.

28.12 Ranking Evolution Roadmap

Ranking mechanisms may evolve while preserving transparency.

Current Principle

Total Cost Ranking

Future Enhancements

Improved Cost Estimation

Improved Context Evaluation

Improved Explainability

Constraint

No hidden ranking logic.

28.13 Recommendation Evolution Roadmap

Recommendation capabilities shall mature gradually.

Future Enhancements

Better Explanations

Improved Contextualization

Improved Traceability Presentation

Enhanced User Understanding

Constraint

Recommendations remain subordinate to rankings.

28.14 Opportunity Creation Evolution

Creation workflows shall expand.

Future Capabilities

Advanced Similarity Analysis

Pattern Intelligence

Gap Intelligence

Opportunity Design Assistance

Objective

Support innovation.

28.15 Service Evolution Framework

Services shall evolve independently.

Objectives

Independent deployment.

Independent scaling.

Independent replacement.

Principle

Loose coupling.

28.16 API Evolution Framework

APIs shall evolve through versioning.

Objectives

Backward compatibility.

Controlled change.

Consumer protection.

Principle

Version before breaking.

28.17 Event Evolution Framework

Events shall support long-term evolution.

Objectives

Version management.

Consumer compatibility.

Traceability preservation.

Principle

Events remain governed assets.

28.18 Data Evolution Framework

Data structures shall support evolution.

Techniques

Versioning

Schema Migration

Historical Preservation

Compatibility Controls

Objective

Protect information continuity.

28.19 Database Evolution Roadmap

Persistence architecture may evolve.

Current

PostgreSQL + pgvector

Future Possibilities

Distributed PostgreSQL

Dedicated Vector Infrastructure

Graph Databases

Analytical Platforms

Constraint

Governance requirements remain unchanged.

28.20 Infrastructure Evolution Roadmap

Infrastructure capabilities shall evolve.

Current

Container-Based Deployment

Future Possibilities

Multi-Region Infrastructure

Global Infrastructure

Advanced Automation

Self-Healing Systems

Objective

Support growth.

28.21 Observability Evolution Roadmap

Observability shall mature continuously.

Future Capabilities

Operational Intelligence

Governance Intelligence

Trust Intelligence

Predictive Monitoring

Objective

Increase visibility.

28.22 Security Evolution Roadmap

Security capabilities shall evolve with threats.

Future Areas

Advanced Identity Controls

Adaptive Security

Threat Intelligence

Continuous Verification

Objective

Protect trust.

28.23 Governance Evolution Roadmap

Governance frameworks shall mature.

Future Capabilities

Advanced Governance Dashboards

Governance Intelligence

Continuous Assurance

Automated Governance Validation

Objective

Improve accountability.

28.24 AI Evolution Framework

AI capabilities shall evolve cautiously.

Permitted Areas

Summarization

Explanation

Classification Assistance

Discovery Assistance

Analysis Assistance

Restricted Areas

Authority Decisions

Evidence Validation

Ranking Authority

Governance Decisions

Trust Determination

Objective

Preserve accountability.

28.25 Human-AI Balance Roadmap

Future evolution shall preserve human responsibility.

Principle

AI Assists

↓

Humans Decide

Objective

Prevent authority displacement.

28.26 Extensibility Design Principles

All future capabilities shall follow extensibility rules.

Principles

Modularity

Loose Coupling

Versioning

Traceability

Auditability

Reproducibility

Objective

Support sustainable growth.

28.27 Technology Replacement Strategy

Technology components may be replaced.

Replacement Conditions

Performance Limitations

Operational Limitations

Cost Limitations

Strategic Requirements

Requirement

Business principles remain unchanged.

28.28 Architectural Review Framework

Architecture shall undergo periodic review.

Review Areas

Scalability

Security

Governance

Performance

Reliability

Business Alignment

Objective

Maintain architectural health.

28.29 Evolution Governance Framework

Architectural evolution shall be governed.

Governance Inputs

Business Needs

Technical Needs

Risk Assessment

Trust Impact Assessment

Objective

Prevent uncontrolled evolution.

28.30 Innovation Governance Framework

Innovation shall remain accountable.

Requirements

Traceability

Auditability

Risk Evaluation

Governance Review

Objective

Balance innovation and trust.

28.31 Future Capability Domains

Potential future capability areas include:

Domain 1

Opportunity Intelligence

Domain 2

Authority Intelligence

Domain 3

Evidence Intelligence

Domain 4

Trust Intelligence

Domain 5

Governance Intelligence

Domain 6

Global Opportunity Networks

28.32 Evolution Metrics

Architectural evolution shall be measurable.

Example Metrics

Platform Extensibility

Deployment Frequency

Technology Adoption

Technical Debt

Governance Compliance

Objective

Monitor architectural health.

28.33 AI Usage Policy

AI may assist:

- Architectural analysis
 - Roadmap planning
 - Capability exploration
 - Impact assessment
-

AI May Not

Approve architectural changes.

Override governance reviews.

Modify core principles.

Change trust requirements.

Reason

Platform evolution must remain accountable.

28.34 Evolution Governance Rules

The following rules are mandatory.

EVO-01

Core principles must remain preserved.

EVO-02

Trust mechanisms must remain preserved.

EVO-03

Architectural changes must be traceable.

EVO-04

Architectural changes must be auditable.

EVO-05

Technology changes must not alter governance requirements.

EVO-06

Innovation must remain accountable.

EVO-07

Human responsibility must remain preserved.

28.35 Architectural Constraints

The following constraints are non-negotiable.

AC-EVO-01

Evidence First remains unchanged.

AC-EVO-02

Authority Before Recommendation remains unchanged.

AC-EVO-03

Ranking Before Recommendation remains unchanged.

AC-EVO-04

No Hidden Ranking Logic remains unchanged.

AC-EVO-05

AI Is Not An Authority remains unchanged.

AC-EVO-06

Human Creates Novelty remains unchanged.

28.36 Long-Term Architectural Vision

The long-term vision of TODP is to become a trusted opportunity infrastructure platform supporting:

Discovery



Verification



Ranking



Recommendation



Creation

across multiple domains, geographies, and organizational ecosystems.

This vision shall be achieved through controlled evolution rather than disruptive redesign.

28.37 Future Evolution, Extensibility & Architectural Roadmap Summary

The Future Evolution Architecture provides a structured roadmap for growing TODP while preserving its foundational trust model.

The framework enables:

Business Evolution



Technology Evolution



Governance Evolution



Capability Expansion



Innovation

while ensuring that core principles remain stable.

This architecture guarantees that TODP can evolve for years without compromising transparency, accountability, traceability, auditability, reproducibility, or human responsibility.

SDD Progress

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SDD V1 — Section 29

Architecture Decision Records (ADR) & Foundational Architectural Decisions

29.1 Purpose

This section documents the major Architectural Decision Records (ADRs) that define the foundation of TODP.

Architectural decisions are among the most important assets of a system because they explain:

- Why a decision was made
- Which alternatives were considered
- Which trade-offs were accepted
- Which constraints influenced the decision
- Which principles the decision supports

Without ADRs, future teams may understand what was built but not why it was built.

The purpose of this section is to preserve architectural intent and institutional knowledge.

29.2 ADR Philosophy

Architectures are not collections of technologies.

Architectures are collections of decisions.

Every major architectural decision should be:

- Explicit
 - Traceable
 - Reviewable
 - Defensible
 - Reproducible
-

Core Principle

Important architectural decisions must be documented.

29.3 ADR Structure

Every ADR shall contain:

Decision ID

Unique identifier.

Decision

What was decided.

Context

Why the decision was needed.

Alternatives

Options considered.

Consequences

Impacts and trade-offs.

Status

Approved, Deprecated, Replaced, or Proposed.

29.4 ADR-001

Evidence First Architecture

Status

Approved

Decision

Evidence is the primary foundation of trust.

Context

Recommendations without evidence create unverifiable outcomes.

Alternatives Considered

Expert Opinion First

AI Recommendation First

Popularity-Based Systems

Decision Rationale

Evidence provides objective support for claims.

Consequences

Evidence management becomes a core platform capability.

29.5 ADR-002

Authority Before Recommendation

Status

Approved

Decision

Recommendations must reference validated authorities.

Context

Users require trustworthy sources.

Alternatives Considered

AI-Only Recommendations

Crowdsourced Recommendations

Popularity-Based Recommendations

Decision Rationale

Authorities improve credibility.

Consequences

Authority verification becomes mandatory.

29.6 ADR-003

Ranking Before Recommendation

Status

Approved

Decision

Ranking must occur before recommendations are generated.

Context

Recommendations require an objective foundation.

Alternatives Considered

Recommendation First

AI Ranking

Popularity Ranking

Decision Rationale

Ranking provides deterministic prioritization.

Consequences

Ranking becomes a core platform service.

29.7 ADR-004

Total Cost Ranking Model

Status

Approved

Decision

Ranking shall be based on:

Money Cost

-

Time Cost

-

Distance Cost

Context

Users allocate finite resources.

Alternatives Considered

Popularity-Based Ranking

Engagement-Based Ranking

Sponsored Ranking

Manual Ranking

Decision Rationale

Resource expenditure provides a universal comparison model.

Consequences

Ranking remains explainable and measurable.

29.8 ADR-005

No Sponsored Content

Status

Approved

Decision

Sponsored content shall not influence ranking.

Context

Sponsored placement creates conflicts of interest.

Alternatives Considered

Sponsored Opportunities

Promoted Results

Advertising-Based Ranking

Decision Rationale

Trust requires independence.

Consequences

Revenue models cannot depend upon ranking manipulation.

29.9 ADR-006

No Hidden Ranking Logic

Status

Approved

Decision

Ranking logic shall remain transparent.

Context

Opaque rankings undermine trust.

Alternatives Considered

Proprietary Ranking Logic

Black-Box Ranking

Hidden Weighting Systems

Decision Rationale

Transparency improves accountability.

Consequences

Ranking explanations become mandatory.

29.10 ADR-007

AI Is Not An Authority

Status

Approved

Decision

AI cannot act as an authority source.

Context

AI generates outputs but does not establish truth.

Alternatives Considered

AI Authority Systems

AI Validation Systems

AI Governance Systems

Decision Rationale

Authority must remain independently verifiable.

Consequences

Authority frameworks remain human-verifiable.

29.11 ADR-008

Human Creates Novelty

Status

Approved

Decision

Novel opportunity creation remains a human responsibility.

Context

AI recombines information.

Human creativity introduces new concepts.

Alternatives Considered

Fully Autonomous Innovation

AI-Led Opportunity Creation

Decision Rationale

Human judgment remains accountable.

Consequences

Creation workflows remain human-centered.

29.12 ADR-009

Hybrid Retrieval Architecture

Status

Approved

Decision

Retrieval shall combine BM25 and Vector Search.

Context

Single retrieval approaches have limitations.

Alternatives Considered

BM25 Only

Vector Only

Rule-Based Retrieval

Decision Rationale

Hybrid retrieval improves discovery coverage.

Consequences

Retrieval complexity increases.

29.13 ADR-010

PostgreSQL as System of Record

Status

Approved

Decision

PostgreSQL serves as the primary persistence platform.

Context

Need for reliability, maturity, and governance.

Alternatives Considered

NoSQL Databases

Graph Databases

Distributed Databases

Decision Rationale

Mature ecosystem and governance support.

Consequences

Structured data management emphasis.

29.14 ADR-011

pgvector for Semantic Retrieval

Status

Approved

Decision

Semantic retrieval shall use pgvector initially.

Context

Need semantic search without separate infrastructure.

Alternatives Considered

Dedicated Vector Databases

External Retrieval Systems

Decision Rationale

Operational simplicity.

Consequences

Future migration path remains open.

29.15 ADR-012

FastAPI Backend Architecture

Status

Approved

Decision

Backend services use FastAPI.

Context

Need high-performance API services.

Alternatives Considered

Django

Flask

Node.js Services

Decision Rationale

Performance, simplicity, and API focus.

Consequences

API-first architecture emphasis.

29.16 ADR-013

React Frontend Architecture

Status

Approved

Decision

Frontend shall use React.

Context

Need flexible user interfaces.

Alternatives Considered

Angular

Vue

Server-Rendered Applications

Decision Rationale

Mature ecosystem and flexibility.

Consequences

Component-driven UI architecture.

29.17 ADR-014

Event-Driven Communication

Status

Approved

Decision

Use hybrid synchronous and asynchronous communication.

Context

Need scalability without excessive coupling.

Alternatives Considered

Synchronous Only

Asynchronous Only

Monolithic Communication

Decision Rationale

Balance consistency and scalability.

Consequences

Event governance required.

29.18 ADR-015

Auditability as Product Capability

Status

Approved

Decision

Auditability is a core platform capability.

Context

Trust requires accountability.

Alternatives Considered

Compliance-Only Auditing

Limited Logging

Operational Auditing Only

Decision Rationale

Users require defensible decisions.

Consequences

Audit systems become first-class architecture.

29.19 ADR-016

Traceability as Core Principle

Status

Approved

Decision

All important workflows must be traceable.

Context

Users need to understand outcomes.

Alternatives Considered

Partial Traceability

Operational Traceability Only

Decision Rationale

Trust requires lineage visibility.

Consequences

Lineage becomes mandatory.

29.20 ADR-017

Reproducibility Requirement

Status

Approved

Decision

Important outcomes must be reproducible.

Context

Historical verification is required.

Alternatives Considered

Current-State-Only Systems

Audit-Only Systems

Decision Rationale

Trust requires verification.

Consequences

Historical state preservation required.

29.21 ADR-018

Security by Architecture

Status

Approved

Decision

Security shall be integrated into architecture.

Context

Trust depends on protection.

Alternatives Considered

Perimeter Security Only

Reactive Security

Decision Rationale

Security must be systemic.

Consequences

Security becomes a cross-cutting concern.

29.22 ADR-019

Observability as First-Class Capability

Status

Approved

Decision

Observability is a core architectural capability.

Context

Traceability and governance require visibility.

Alternatives Considered

Basic Monitoring

Infrastructure Monitoring Only

Decision Rationale

Operational intelligence improves trust.

Consequences

Telemetry becomes mandatory.

29.23 ADR-020

Governance-Centric Architecture

Status

Approved

Decision

Governance is embedded into platform architecture.

Context

Trust must be operationalized.

Alternatives Considered

Minimal Governance

Administrative Governance Only

Decision Rationale

Governance preserves accountability.

Consequences

Governance services become architectural components.

29.24 Decision Relationship Model

Architectural decisions are interconnected.

Example Chain

Evidence First

↓

Authority Before Recommendation

↓

Ranking Before Recommendation

↓

Recommendation Architecture

↓

Traceability Requirements

↓

Auditability Requirements

Objective

Preserve decision coherence.

29.25 Decision Review Framework

ADRs shall be reviewed periodically.

Review Triggers

Major Releases

Architecture Changes

Technology Changes

Governance Changes

Objective

Maintain architectural relevance.

29.26 ADR Lifecycle

Architectural decisions follow a lifecycle.

Proposed

Decision under consideration.

Approved

Decision accepted.

Deprecated

Decision no longer recommended.

Replaced

Decision superseded.

Objective

Support architectural evolution.

29.27 ADR Traceability Framework

ADRs shall remain linked to architecture artifacts.

Traceability Targets

PRD

ADD

DDD

SDD

Policies

Governance Controls

Objective

Preserve decision lineage.

29.28 ADR Governance Rules

The following rules are mandatory.

ADR-01

Major architectural decisions must be documented.

ADR-02

Architectural rationale must be preserved.

ADR-03

Decision alternatives must be recorded.

ADR-04

Decision impacts must be documented.

ADR-05

Decision reviews must be traceable.

ADR-06

Architectural evolution must update ADRs.

29.29 Architectural Constraints

The following constraints are non-negotiable.

AC-ADR-01

Core principles require ADR protection.

AC-ADR-02

Architectural intent must remain preserved.

AC-ADR-03

ADR history must remain available.

AC-ADR-04

Decision rationale must remain auditable.

AC-ADR-05

Architectural evolution must remain traceable.

AC-ADR-06

Foundational decisions may only change through governance review.

29.30 ADR & Foundational Architectural Decisions Summary

The Architecture Decision Records framework preserves the reasoning behind TODP's most important architectural choices.

The framework documents:

Core Principles



Architectural Decisions



Decision Rationale



Trade-Offs



Governance Requirements



Evolution History

ensuring that future teams understand not only what was built, but why it was built.

This section serves as the permanent architectural memory of TODP and protects the platform from unintended drift away from its foundational principles.

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SDD V1 — Section 30

SDD Conclusion, Architectural Integrity Statement & Final Approval Baseline

30.1 Purpose

This section formally concludes System Design Document (SDD) V1 for TODP.

The purpose of this section is to:

- Establish architectural closure
- Define architectural integrity requirements
- Define implementation boundaries
- Establish approval authority
- Define governance obligations
- Establish the official architectural baseline

Upon approval, SDD V1 becomes the authoritative technical architecture reference for TODP implementation.

30.2 SDD V1 Scope Summary

SDD V1 defines the complete system architecture for TODP.

The architecture covers:

Business Architecture

Opportunity Discovery

Authority Verification

Evidence Validation

Ranking

Recommendation

Opportunity Creation

Application Architecture

Frontend

Backend

Services

APIs

Workflows

Data Architecture

Data Models

Persistence

Governance

Lifecycle Management

Operational Architecture

Security

Observability

Reliability

Deployment

Testing

Governance

Trust Architecture

Auditability

Traceability

Reproducibility

Compliance

Trust Assurance

30.3 Architectural Vision Statement

TODP exists to become a trusted opportunity infrastructure platform.

The platform shall support:

Discovery

↓

Verification

↓

Ranking

↓

Recommendation

↓

Creation

while preserving:

- Transparency
- Accountability

- Explainability
- Auditability
- Traceability
- Reproducibility

for all important platform outcomes.

30.4 Architectural Integrity Statement

Architectural integrity means preserving consistency between:

Business Principles

↓

Architecture

↓

Implementation

↓

Operations

↓

Governance

Objective

Prevent architectural drift.

30.5 Architectural Integrity Principles

The following principles define TODP's architectural identity.

Principle 1

Evidence First

Principle 2

Authority Before Recommendation

Principle 3

Ranking Before Recommendation

Principle 4

Traceability

Principle 5

Auditability

Principle 6

Reproducibility

Principle 7

No Sponsored Content

Principle 8

No Paid Placement

Principle 9

No Hidden Ranking Logic

Principle 10

AI Is Not An Authority

Principle 11

Human Creates Novelty

30.6 Architectural Integrity Rules

All future implementation activities shall preserve architectural integrity.

AI Systems

Must comply with architecture.

User Interfaces

Must comply with architecture.

Services

Must comply with architecture.

Data Models

Must comply with architecture.

Governance Controls

Must comply with architecture.

30.7 Architecture-to-Implementation Alignment

Implementation shall align with architecture.

Requirement

No implementation may knowingly violate approved architecture.

Exception

Only approved governance processes may authorize changes.

30.8 Approved Technology Baseline

The following technologies constitute the approved V1 implementation baseline.

Frontend

React

Backend

FastAPI

Database

PostgreSQL

Vector Search

pgvector

Cache

Redis

Observability

OpenTelemetry

Prometheus

Grafana

Retrieval

BM25

Vector Search

Hybrid Retrieval

30.9 Approved Architectural Baseline

The following architectural domains are approved.

Discovery Architecture

Approved

Authority Architecture

Approved

Evidence Architecture

Approved

Ranking Architecture

Approved

Recommendation Architecture

Approved

Opportunity Creation Architecture

Approved

Governance Architecture

Approved

30.10 Trust Preservation Baseline

The following trust mechanisms are mandatory.

Evidence Validation

Required

Authority Verification

Required

Ranking Transparency

Required

Recommendation Traceability

Required

Audit Recording

Required

Historical Reproducibility

Required

30.11 Governance Preservation Baseline

Governance requirements shall remain enforceable.

Administrative Governance

Mandatory

Operational Governance

Mandatory

Trust Governance

Mandatory

Compliance Governance

Mandatory

Information Governance

Mandatory

30.12 Architectural Ownership

Architecture ownership shall be assigned.

Responsibilities

Architecture Maintenance

Architecture Evolution

Architecture Reviews

Architecture Governance

Architecture Integrity

Objective

Preserve architectural accountability.

30.13 Architectural Change Framework

Architecture changes require governance.

Trigger Examples

Business Changes

Technology Changes

Governance Changes

Risk Changes

Scalability Requirements

Objective

Prevent uncontrolled evolution.

30.14 Architectural Review Requirements

Architecture shall undergo periodic review.

Review Areas

Business Alignment

Security

Performance

Scalability

Governance

Trust Preservation

Objective

Maintain relevance.

30.15 Architectural Compliance Requirements

Implementation shall be assessed for compliance.

Compliance Areas

Architecture

Security

Governance

Trust

Operations

Objective

Validate adherence.

30.16 Architecture Quality Attributes

The architecture is designed to achieve the following quality attributes.

Trustworthiness

Scalability

Reliability

Security

Maintainability

Extensibility

Auditability

Traceability

Reproducibility

30.17 Architectural Success Criteria

The architecture shall be considered successful if it enables:

Trusted Discovery

Trusted Ranking

Trusted Recommendation

Trusted Governance

Trusted Operations

Sustainable Evolution

30.18 Implementation Governance Requirements

Implementation teams shall follow architecture governance.

Requirements

Architecture Reviews

Design Reviews

Testing Reviews

Security Reviews

Governance Reviews

Objective

Maintain alignment.

30.19 Architectural Risk Statement

Architectural risks shall be monitored continuously.

Risk Categories

Technology Risks

Operational Risks

Governance Risks

Trust Risks

Scalability Risks

Objective

Protect architectural integrity.

30.20 Future Evolution Commitment

Future evolution shall preserve foundational principles.

Requirement

Evolution may modify implementation.

Evolution may modify technologies.

Evolution may modify workflows.

Constraint

Evolution shall not violate core principles.

30.21 AI Governance Commitment

AI usage shall remain bounded by governance controls.

AI May Assist

Discovery

Analysis

Explanation

Summarization

Classification

AI May Not

Act as authority.

Replace evidence.

Override rankings.

Override governance.

Approve trust decisions.

Objective

Maintain accountability.

30.22 Human Responsibility Commitment

Human responsibility remains fundamental.

Responsibilities

Governance

Authority Validation

Trust Decisions

Architectural Decisions

Novel Opportunity Creation

Objective

Preserve accountability.

30.23 SDD Approval Baseline

Approval of SDD V1 signifies acceptance of:

Architectural Scope

Approved

Architectural Principles

Approved

Architectural Constraints

Approved

Technology Baseline

Approved

Governance Requirements

Approved

Trust Requirements

Approved

30.24 Frozen Baseline Definition

Upon approval, SDD V1 becomes a frozen baseline.

Frozen Elements

Architecture

Principles

Constraints

Governance Requirements

Trust Requirements

Change Requirement

Formal governance review required.

30.25 Relationship to Other Documents

SDD V1 shall be interpreted together with:

PRD V1

Product Requirements

ADD V1

Architectural Definition

DDD V1

Domain-Driven Design

SDD V1

System Design

Objective

Maintain architectural consistency.

30.26 Architectural Authority Statement

SDD V1 is the authoritative system architecture reference for TODP.

Scope

Implementation

Operations

Governance

Evolution

Review Activities

Objective

Provide a single source of architectural truth.

30.27 Architectural Integrity Oath

All future contributors to TODP shall strive to preserve:

- Transparency
- Accountability
- Traceability
- Auditability
- Reproducibility
- Trustworthiness

while evolving platform capabilities.

Objective

Protect long-term platform integrity.

30.28 Final Architectural Declaration

TODP is designed as a trust-centered opportunity infrastructure platform.

Its architecture is intentionally built upon:

Evidence

↓

Authority

↓

Ranking

↓

Recommendation

↓

Creation

rather than:

Advertising

↓

Popularity

↓

Engagement Manipulation

↓

Opaque Algorithms

This distinction defines the identity of the platform.

30.29 SDD V1 Completion Statement

System Design Document Version 1 is hereby considered complete.

The document establishes:

- Architectural foundations
- Technology foundations
- Governance foundations
- Trust foundations
- Operational foundations

required for implementation and future evolution.

30.30 Final Approval Baseline

The following baseline is formally established.

Document Status

APPROVED

Version

SDD V1

State

FROZEN

Authority

Official TODP System Architecture

Change Policy

Governed Change Only

Effective Date

Upon Approval

Supersession Rule

Future SDD versions shall explicitly supersede this baseline.

SDD V1 Final Summary

The TODP System Design Document defines a complete enterprise-grade architecture for a trust-centered opportunity discovery platform.

The architecture encompasses:

Business Architecture



Application Architecture



Data Architecture



Security Architecture



Observability Architecture



Governance Architecture



Trust Architecture



Operational Architecture



Evolution Architecture

while preserving the foundational principles that define TODP.

SDD V1 now serves as the official implementation and governance blueprint for all future TODP development activities.

SDD V1 Status

- ☐ Section 1 — Document Control & Purpose
- ☐ Section 2 — System Context & High-Level Architecture
- ☐ Section 3 — Logical Architecture & Service Decomposition
- ☐ Section 4 — Physical Architecture & Deployment Topology
- ☐ Section 5 — Infrastructure Architecture & Environment Design
- ☐ Section 6 — Data Architecture & Data Management Strategy
- ☐ Section 7 — Database Design & Persistence Architecture
- ☐ Section 8 — Retrieval Architecture & Search Strategy
- ☐ Section 9 — Ranking Architecture & Decision Engine
- ☐ Section 10 — Recommendation Architecture & Explanation Engine
- ☐ Section 11 — Authority Architecture & Verification Framework
- ☐ Section 12 — Evidence Architecture & Validation Framework
- ☐ Section 13 — Opportunity Architecture & Lifecycle Management
- ☐ Section 14 — User Architecture & Context Management Framework
- ☐ Section 15 — Opportunity Creation Architecture & Innovation Framework
- ☐ Section 16 — API Architecture & Service Contracts
- ☐ Section 17 — Security Architecture & Trust Framework
- ☐ Section 18 — Observability Architecture & Operational Intelligence Framework
- ☐ Section 19 — Auditability, Traceability & Reproducibility Architecture
- ☐ Section 20 — Event-Driven Architecture & Inter-Service Communication Framework
- ☐ Section 21 — Scalability, Performance & Capacity Architecture
- ☐ Section 22 — Reliability, Resilience & Disaster Recovery Architecture
- ☐ Section 23 — Deployment Architecture, Release Management & CI/CD Framework

- Section 24 — Testing, Quality Assurance & Validation Architecture
 - Section 25 — Operational Governance & Platform Administration Architecture
 - Section 26 — Compliance, Risk Management & Trust Assurance Architecture
 - Section 27 — Data Lifecycle, Retention & Information Governance Architecture
 - Section 28 — Future Evolution, Extensibility & Architectural Roadmap
 - Section 29 — Architecture Decision Records (ADR) & Foundational Architectural Decisions
 - Section 30 — SDD Conclusion, Architectural Integrity Statement & Final Approval Baseline
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SYSTEM DESIGN DOCUMENT (SDD V1)

STATUS: COMPLETE

VERSION: FROZEN

APPROVAL STATE: READY FOR IMPLEMENTATION